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CHAPTER ONE

Population Growth and Economic Development: Causes, Consequences, and Controversies

Introduction

In this unit, we examine many of the issues relating population growth to economic development. After explaining basic demographic concepts, we present some well-known economic models and hypotheses regarding the causes and consequences of rapid population growth in contemporary developing countries. Controversies surrounding the significance of the population factor in general and these models and hypotheses in particular are then explored. Finally, we evaluate a range of alternative policy options that developing countries may wish to adapt to influence growth of their population, as well as ways in which industrialized countries can contribute to a more manageable global population and resource environment.

Unit Objectives:

After completing this unit, you should be able to:

- explain the basic demographic concepts
- Understand the linkage between population growth and economic development have deep knowledge in areas of economic models and hypotheses regarding what cause high population growth and its consequences in the contemporary developing nations
- Examine controversies surrounding the significance of population growth
- Analyze and evaluate alternative policy options that developing countries should pursue
- Be able to analyze what developed nations should do in relation to fair and equitable global resource utilization

1.1 The Basic Issue: Population Growth and the Quality of Life

Every year approximately above 110 million people are being added to the population. Almost all this net population increase (97%) is in developing countries. Increases of such magnitude are unprecedented. But the problem of population growth is not simply a problem of numbers. It is a problem of human welfare and of development. Rapid population growth can have serious consequences for the well-being of all of humanity. If development entails the improvement in peoples levels of living, then the really important question about population growth is this: How does the contemporary population situation in many LDC contribute to or detract from their chances of realizing the goals of development, not only for the current generation but also for future generations? Conversely, how does development affect population growth? Among the major issues relating to this basic question are the following:

1. Will developing countries be capable of improving the levels of living for their people with the current and anticipated levels of population growth? To what extent does rapid population increase make it more difficult to provide essential social services, including housing, transport, sanitation, and security?
2. How will the developing countries be able to cope with the vast increases in their labor forces over the coming decades? Will employment opportunities be plentiful, or will it be a major achievement just to keep unemployment levels from rising?
3. What are the implications of higher population growth rates among the worlds poor for their chances of overcoming the human misery of absolute poverty? Will world food supply and its distribution be sufficient not only to meet the anticipated population increase in the coming decades but also to improve nutritional levels to the point where all humans can have an adequate diet?
4. Given the anticipated population growth, will developing countries be able to extend the coverage and improve the quality of their health and educational systems so that everyone can at least have the chance to secure adequate health care and a basic education?
5. To what extent are low levels of living an important factor in limiting the freedom of parents to choose a desired family size? Is there a relationship between poverty and family size?
6. To what extent is the governing affluence among the economically more developed nations an important factor preventing poor nations from accommodating their growing populations? Is the inevitable pursuit of increasing affluence among the rich more detrimental to the global environment and to rising living standards among the poor than the absolute increase in their numbers? We will try to explore and explain these and other problems in this course.

1.2 Population Growth — Past, Present, and Future: A Review

1.2.1 World Population Growth through History

Throughout most of the more than 2 million years of human existence on earth, humanity's numbers have been few. When people first started to cultivate food through agriculture some 12,000 years ago, the estimated world population was no more than 5 million. At the beginning of the Common Era 2,000 years ago, world population had grown to nearly 250 million, less than a fifth of the population of China today. From A.D. to the beginning of the industrial revolution around 1750, it tripled to 728 million people, well under the total number living in India today. During the next 200 years (1750-1950), an additional 1.7 billion people were added to the earth's

numbers. But in just four decades thereafter (1950-1990), world population more than doubled again, bringing the total figure to around 5.3 billion. The world entered the twenty-first century with almost 6.1 billion people.

Turning from absolute numbers to percentage growth rates for almost the whole of human existence on earth until approximately 300 years ago, population grew at an annual rate not much greater than zero (0.002%). Naturally, this overall rate was not steady; there were many ups and downs as a result of natural catastrophes and variations in growth rates among regions. By 1750, the population growth rate had accelerated by 150 times, from 0.002% to 0.3% per year. By the 1950s, the rate had again accelerated, tripling to about 1.0% per year. It continued to accelerate until around 1970, when it peaked at 2.35%. Today the world's population growth rate remains at a historically high rate of 1.3% per year, but the rate of increase is slowing. However, the population growth rate in Africa is still extremely high 2.4% per year.

The reason for the sudden change in overall population trends is that for almost all of recorded history, the rate of population changes, whether up or down, had been strongly influenced by the combined effects of famine, disease, malnutrition, plague, and war-conditions that resulted in high and fluctuating death rates. In the twentieth century, such conditions came increasingly under technological and economic control. As a result, human mortality is now lower than at any other point in human existence. The decline in mortality resulting from rapid technological advances in modern medicine and the spread of modern sanitation measures throughout the world, particularly within the past half century that has resulted in the unprecedented increases in world population growth, especially in developing countries.

In short, population growth today is primarily the result of a rapid transition from a long historical era characterized by high birth and death rates to one in which death rates have fallen sharply but birth rates, especially in developing countries, are only just beginning to fall from their historically high levels.

1.2.2 Structure of the World's Population

The world's population is very unequally distributed by geographic region, by fertility and mortality levels, and by age structures.

Geographic Region More than three-quarters of the world's people live in developing countries; less than one person in four lives in an economically developed nation. Given current population growth rates in different parts of the world, the regional distribution of the world's population will inevitably change by 2050. By that time, it is likely that there will be 6.6 billion more people on the earth than in 1950 and 3.1 billion more than in 2000. Africa will experience the largest

percentage increase (84%), and its projected population of 2.1 billion will be almost three times its 1998 population and almost 10 times its 1950 population.

Fertility and Mortality Trends: the **rate of population increase** is quantitatively measured as the percentage yearly net relative increase in population size due to **natural increase** and net **international migration**. Natural increase simply measures the excess of births over deaths or, in more technical terms, the difference between fertility and mortality rates.

- The difference between developing and developed nations in terms of rates of population growth can be explained simply by the fact the **birth rates (fertility rates)**. In developing countries birth rate are generally much higher than in the rich nations. Death rates (mortality rates) are also higher. However, these death rate differences are substantially smaller than the differences in birthrates.
- While fertility has been declining in many Developing countries, there has been a rapid narrowing of the gap in mortality rates between developed and less developed countries. The primary reason is undoubtedly the improvement in health conditions throughout the developing world. Modern vaccination campaigns against diseases as well as the proliferation of public health facilities, clean water supplies, improved nutrition, and public education have all worked together over the past three decades to lower death rates by as much as 50% in parts of Asia and Latin America and by over 30 % in much of Africa and the Middle East. Nevertheless, the average life span remains almost 13 years greater in the developed countries. But even this gap has been sharply reduced in recent decades.

Age Structure and dependency Burdens: World population today is very youthful, particularly in the developing world. Children under the age of 15 constitute more than 30 % of the total population of developing countries but just 18% of developed nations.

For example, 46% of Ethiopians population is young (less than 15 years old). In countries with such age structure, the **youth dependency ratio** — the proportion of youths (under age 15) to economically active adults (ages 15 to 64) — is very high. Thus, the workforce in developing countries must support almost twice as many children as it does in the wealthier countries. For example, in Sweden and the United Kingdom, the workforce age group (15 to 64) amounts to about 65% of the total population. This workforce has to support only about 18% of the population as youthful dependents. Their main problems relate more to their low population growth and old-age dependents (over age 65), who represent 18% to 16% of their populations.

By contrast, in countries like Ethiopia, the economically active workforces and the child dependents each make up about 46% of the total population. In general, the more rapid the

population growth rate, the greater the proportion of dependent children in the total population and the more difficult it is for people who are working to support those who are not. This phenomenon of youth dependency also leads to an important concept, the **hidden momentum of population growth**.

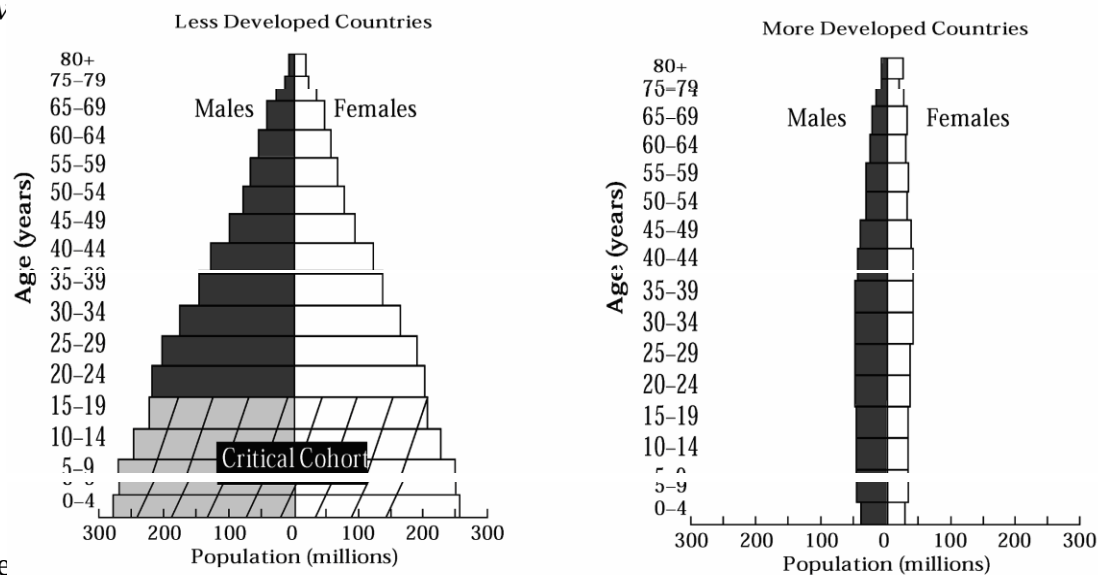
1.3 The Hidden Momentum of Population Growth

Perhaps the least understood aspect of population growth is its tendency to continue even after birthrates have declined substantially. Population growth has a **built-in tendency** to continue, a powerful momentum that, like a speeding automobile when the brakes are applied, tends to keep going for some time before coming to a stop. **In the case of population growth, this momentum can persist for decades after birthrates drop.** There are two basic reasons for this.

First, high birthrates cannot be altered substantially overnight. The social, economic, and institutional forces that have influenced fertility rates over the course of centuries do not simply evaporate at the urging of national leaders. Consequently, even if developing countries assign top priority to the limitation of population growth, it will still take many years to lower national fertility to desired levels. The second and less obvious reason for the hidden momentum of population growth rate is related to the age structure of LDC (Less Developed Countries) populations.

Figure 1.1 illustrates the great difference between age structures in less developed and more developed countries by means of two **population pyramids** for 1998. Each pyramid rises by five-year age intervals from 0 to 80 + years for both males and females, with the total number in each age cohort measured on the horizontal axis. We see from the LDC pyramid that in 1998 there were over 2 billion young people below the age of 20 and over 400 million youths aged 15 to 19. By contrast, in more developed countries there were fewer than 75 million youths aged 15 to 19. In terms of future population growth, this enormous numerical difference is compounded by the fact that the number of teens giving birth can run as high as 18% to 24% of all teenagers in many Developing countries, while in some the ratio is less than 1%.

Figure 1:1 Population by age and sex (Population pyramids): Less Developed and More Dev



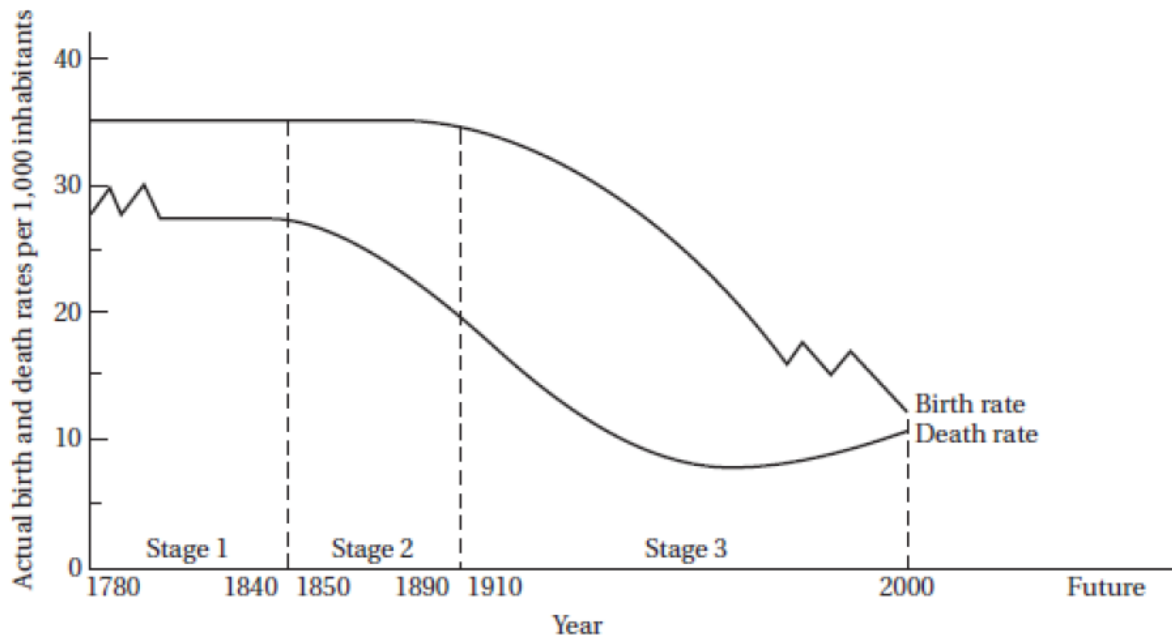
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young people greatly outnumber their parents, and when their generation reaches adulthood, the number of potential parents will inevitably be much larger than at present. It follows that even if these new parents have only enough children to replace themselves (two per couple, as compared with their parents, who may have had four children), the fact that the total number of couples having two children is much greater than the number of couples who previously had four children means that the total population will still increase substantially before leveling off.

As a result, developing countries should set goals for desirable future population sizes, they must accept the fact that increases on the order to 60% to 125% are coming regardless of the policy strategies they adopt. But this should not be a cause for despair or a diminished commitment on the part of countries that believe that slowing population growth is in their best national interest. The important message of population momentum is that every year that passes without a reduction in fertility means a larger multiple of the present total population size before it can eventually level off.

1.4 The Demographic Transition

The process by which fertility rates eventually decline to replacement levels has been portrayed by a famous concept in economic demography called the **demographic transition**. The demographic transition attempts to explain why all contemporary developed nations have more or less passed through the same three stages of modern population history.

Figure: 1. 2: Demographic Transition in Western Europe



Before their economic modernization, these countries for centuries had stable or very slow growing populations as a result of a combination of high birthrates and almost equally high death rates. This was stage 1. Stage 2 began when modernization, associated with better public-health methods healthier diets, higher incomes, and other improvements, led to a marked reduction in mortality that gradually raised life expectancy from under 40 years to over 60 years.

However, the decline in death rates was not immediately accompanied by a decline in fertility. As a result, the growing divergence between high birthrates and falling death rates led to sharp increases in population growth compared to past centuries. Stage 2 thus marks the beginning of the demographic transition (the transition from stable or slow-growing population first to rapidly increasing numbers and then to declining rates). Finally, stage 3 was entered when the forces and influences of modernization and development caused the beginning of a decline in fertility; eventually, falling birthrates converged with lower death rates, leaving little or no population growth.

Figure 1:3: Beginning of Demographic Transition in Developing countries

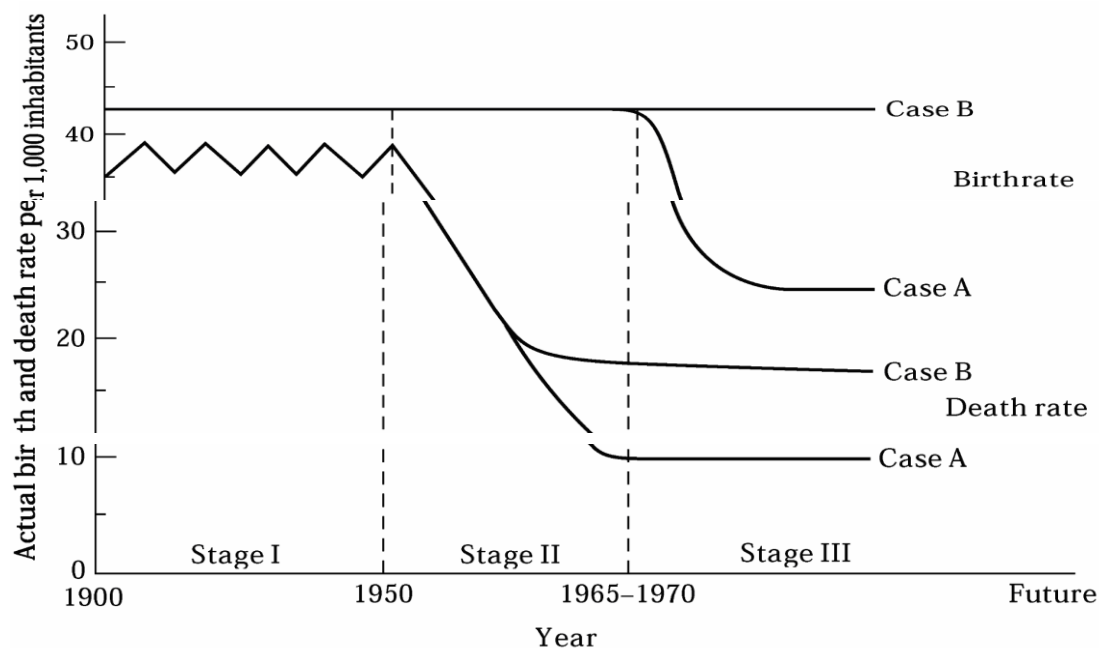


Figure 1.3 shows the population histories of contemporary developing countries, which contrast with those of Western Europe and fall into two patterns. Birthrates in many underdeveloped countries today are considerably higher than they were in pre-industrial Western Europe. This is because women tend to marry at an earlier age. As a result, there are both more families for a given population size and more years in which to have children. Beginning in the 1940s and especially in the 1950s and 1960s, stage 2 of the demographic transition occurred throughout most of the developing world. The application of highly effective imported modern medical and public-health technologies caused developing countries death rates to fall much more rapidly than in nineteenth-century European countries. Given their historically high birthrates (over 40 per 1,000 in many countries), this has meant that stage 2 of the developing countries demographic transition has been characterized by population growth rates well in excess of 2.0% per annum.

With regard to stage 3, we can distinguish between two broad classes of developing countries. In case A in figure 1.3, modern methods of death control combined with rapid and widely distributed rises in levels of living have resulted in death rates falling as low as 10 per 1,000 and birthrates also falling rapidly, to levels between 20 and 30 per 1,000. In the 1980s and 1990s, countries such as: Colombia, Indonesia, the Dominican Republic, Thailand, Malaysia, Mexico, Kenya, South Africa, and Brazil, appeared to be entering a period of sustained fertility decline consistent with case A.

But some developing countries fall into case B of figure 1.3. After an initial period of rapid decline, death rates have failed to drop further, largely because of the persistence of widespread poverty, and low levels of living more recently because of the AIDS epidemic. Moreover, the continuance of high birthrates as a result of these low levels of living causes overall population growth rates to remain relatively high. These countries, including many of those in sub Saharan Africa and the Middle East, are still in stage 2 of their demographic transition. Though fertility is declining, it remains very high in these parts of the world.

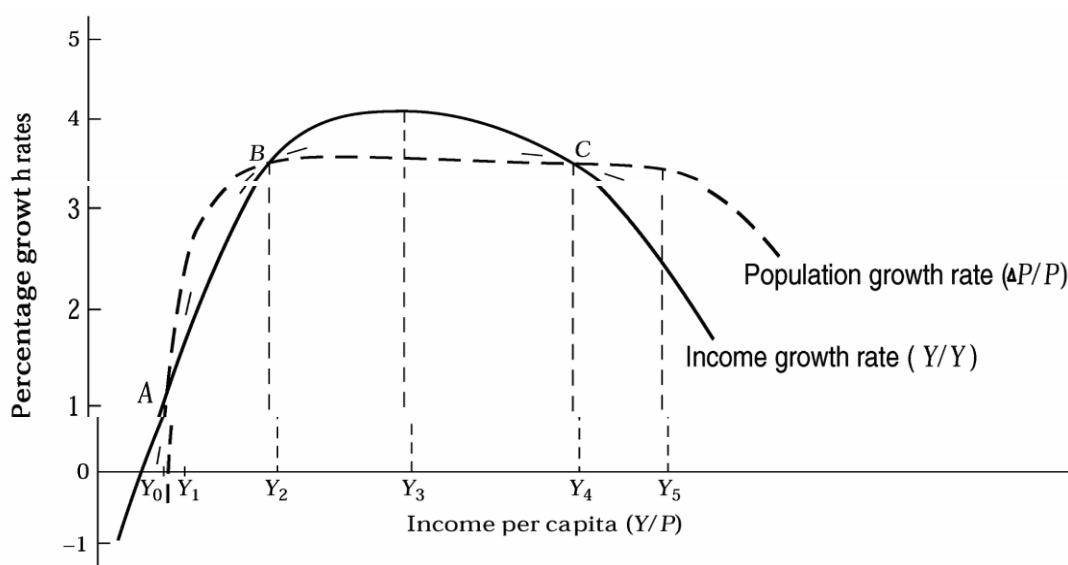
The important question, therefore, is this: When and under what conditions are developing nations likely to experience falling birthrates and a slower expansion of population? To answer this question, we need to ask a prior one. What are the principal causes of high fertility rates in developing countries, and can these determinants of the “demand” for children be influenced by government policy? To try to answer this critical question, we turn to a very old and famous classical macroeconomic and demographic model, the Malthusian “**population trap**,” and the **household theory of fertility**.

1.5 The Causes of High Fertility in Developing Countries: The Malthusian and Household Models

1.5.1 The Malthusian Population Trap

Two centuries ago, the Reverend Thomas Malthus put forward a theory of the relationship between population growth and economic development that still survives today. Writing in 1798 in his *Essay on the principle of population* and drawing on the concept of diminishing return, Malthus postulated a universal tendency for the population of a country, unless checked by dwindling food supplies, to grow at a geometric rate, doubling every 30 to 40 years. At the same time, because of diminishing returns to the fixed factor, land, food supplies could expand only at a roughly arithmetic rate. In fact, as each member of the population would have less land to work, his or her marginal contribution to food production would actually start to decline. Because the growth in food supplies could not keep pace with the burgeoning population, per capita incomes would tend to fall so low as to lead to a stable population existing barely at or slightly above the subsistence level. Malthus therefore contended that the only way to avoid this condition of chronic low levels of living or absolute poverty was for people to engage in “moral restraint” and limit the number of their children. Hence, we might regard Malthus, indirectly and inadvertently, as the father of the modern birth control movement.

Figure 1:4: The Malthusian population Trap



Modern economists have given a name to the Malthusian idea of a population inexorably forced to live at subsistence levels of income. They have called it the low-level equilibrium population traps or the Malthusian population traps. Diagrammatically, the basic Malthusian model can be illustrated by comparing the shape and position of curves representing population growth rates and aggregate income growth rates when these two curves are each plotted against level of per capita income. This is done in figure 1.4.

On the vertical axis, we plot numerical percentage changes, both positive and negative, in the two principal variables under consideration (total population and aggregate income). On the horizontal axis are levels of per capita income. Look first at the dashed curve portraying the assumed relationship between rates of population growth P/P (measured vertically) and levels of per capita income, Y/P (measured horizontally). At a very low level of per capita income, Y_0 , the rate of population change will be nil, and a stable population will exist. Thus, Y_0 might represent our concept of absolute poverty. Birth and death rates are equal, and the population is barely holding its own absolute level. The situation is analogous to stage 1 of the demographic transition theory. At per capita income levels beyond (to the right of) Y_0 , it is assumed that population size will begin to increase under the pressure of falling death rates. Higher incomes mean less starvation and disease. And with birthrates always assumed to be at the biological maximum, falling death rates provide the impetus for an expanding population (i.e., stage 2).

In figure 1.4 population growth achieves its maximum rate, roughly 3.3%, at a per capita income level of Y_2 . It is assumed to remain at that level until much higher per capita income levels are realized. Thereafter (beyond Y_5), in accordance with stage 3 of the demographic transition, birth rates will begin to decline, and the population growth rate curve becomes negatively sloped and once again approaches the horizontal axis.

The other part of the Malthusian theory requires us to plot a relationship between the growth rate of aggregate income (in the absence of population growth) and levels of per capita income. We can then compare the two rates (aggregate income and total population). If aggregate income (total product) is rising faster, per capita income by definition must be increasing; if total population is growing faster than total income, per capita income must be falling. In figure 1.4, the rate of aggregate income growth (also measured vertically) is assumed at first to be positively related to levels of per capita income; that is, the higher the level of per capita income, the higher rate of increase in aggregate income will be. The economic reason for this positive relationship is the assumption that savings vary positively with income per capita. Countries with higher per capita incomes are assumed to be capable of generating higher savings rates and thus more investment. Beyond a certain per capita income point (Y_3), however, the income growth rate curve is assumed to level off and then begin to decline as new investments and more people are required to work with fixed quantities of land and natural resources. This is the point of diminishing returns in the Malthusian model (note that the possibility of technological progress is not considered). The aggregate income growth curve is therefore conceptually analogous to the total product curve in the basic theory of production.

In Figure 1.4 observe that, the curves are drawn so that they intersect at three points, A, B and C. Point A represents the point at which the Malthusian population trap level of per capita income (Y_1) is attained. It is a stable equilibrium point—any small movement to the left or right of point A will cause the per capita income equilibrium point to return to Y_1 . For example, as per capita income rises from Y_1 toward Y_2 , the rate of population increase will exceed the rate of aggregate income growth (the $\Delta P/P$ curve is vertically higher than the Y/Y curve). We know that whenever population is growing faster than income, per capita income must fall. The arrow pointing in the direction of A from the right therefore shows that per capita income must fall back to its very low level at Y_1 for all points between Y_1 and Y_2 . Similarly, to the left of point A, incomes grow faster than population, causing the equilibrium per capita income level to rise to Y_1 .

According to the neo-Malthusians, poor nations will never be able to rise much above their subsistence levels of per capita income unless they initiate **preventive checks** (birth control) on

their population growth. In the absence of such preventive checks, Malthusian **positive checks** (starvation, disease, wars) on population growth will inevitably provide the restraining force.

Completing our description of the population trap portrayed in Figure 1.4, we see that point B is an “unstable” equilibrium point. If per capita income can somehow jump rapidly from Y_1 to Y_2 (e.g., as a result of “big push” investment and industrialization programs) before Malthusian positive checks take their toll, it will continue to grow until the other stable equilibrium point C at capita income level Y_4 is reached. Point B is an unstable equilibrium point in the sense that any movement to the left or right will continue until either A or C is reached.

1.5.1.1 Criticisms of the Malthusian Model

The Malthusian population trap provides a theory of the relationship between population growth and economic development. Unfortunately, it is based on a number of simplistic assumptions and hypotheses that do not stand the test of empirical verification. We can thus largely reject the Malthusian and neo-Malthusian theories as applied to contemporary developing nations on the following grounds:

1. They do not take adequate account of the role and impact of technological progress (diminishing marginal return could be avoided by technological progress).
2. They are based on a hypothesis about a macro relationship between population growth and levels of per capita income that does not stand up to empirical testing.
3. They focus on the wrong variable, per capita income, as the principal determinant of population growth rates.

A much better and more valid approach to the question of population and development centers on the microeconomics of family size decision making in which individual, and not aggregate, levels of living become the principal determinant of a family's decision to have more or fewer children.

1.5.2 The Microeconomic Household Theory of Fertility

In recent years, economists have begun to look more closely at the microeconomic determinants of family fertility in an attempt to provide a better theoretical and empirical explanation for the observed falling birthrates associated with stage 3 of the demographic transition. In doing this, they have drawn on the traditional neoclassical theory of household and consumer behavior for their basic analytical model and have used the principles of economy and optimization to explain family size decisions.

The conventional theory of consumer behavior assumes that an individual with a given set of tastes or preferences for a range of goods (a “utility function”) tries to maximize

The satisfaction derived from consuming these goods subject to his or her own income constraint and the relative prices of all goods. In the application of this theory to fertility analysis, children are considered as a special kind of consumption (and in Developing countries, investment) good so that fertility becomes a rational economic response to the consumers (families) demand for children relative to other goods. The usual income and substitution effects are assumed to apply. That is, if other factors are held constant, the desired number of children can be expected to vary directly with household income (this direct relationship may not hold for poor societies; it depends on the strength of demand for children relative to other consumer goods and to the sources of increased income, such as female employment), inversely with the price (cost) of children, and inversely with the strength of tastes for other goods relative to children.

Mathematically, these relationships can be expressed as follows:

$$Cd = f(Y, Pc, Px, Tx) \text{ -----1}$$

Where:

Cd = demand for surviving children , Y = level of household income (+) , Pc = net price of children (-) , PX = Price of all other goods (+) and TX = tastes for goods relative to children (-)

It is also possible to express the expected signs of each of the independent variables in the following forms.

$$dCd/dY > 0 \qquad dCd/dPx > 0$$

$$dCd/dTx < 0 \qquad dCd/dPc < 0$$

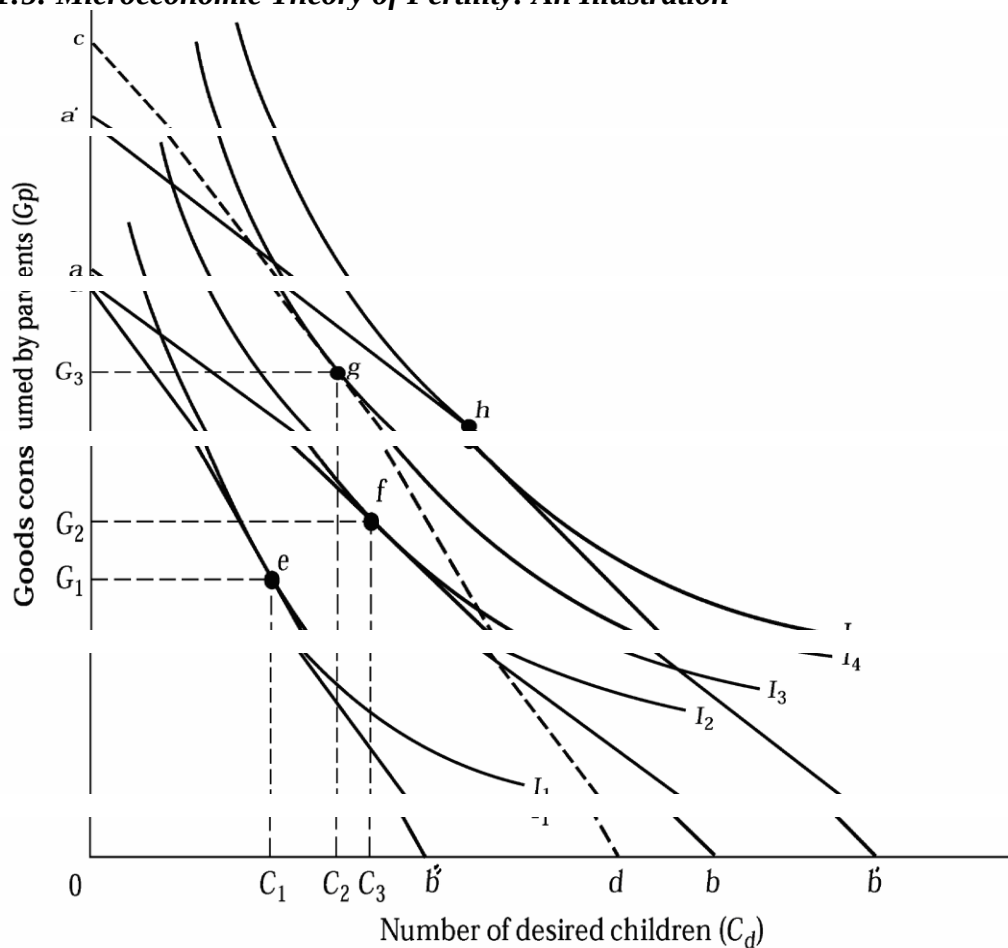
Where d is the partial derivative of demand for children function with respect to the explanatory variables. It measures the effect of an increase in explanatory variable on the demand of child by the households.

- The higher the household income, the greater the demand for children
- The higher the net price of children, the lower the quantity demanded
- The higher the prices of all other goods relative to children, the greater the quantity of children demanded
- The greater the strength of tastes for goods relative to children, the fewer children demanded.

Figure 1.5 provides a simplified diagrammatic presentation of the **microeconomic theory of fertility**. The number of desired (surviving) children, Cd , is measured along the horizontal axis, and the total quantity of goods consumed by the parents, G_p is measured on the vertical axis.

Household desires for children are expressed in terms of an indifference map representing the subjective degree of satisfaction derived by the parents for all possible combinations of commodities and children. Each individual indifference curve portrays a locus of commodity-child combinations that yield the same amount of satisfaction. Any point (or combination of goods and children) on a “higher” indifference curve that is, on a curve farther out from the origin—represents a higher level of satisfaction than any point on a lower indifference curve. But each indifference curve is a “constant satisfaction” locus.

Figure 1:5: Microeconomic Theory of Fertility: An Illustration



In figure 1.5, only four indifference curves, I_1 to I_4 , are shown; in theory, there is an infinite set of such curves, filling the whole quadrant and covering all possible commodity-child combinations. The household's ability to “purchase” alternative combinations of goods and children is shown on line ab (within the triangular area $0ab$) are financially attainable by the household on the basis of its perceived income prospects and the relative prices of children and goods, as represented by the

slope of the ab budget constraint. The steeper the slope of the budget line, the higher the price of children relative to goods.

According to the demand-based theory of fertility, the household chooses from among all attainable combinations the one combination of goods and children that maximizes family satisfaction on the basis of its subjectively determined preferences. Diagrammatically, this optimal combination is represented by point f, the tangency points between the budget constraint, ab, and indifference curve, I_2 . Therefore, C_3 children and G_2 good will be demanded.

A rise in family income, represented in Figure 1.5 by the parallel outward shift of the budget line from ab to a b, enables the household to attain a higher level of satisfaction (point h on curve I_4 by consuming more of both commodities and children- that is, if children, like most commodities, are assumed to be normal goods (demand for them rises with income), an important if in low-income countries where children are often in demand primarily as a source of future financial security. Note that as income rises, parents may spend more on each child, preferring a smaller number of children each of higher “quality,” for example, healthier and better-educated.

Similarly, an increase in the price (opportunity cost) of children relative to other goods will cause households to substitute commodities for children. Other factors (namely, income and tastes) being constant, a rise in the relative price of children causes the household utility-maximizing consumption combination to occur on a lower indifference curve, as shown by the movement of the equilibrium point from f to e when the budget line rotates around point a to ab”

Note, finally, that if there is a simultaneous increase in household income and net child price as a result of, say, expanding female employment opportunities and a rise in wages coupled with a tax on children beyond a certain number per family, there will be both an outward shift and downward rotation of the budget constraint line of Figure 1.5 to, say, dashed line cd. The result is a new utility-maximizing combination that includes fewer children per family) point g compared with point f). In other words, higher levels of living for low-income families in combination with a relative increase in the price of children (whether brought about directly by fiscal measures or indirectly by expanded female employment opportunities) will motivate households to have fewer children while still improving their welfare. This is just one example of how the economic theory of fertility can shed light on the relationship between economic development and population growth as well as suggest possible lines of policy.

1.5.2.1 The Demand for Children in Developing Countries

As we have seen, the economic theory of fertility assumes that the household demand for children is determined by family preferences for a certain number of surviving (usually male) children (i.e., in regions of high mortality, parents may produce more children than they actually desire in the

expectation that some will not survive), by the price or “opportunity cost” of rearing these children, and by levels of family income. Children in poor societies are seen partly as economic investment goods in that there is an expected return in the form of both child labor and the provision of financial support for parents in old age. However, in many developing countries, there is a strong intrinsic psychological and cultural determinant of family size, so that the first two or three children should be viewed as “consumer” goods for which demand may not be very responsive to relative price changes.

The choice mechanism in the economic theory of fertility as applied to Developing countries is assumed, therefore, to exist primarily with regard to the additional or marginal children who are considered as investment. In deciding whether or not to have additional children, parents are assumed to weigh private economic benefits against private costs, where the principal benefits are, as we have seen, the expected income from child labor, usually on the farm, and eventual financial support for elderly parents. Balanced against these benefits are the two principal elements of cost: the opportunity cost of the mothers time (the income she could earn if she were not at home caring for her children) and the opportunity and actual cost of educating children the financial trade-off between having fewer “high-quality,” high-cost, educated children with high-income-earning potentials versus more “low-quality,” low-cost, uneducated children with much lower earning prospects.

Using the same thought processes as in the traditional theory of consumer behavior, the theory of family fertility as applied to Developing countries concludes that when the price or cost of children rises as a result of, say, increased educational and employment opportunities for women or a rise in school fees or the establishment of minimum-age child labor laws or the provision of publicly financed old-age social security schemes, parents will demand fewer additional children, substituting, perhaps, quality for quantity or a mothers employment income for her child-rearing activities. It follows that one way to induce families to desire fewer children is to raise the price of child rearing by, say, providing greater educational opportunities and a wider range of higher-paying jobs for young women.

1.5.2.2 Implications of Womens Education for Development and Fertility

Implications of womens education for fertility and development can be summarized by saying that the effect of social and economic progress in lowering fertility in developing countries will be the greatest when the majority of the population and especially the very poor share in its benefits. Specifically, birthrates among the very poor are likely to fall where there is:

1. An increase in the education of women and a consequent change in their role and status.

2. An increase in female nonagricultural wage employment opportunities, which raises the price or cost of their traditional child-rearing activities
3. A rise in family income levels through the increased direct employment and earnings of a husband and wife or through the redistribution of income and assets from rich to poor.
4. A reduction in infant mortality through expanded public-health programs and better nutritional status for both mother and child and better medical care
5. The development of old-age and other social security systems outside the extended family network to lessen the economic dependence of parents, especially women, on their offspring.
6. Expanded schooling opportunities so parents can better substitute child “quality” for large numbers of children.

1.6 The Consequences of High fertility: Some Conflicting Opinions

For many years, development economists and other social scientists have debated the seriousness of the consequences of rapid population growth. On the one hand, we must recognize that population growth is not the only, or even the primary, source of low levels of living, eroding self-esteem, and limited freedom in developing nations. On the other hand, it would be equally naive to think that rapid population growth in many countries and regions is not a serious intensifier and multiplier of those integral components of underdevelopment. The following discussion summarizes some of the main arguments for and against the idea that the consequences of rapid population growth lead to serious development problems. It then considers whether some consensus can be reached so that specific policy goals and objectives can be postulated.

1.6.1 Population Growth is Not a Real Program

We can identify three general lines of argument on the part of people who assert that population growth is not a cause for concern:

1. The problem is not population growth but other issues.
2. Population growth is a false issue deliberately created by dominant rich-country agencies and institutions to keep developing countries in their underdeveloped, dependent condition.
3. For many developing countries and regions, population growth is in fact desirable.

The Problem is population growth but others Issues: Many knowledgeable people from both rich and poor nations argue that the real problem is not population growth per se but one or all of the following four issues.

(i) Underdevelopment: If correct strategies are pursued and lead to higher levels of living, greater self-esteem, and expanded freedom, population will take care of itself. Eventually, it will disappear as a problem, as it has in all of the present economically advanced nations. According to this argument, underdevelopment is the real problem, and development should be the only goal. With it will become economic progress and social mechanisms that will more or less automatically regulate population growth and distribution. As long as the vast majority of people in developing countries remain impoverished, uneducated, and physically and psychologically weak, the large family will constitute the only real source of social security. Proponents of the underdevelopment argument then conclude that birth control programs will surely fail, as they have in the past, when there is no motivation on the part of poor families to limit their size.

(ii) World Resource Depletion and Environmental Destruction: Population can only be an economic problem in relation to the availability and utilization of scarce natural and material resources. The fact is that developed countries, with less than one-quarter of the world's population, consume almost 80% of the world's resources. In terms of the depletion of the world's limited resources, therefore, the addition of another. This combination of rising affluence and extravagant consumption habits in rich countries and among rich people in poor countries, and not population growth, should be the major world concern.

(iii) Population Distribution: According to this third argument, it is not the number of people per se that is causing population problems but their distribution in space. Many regions of the world (e.g. parts of sub-Saharan Africa) and many regions within countries (e.g., the northeastern and Amazon regions of Brazil) are in fact under populated in terms of available or potential resources. Others simply have too many people concentrated in too small an area e.g., most urban concentrations in Developing countries. Governments should therefore strive not to moderate the rate of population growth but rather to reduce rural- urban migration and to bring about a more natural spatial distribution of the population in terms of available land and other productive resources.

(iv). Subordination of Women: women often bear the disproportionate burdens of poverty, poor education, lack of jobs, and limited social mobility. In many cases, their inferior roles, low status, and restricted access to birth control are manifested in their high fertility. According to this argument, population growth is a natural outcome of women's lack of economic opportunity. If women's health, education, and economic well-being are improved along with their role and status

in both the family and the community, this empowerment of women will inevitably lead to smaller families and lower population growth.

Deliberately Contrived False Issue: The second main line of argument denying the significance of population growth as a major development problem is closely allied to the neocolonial dependence theory of underdevelopment. Basically, it is argued that the chaotic over concern in the rich nations with the population growth of poor nations is really an attempt by the former to hold down the development of the latter in order to maintain an international status quo that is favorable to their self-interests. Rich nations are pressuring poor nations to adopt aggressive population control programs even though they themselves went through a period of sizable population increase that accelerated their own development processes.

A Desirable Phenomenon: A more conventional economic argument is that of population growth as an essential ingredient to stimulate economic development. Larger populations provide the needed consumer demand to generate favorable economies of scale in production, to lower production costs, and to provide a sufficient and low-cost labor supply to achieve higher output levels. Population "revisionist" economists of the neoclassical counterrevolution school argue that free markets will always adjust to any scarcities created by population pressures. Such scarcities will drive up prices and signal the need for new cost -saving production technologies. In the end, free markets and human ingenuity will solve any and all problems arising from population growth. This revisionist viewpoint was clearly in contrast with the traditional "orthodox" argument that rapid population growth had serious economic consequences that, if left uncorrected, would slow economic development.

At the other end of the political spectrum, it has been argued by some developing-world neo-Marxist pronatalist that many rural regions in developing countries are in reality under populated in the sense that much unused but arable land could yield large increases in agricultural output if only more people were available to cultivate it.

Three other non-economic arguments, each found to some degree in a wide range of developing countries, complete the "population growth is desirable" viewpoint. First, many countries claim a need for population growth to protect currently underpopulated border regions against the expansionist intentions of neighboring nations. Second, there are many ethnic, racial, and religious groups in less developed countries whose attitudes favoring large family size have to be protected for both moral and political reasons. Finally, military and political powers are often seen as dependent on a large and youthful population.

The important point is that the arguments represent a considerable range of opinions and viewpoints and therefore need to be seriously weighed against the counter arguments of theorists

who believe that rapid population growth is indeed a real and important problem for underdeveloped countries. Let us now look at some of these counterarguments.

1.6.2 Population Growth is a Real Problem

Positions supporting the need to curtail population growth because of the negative economic, social, and environmental consequences are typically based on one of the following three arguments.

(i) The Extremist Argument: Population and the Global Crisis: The extreme version of the population-as-problem position attempts to attribute almost all of the world's economic and social evils to excessive population growth. Unrestrained population increase is seen as the major crisis facing humankind today. It is regarded as the principal cause of poverty, low levels of living, malnutrition, ill health, environmental degradation, and a wide array of other social problems. Indeed, direct predictions of world food catastrophes and ecological disaster are attributed almost entirely to the growth in world numbers. Such an extreme position leads some of its advocates to assert that developing countries population stabilization or even decline is the most urgent contemporary task even if it requires severe and coercive measures such as compulsory sterilization to control family size in some of the most populated developing countries.

(ii) The Theoretical Argument: Population-Poverty Cycles and the Need for Family Planning Programs: The population-poverty cycle theory is the main argument advanced by economists who hold that too rapid population growth yields negative economic consequences and thus should be a real concern for developing countries. Advocates start from the basic proposition that population growth intensifies and exacerbates the economic, social, and psychological problems associated with the condition of underdevelopment. Population growth is believed to retard the prospects for a better life for the already born by reducing savings rates at the household and national levels. It also severely draws down limited government revenues simply to provide the most rudimentary economic, health, and social services to the additional people. This in turn further reduces the prospects for any improvement in the levels of living of the existing generation and helps transmit poverty to future generations of low-income families.

(iii) The Empirical Argument: Seven Negative Consequences of Population Growth: According to the latest empirical research, the potential negative consequences of population growth for economic development can be divided into seven categories:

Economic Growth: Evidence shows that rapid population growth lowers per capita income growth in most Developing countries, especially those that are already poor, dependent on agriculture, and experiencing pressures on land and natural resources.

Poverty and Inequality Even though aggregate statistical correlations between measures of poverty and population growth at the national level are often inconclusive, at the household level the evidence is strong and compelling. The negative consequences of rapid population growth fall most heavily on the poor because they are the ones who are made landless, suffer first from cuts in government health and education programs, bear the brunt of environmental damage, and are the main victims of job cuts due to the slower growth of the economy. Poor women once again bear the greatest burden of government austerity programs, and another vicious cycle is set in motion. To the extent that large families perpetuate poverty, they also exacerbate inequality.

Education : Although the data are sometimes ambiguous on this point, it is generally agreed that large family size and low incomes restrict the opportunities of parents to educate all their children. At the national level, rapid population growth causes given educational expenditures to be spread more thinly, lowering quality for the sake of quantity. This in turn feeds back on economic growth because the stock of human capital is reduced by rapid population growth.

Health: High fertility harms the health of mothers and children. It increases the health risks of pregnancy, and closely spaced births have been shown to reduce birth weight and increase child mortality rates.

Food: Feeding the world's population is made more difficult by rapid population growth—over 90% of additional Developing countries food requirements are caused by population increases. New technologies of production must be introduced more rapidly, as the best lands have already been cultivated. International food relief programs become more widespread.

Environment: Rapid population growth contributes to environmental degradation in the form of forest encroachment, deforestation, fuel-wood depletion, soil erosion, declining fish and animal stocks, inadequate and unsafe water, air pollution, and urban congestion.

International Migration: Many observers consider that rapid increase in international migration, both legal and illegal, to be one of the major consequences of developing countries' population growth. Though many factors cause migration, an excess of job seekers over job opportunities in the Developing countries is surely one of them. However, unlike the first six consequences listed here, some of the economic and social costs of international migration fall on recipient countries—increasingly in the developed world.

1.7 Goals and Objectives: Toward a Consensus

In spite of what may appear to be seriously conflicting arguments about the positive and negative consequences of population growth, during the past decade there emerged a common ground on

which many people on both sides of the debate could agree. The following three propositions constitute the essential components of this intermediate or consensus opinion.

1. Population growth is not the primary cause of low levels of living, gross inequalities, or the limited freedom of choice that characterize much of the developing world. The fundamental causes of these problems must be sought. Rather, in the troubles of poor families, especially women, and failure of other aspects of domestic and international development policy.
2. The problem of population is not simply one of numbers but involves the quality of life and material well-being. Thus, developing countries population size must be viewed in conjunction with developed-country affluence in relation to the quantity, distribution, and utilization of world resources, not just in relation to indigenous resources of the Developing countries.
3. Rapid population growth does serve to intensify problems of underdevelopment and make prospects for development that much more remote.

As we have seen, the momentum of growth means that, barring catastrophe, the population of developing countries will increase dramatically over the coming decades, no matter what fertility control measures are adopted now. It follows that high population growth rates, though not the principal cause of underdevelopment, are nevertheless important contributing factors in specific countries and regions of the world.

In view of these three propositions, we may conclude that the following three policy goals and objectives might be included in any realistic approach to the issue of population growth in developing countries.

In countries or regions where population size, distribution, and growth are viewed as an existing or potential problem, the primary objective of any strategy to limit further growth must deal not only with the population variable by itself but also with the underlying social and economic conditions of underdevelopment. Problems such as absolute poverty, gross inequality, widespread unemployment (especially among women), limited female access to education, malnutrition, and poor health facilities must be given high priority. Their amelioration is both a necessary concomitant of development and a fundamental motivational basis for the expanded freedom of the individual to choose an optimal-and, in many cases, smaller-family size.

To bring about smaller families through development-induced motivations, family- planning programs providing both the education and the technological means to regulate fertility for people who wish to regulate it should be established.

Developed countries should assist developing countries to achieve their lowered fertility and mortality objectives not only by providing contraceptives and funding family-planning clinics but, more important, by curtailing their own excessive depletion of nonrenewable world resources through programs designed to cut back on the unnecessary consumption of products that intensively use such resources; by making genuine commitments to eradicating poverty, illiteracy, disease, and malnutrition in developing countries as well as their own; and by recognizing in both their rhetoric and their international economic and social dealings that development is the real issue, not simply population control.

1.8 Some Policy Approaches

In view of these broad goals and objectives, what kinds of economic and social policies might the developing country and developed-country governments and international assistance agencies consider bringing about long-term reductions in the overall rate of world population growth? Three areas of policy can have important direct and indirect influences on the well-being of present and future world populations:

1. General and specific policies that LDC governments can initiate to influence and perhaps even control their population growth and distribution
2. General and specific policies that developed-country governments can initiate in their own countries to lessen their disproportionate consumption of limited world resources and promote a more equitable distribution of the benefits of global economic progress.
3. General and specific policies that developed-country governments and International assistance agencies can initiate to help developing countries achieve their population objectives. Let us deal with each of these areas in turn.

1.8.1 What Developing Countries Can Do?

Certain development policies are particularly crucial in the transition from a high-growth to a low-growth population. These policies aim at eliminating absolute poverty; lessening income inequalities; expanding educational opportunities, especially for women; providing increased job opportunities for both men and women; bringing the benefits of modern preventive medicine and public-health programs, especially the provision of clean water and sanitation to the rural and urban poor; improving maternal and child health through more food, better diets, and improved nutrition to lower infant mortality; and creating a more equitable provision of other social services to wide segments of the population.

Given the economic and non-economic benefit of lowering population growth, there are some more specific to Developing countries governments might try to adapt to lower birthrates in the short run. Governments can attempt to control fertility in five ways.

First, they can try to persuade people to have smaller families through the media and the educational process.

Second, they can establish family-planning programs to provide health and contraceptive services to encourage the desired behavior. Such publicly sponsored or officially supported programs now exist in most developing countries.

Third, they can deliberately manipulate economic incentives and disincentives for having children-for example, through the elimination or reduction of maternity leaves and benefits, the reduction or elimination of financial incentives, or the imposition of financial penalties for having children beyond a certain number; the establishment of old-age social security provisions and minimum-age child labor laws; the raising of school fees and the elimination of heavy public subsidies for secondary and higher education; and the subsidization of smaller families through direct money payments.

Fourth, governments can attempt to coerce people into having smaller families through the power of state legislation and penalties. For obvious reasons, few governments would attempt to engage in such coercion; not only is it often morally repugnant and politically unacceptable, but it is also almost always extremely difficult to administer.

Finally, no policy measures will be successful in controlling fertility unless efforts are made to raise the social and economic status of women and hence create conditions favorable to delayed marriage and lower marital fertility. A crucial ingredient in any program designed to lower fertility rates is the increased education of women, followed by the creation of jobs for them outside the home. The availability of income-earning opportunities can lead young women to delay marriage by enabling them to become economically self-sufficient and therefore in a better position to exercise control over the choice of partner and the timing of marriage. It can also reduce family pressures for early marriage by allowing women to contribute to parental household income. An independent source of income also secures a stronger position for married women in the household, reducing their dependence on other family members, particularly male offspring, for economic security. Furthermore, it enables women to consider the opportunity costs of additional children when childbearing competes with income-generating activities.

1.8.2 What the Developed Countries Can Do?

When we view the problems of population from the perspective of global resources and the environment, the question of the relationship between population size and distribution and the depletion of many nonrenewable resources in developed and underdeveloped countries assumes major importance.

In a world where 4.6% of the population, located in one country, the United States, accounts for 40% of annual world resource use. We must also be concerned with the impact of rising affluence and the very unequal worldwide distribution of incomes on the depletion of many nonrenewable resources such as petroleum, certain basic metals, and other raw materials essential for economic growth. In terms of food consumption, basic grains like wheat, corn, and rice are by far the most important source of humanity's direct food energy supply. Consumed indirectly, they make up a significant share of the remainder. In resource terms, more than 70% of the world's cropland goes into grain production. Yet the average North American directly and indirectly consumes five times as much grain and the corresponding agricultural resources-land, fertilizer, water-as his or her counterpart in India or Nigeria for example.

With regard to energy, probably the second most essential resource to modern society, the average American's consumption of energy fuels in 1990 was 25 times the average Brazilian's, 60 times the average Indian's, 191 times the average Nigerian's, and 351 times the average Ethiopian's consumption level! This use of fossil fuel energy to power private automobiles, operate home and office air conditioners, and activate electric tooth brushes in the developed nations is by far the major contributor of carbon dioxide gases into the atmosphere and to the phenomenon of ozone depletion and greenhouse global warming. It also means that there is potentially that much less to fertilize small family farms in the less developed nations. Alternatively, it means that poor families will have to pay more to obtain these valuable resource inputs.

The point, therefore, is that any worldwide program designed to engender a better balance between resources and people by limiting LDC population growth through social intervention and family planning must also include the responsibility of rich nations systematically to simplify their own consumption demands and lifestyles. Such changes would free resources that could then be used by poor nation to generate the social& economic development essential to slow population growth.

1.8.3 How Developed Countries Can Assist Developing Countries with Their Population Programs

There are also a number of ways in which the governments of rich countries and multilateral donor agencies can help the governments of developing countries achieve their population policy objectives in shorter periods of time. The most important of these concerns the willingness of rich countries to be of genuine assistance to poor countries in their development efforts. Such genuine support would consist not only of expanded public and private financial assistance but also of improved trade relations, such as tariff- and quota-free access to developed-country markets, more appropriate technology transfers, and assistance in developing indigenous scientific research capacities, better international commodity-pricing policies, and a more equitable sharing of the world's scarce natural resources.

There are two other activities more directly related to fertility moderation in which rich-country governments, international donor agencies, and private non-governmental organizations (NGOs) can play an important assisting role.

The first is the whole area of research into the technology of fertility control, the contraceptive pill, modern intrauterine devices (IUDs), voluntary sterilization procedures, and, particularly for Africa in the age of AIDs, effective barrier contraception.

The second area includes financial assistance from developed countries for family-planning programs, public education, and national population policy research activities in the developing countries. This has traditionally been the primary area of developed-country assistance in the field of population.

Unit Summary

In this unit we reviewed population growth and introduced the basic demographic concepts, the linkage between population growth and quality of life. The structure of the world population, the hidden momentum of population growth, and the demographic transition have been widely discussed in the unit. In order to examine the causes of high fertility rate in the contemporary developing nations, two theories are explored in the current unit: the Malthusian population Trap and the Microeconomic Household Theory of Fertility. Some conflicting opinions on impact of population growth on development are also discussed in the unit. Finally, the unit highlighted some consensus among different views on population growth rate and what developed and developing countries should do with respect to population growth in the developing world and fair and equitable global resource utilization.

CHAPTER TWO

Human Capital Education and Health in Economic Development

Introduction

In the second unit, we examine the roles of education and health in economic development. These two human capital issues are treated together because of their close relationship. We consider the relationships between income on the one hand and health and education on the other. At the same time, the most important root cause of poor health in developing countries will be explored. Finally, we take a systematic look at educational and health systems in developing countries, to understand the sources of the severe inequalities and inefficiencies that continue to plague them. We will find compelling evidence that investments in human capital have to be undertaken with both equity and efficiency for them to successfully realize their potential impact on incomes. The gender gap in human capital development will get due attention in the unit. Finally, the unit closely examines health & education policies and systems in the contemporary developing economies in relation to income generation and inequality.

Unit Objectives

After completing this unit, you will be able

- To understand the role of human capital (mainly contributed by education and health status of the labor force) in development process of a country
- To examine the close relationship between these two major components of human capital: Health and Education
- To understand the causes and consequences of child labor in the contemporary developing countries
- To examine the significance of investing in human capital as a major path to economic development of a country
- To have a clear picture with regards to the gender gap in educational attainment and its costs to an economy
- To have exposure to the educational and health systems and development in the contemporary developing economies and different policy options that they pursue

2.1 Education and Health as Joint Investments for Development

Health and education are closely related in economic development. On the one hand, greater health capital may improve the return to investment in education, in part because health is an important factor in school attendance and in the formal learning process of a child. A longer life raises the return to investments in education; better health at any point during working life may in effect lower the rate of depreciation of education capital. On the other hand, greater education capital may

improve the return to investments in health, because many health programs rely on basic skills often learned at school, including personal hygiene and sanitation, not to mention basic literacy; education is also needed for the formation and training of health personnel. Finally, an improvement in productive efficiency from investments in education raises the return on a life saving investment in health.

2.2 Investing in Education and Health: The Human Capital Approach

The analysis of investments in health and education is unified in the human capital approach. Human capital is the term economists often use for education, health, and other human capacities that can raise productivity when increased. An analogy is made to conventional investments in physical capital: After an initial investment is made, a stream of higher future income can be generated from both expansion of education and improvements in health. As a result, a rate of return can be deduced and compared with returns to other investments. This is done by estimating the present discounted value of the increased income stream made possible by these investments and then comparing it with their direct and indirect costs. Of course, health and education also contribute directly to well-being, but the human capital approach focuses on their indirect ability to increase utility by increasing incomes. Formally, the income gains can be written as follows, where E is income with education, N is income without the extra education, t is year, and the summation is over expected years of working life:

$$\frac{\sum (E_t - N_t)}{(1+i)^t}$$

An analogous formula applies to health, with the direct and indirect cost of resources devoted to health compared with the extra income gained in the future as a result of higher health (such as improved nutritional status).

Figure 2:1: Financial Trade — Offs in the Decision to Continue in School

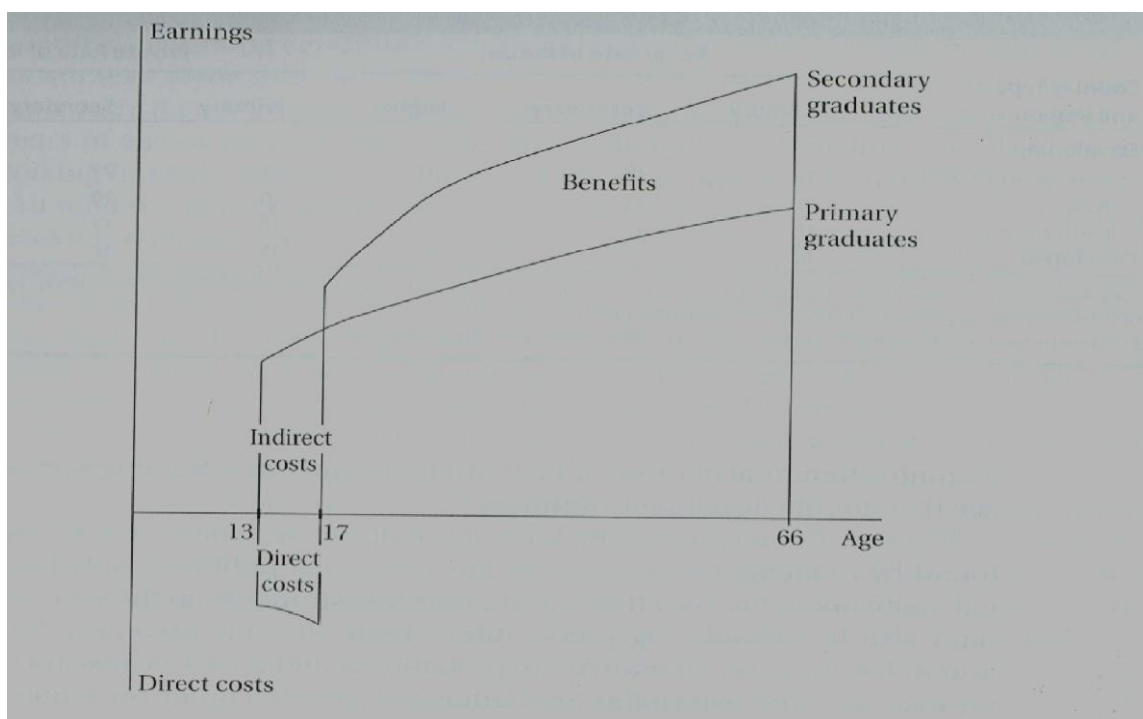


Figure 2.1 provide a schematic representation of the trade-offs involved in the decision to continue in school. It is assumed that the individual works from the time he or she finishes school until he or she is unable to work or dies. This is taken to be the current world life expectancy of 66 years. Two earnings profiles are presented—for those with primary school but no secondary education and for those with a full secondary (but no higher) education. Primary graduates are assumed to begin work at age 13, secondary graduates at age 17. For an individual in a developing country deciding whether to go on from primary to secondary education, four years of income are foregone. This is the indirect cost, as labeled in the diagram. The child may work part time, a possibility ignored here for simplicity, but if so, only part of the indirect-cost area applies.

There is also a direct cost, such as fees, school uniforms, books, and other expenditures that would not have been made if the individual had left school at the end of the primary grades. Over the rest of the person's life, he or she makes more money each year than would have been earned with only a primary education. This differential is labeled "Benefits" in the diagram. Before comparing costs with benefits, note that a dollar today is worth more to an individual than a dollar in the future, so those future income gains must be discounted accordingly. Although a simple analytical formula for the rate of return would rely on unrealistic assumptions, the rate of return

will be higher whenever the discount rate is lower, the direct or indirect costs are lower, or the benefits are higher.

2.3 Improving Health and Education: Why Increasing Income Is Not Sufficient?

Health and education levels are much higher in high-income countries. There are good reasons to believe that the causality runs in both directions: With higher income, people and governments can afford to spend more on education and health; with greater health and education, higher productivity and incomes are possible.

Because of these relationships, development policy needs to focus on income, health, and education simultaneously. This conclusion is parallel to our finding in the previous module that we need all-round strategies to address the stubborn problems of absolute poverty.

People will spend more on human capital when income is higher. But some evidences show clearly that even if we were able to raise incomes without a large improvement in health and education, we could not count on that income increase being used to adequately invest in children's education and health. The market will not solve this problem automatically, and in many cases, household consumption choices themselves may lead to a surprisingly small link between income and nutrition, especially for children. This small proportional response is due to two factors: Income is spent on other goods besides food, and part of the increased food expenditures is used to increase food variety without necessarily increasing the consumption of calories. If the relationship between income and nutrition is indeed quite low, as some studies suggest, then development policies that emphasize increasing incomes of the poor, without attention to the way these additional resources are expended within the family, may not lead to improved health and successful development more generally, at least not very quickly.

Health status, once attained, also affects school performance, as has shown in studies of many developing countries. Better health and nutrition lead to earlier and longer school enrollment, better school attendance, and more effective learning. Children with low height for their age, an indicator of under nutrition, have been found to lag in school grade attainment in many parts of the world. Thus, to improve the effectiveness of schooling, we must improve the health of children in developing countries. Indeed, advances in statistical methods are showing that the links from health to educational attainment in developing countries are stronger than had been believed until very recently. These effects are large for both boys and girls but especially so for girls.

Finally, there are other important spillover benefits to investment in one's health or education. An educated person provides benefits to people around him or her, such as reading for them or

coming up with innovations that benefit the community. As a result, there are significant market failures in education. Moreover, a healthy person is not only less contagious but can also benefit the community in many ways that a sick person cannot. Because of such spillover effects, the market cannot be counted on to deliver the socially efficient levels of health and education. Thus, as the World Health Organization (WHO) concluded in its 2000 World Health Report on health systems, "Ultimate responsibility for the performance of a country's health system lies with government." Developing-country officials are drawing lessons from many studies that show the interrelationships among health, education, and incomes and are devising integrated strategies.

2.4. Educational Systems and Development

Much of the literature and public discussion about education and economic development in general, and education and employment in particular, revolve around two fundamental economic processes. First, the interaction between economically motivated demands and politically responsive supplies in determining how many school places are provided, who gets access to these places, and what kind of instruction they receive. Second the important distinction between social and private benefits and costs of different levels of education and the implications of these differentials for educational investment strategy.

2.4.1 Educational Supply and Demand:

The Relationship between Employment Opportunities and Educational Demands

The amount of schooling received by an individual, although affected by many nonmarket factors, can be regarded as largely determined by demand and supply, like any other commodity or service. However, because most education is publicly provided in less developed countries, the determinants of the amount demanded turn out to be much more important than the determinants of supply.

On the demand side, the two principal influences on the amount of schooling desired are (1) a more educated student's prospects of earning considerably more income through future modern-sector employment (the family's private benefits of education) and (2) the educational costs, both direct and indirect, that a student or family must bear. The amount of education demanded is thus in reality a derived demand for high-wage employment opportunities in the modern sector. This is because access to such jobs is largely determined by an individual's education. Most people (especially the poor) in less developed nations do not demand education for its intrinsic non-economic benefits but simply because it is the only means of securing modern-sector employment. These derived benefits must in turn be weighed against the costs of education.

On the supply side, the quantity of school places at the primary, secondary, and university levels is determined largely by political processes, often unrelated to economic criteria. Given mounting political pressure throughout the developing world for greater numbers of school places, we can for

convenience assume that the public supply of these places is fixed by the level of government educational expenditures. These are in turn influenced by the level of aggregate private demand for education.

Because it is the amount of education that is demanded that largely determines the supply let us look more closely at the economic determinants of this derived demand. The amount of schooling demanded that is sufficient to qualify an individual for modern- sector jobs appears to be related to or determined by the combined influence of four variables: the wage or income differential, the probability of success in finding modern- sector employment, the direct private costs of education, and the indirect or opportunity costs of education.

Although several other important variables, many of them non-economic (e.g., cultural traditions, gender, social status, education of parents, and size of family), certainly influence the amount of education demanded, concentrating on the four variables just identified can give important insights into the relationship between the quantity of education demanded and the supply of employment opportunities.

2.4.2 Social versus Private Benefits and Costs

The inevitable attraction of ever-higher levels of education is even more costly than this simple picture suggests (see the following figure). Typically, in developing countries, the social costs of education increase rapidly as students climb the educational ladder. The private costs of education increase more slowly or may even decline.

This widening gap between social and private costs provides an even greater stimulus to the demand for higher education than it does for education at lower levels. But educational opportunities can be accommodated to these distorted demands only at full social cost. As demands are generated progressively through the system, the social cost of accommodation grows much more rapidly than the places provided. More and more resources may be misallocated to educational expansion in terms of social costs, and the potential for creating new jobs will consequently diminish for lack of public financial resources.

Figure 2.2: Private versus Social Benefits and Costs of Education: An illustration

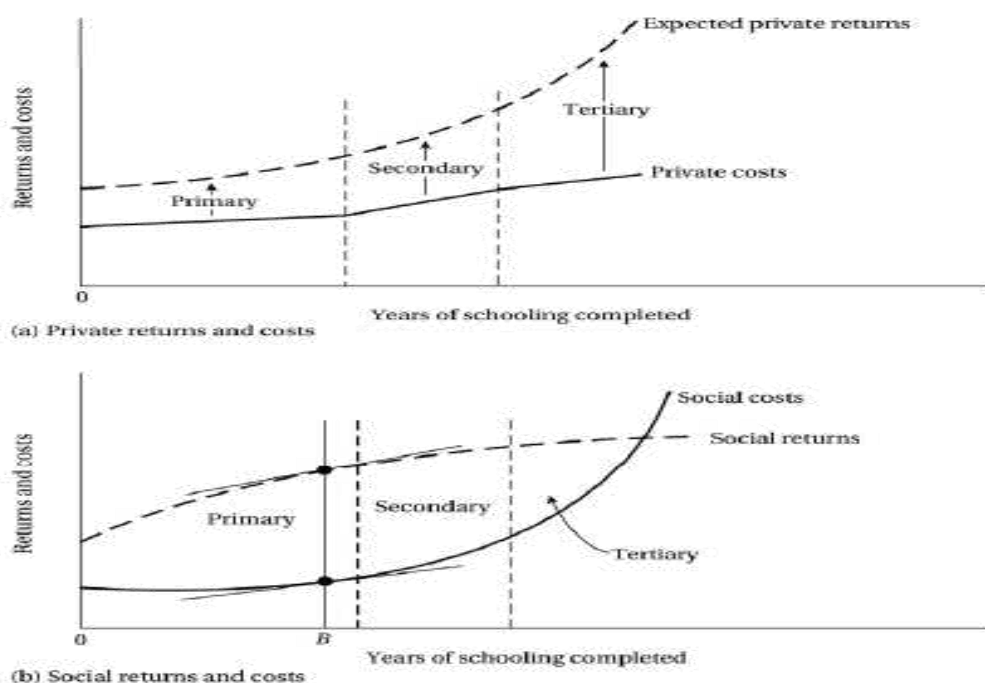


Figure 2.2 provides an illustration of this divergence between private and social benefits and costs. It also demonstrates how this divergence can lead to a misallocation of resources when private interests supersede social investment criteria. In Figure 2.2a, expected private returns and actual private costs are plotted against years of completed schooling. As a student completes more and more years of schooling, expected private returns grow at a much faster rate than private costs, for reasons explained earlier. To maximize the difference between expected benefits and costs (and thereby the private rate of return to investment in education), the optimal strategy for a student would be to secure as much schooling as possible.

Now consider Figure 2.2b, where social returns and social costs are plotted against years of schooling. The social benefits curve rises sharply at first, reflecting the improved levels of productivity of, say, small farmers and the self-employed that result from receipt of a basic education and the attainment of literacy, arithmetic skills, and elementary vocational skills. Thereafter, the marginal social benefit of additional years of schooling rises more slowly, and the social returns curve begins to level off. By contrast, the social cost curve shows a slow rate of growth for early years of schooling (basic education) and then a much more rapid growth for higher levels of education. This rapid increase in the marginal social costs of post-primary education is the result of both the much more expensive capital and recurrent costs of higher

education (buildings and equipment) and, more important, of the fact that much post-primary education in developing countries is heavily subsidized.

It follows from Figure 2.2b that the optimal strategy from a social viewpoint, the one that maximizes the net social rate of return to educational investment, would be one that focuses on providing all students with at least B years of schooling. Beyond B years, marginal social costs exceed marginal social benefits, so additional public educational investment in new, higher-level school places will yield a negative net social rate of return.

Figure 2.2 therefore illustrates the inherent conflict between optimal private and social investment strategies—a conflict that will continue to exist as long as private and social valuations of investment in education continue to diverge as students climb the educational ladder. To a large degree, the problem of divergent social versus private benefits and costs has been artificially created by inappropriate public and private policies with regard to wage differentials, educational selectivity, and the pricing of educational services. As a result, private calculations of the value of education exceed its social value, which must take account of unemployment. As long as artificial and nonmarket incentives in the form of disproportionate expected benefits and subsidized costs continue to exist and place a premium on the number of years one spends getting an education, the individual will decide that it is in his or her best private interests to pursue a lengthy formal education process even though he or she may be aware that modern-sector jobs are becoming scarcer and unemployment rates are rising. Unless these various price signals are made to conform more closely to social realities, the misallocation of national resources will persist and possibly increase.

What is needed is a properly functioning reward and cost structure that develops and allocates human resources in accordance with requirements and opportunities in various segments of the economy. Where this is absent, two obvious misallocations of human resources are likely to occur.

First, with the output of the educational system at higher levels greatly in excess of what the economy can absorb, many students will emerge seeking jobs for which they may be educationally qualified but which have been preempted by others with even more education. They become temporarily unemployed for as long as it takes for their aspirations and status requirements, partly perhaps instilled in them by the educational system itself, to adjust to the existing realities of unemployment in the modern sector.

Second, those who adjust their sights downward and secure modern-sector employment normally have to take jobs for which they are overeducated in terms of the number of years spent in school. Those who fail to get modern-sector jobs at all swell the ranks of the permanently unemployed or

become self-employed in the informal sector. They are thus denied the opportunity to contribute productively to the society that invested so heavily in their education. This combination of the overpaid and, in many cases, overeducated employed and the impoverished and unproductive educated unemployed reflect a serious misallocation of scarce national resources. The resources allocated to the expansion of the educational system might alternatively have been spent on needed rural public works projects or on increasing the quality of basic primary education in rural areas. Such investment would provide emergency employment opportunities for recent graduates as well as for people with less education.

It seems clear that the expansion of educational opportunities at all levels has contributed to aggregate economic growth by (1) creating a more productive labor force and endowing it with increased knowledge and skills; (2) providing widespread employment and income-earning opportunities for teachers, school and construction workers, textbook and paper printers, school uniform manufacturers, and related workers; (3) creating a class of educated leaders to fill vacancies left by departing expatriates or otherwise vacant positions in governmental services, public corporations, private businesses, and professions; and (4) providing the kind of training and education that would promote literacy and basic skills while encouraging "modern" attitudes on the part of diverse segments of the population. Even if alternative investments in the economy could have generated greater growth, this would not detract from the important contributions, non-economic as well as economic, that education can make and has made to promoting aggregate economic growth. That an educated and skilled labor force is a necessary condition of sustained economic growth cannot be denied. However, any evaluation of the role of education in the process of economic development should go beyond the analysis of the single statistic of aggregate growth. We must also consider the structure and pattern of that economic growth and its distribution implications that benefits.

2.4.3 Distribution of Education

The preceding analysis of forces operating for over education in developing countries should not lead us to despair over the possibility of fostering development through greater education. Countries that have developed successfully have generally ensured that educational benefits are more broadly available in the economy to the poor as well as the rich, in the rural areas as well as the urban. Thus, we turn attention to examining the distribution of educational benefits in developing countries.

The implication is that governments should spend more to upgrade existing schools and less to expand the number of school places—that is, they should deepen the investment in human capital rather than extend it to more people. Unfortunately, this raises serious equity questions, and their

findings remain controversial. A balanced policy would be to emphasize the extension of quality primary education to all before embarking on rapid expansion of the quantity of secondary schools.

2.4.4 Education, Inequality, and Poverty

Recent studies have also demonstrated that contrary to what might have been assumed, the educational systems of many developing nations sometimes act to increase rather than to decrease income inequalities.

The basic reason for this perverse effect of formal education on income distribution is the positive correlation between level of education and level of lifetime earnings. This correlation holds especially for workers who are able to complete secondary and university education where income differentials over workers who have completed only part or all of their primary education can be on the order of 300% to 800%. And as levels of earned income are clearly dependent on years of completed schooling, it follows that large income inequalities will be reinforced if students from the middle- and upper- income brackets are represented disproportionately in secondary and university enrollments. In short, if for financial or other reasons the poor are effectively denied access to secondary and higher educational opportunities, the educational system can actually perpetuate and even increase inequality in developing countries.

There are two fundamental economic reasons why one might suspect that many developing countries educational systems are inherently inegalitarian, in the sense that poor students have less chance of completing any given educational cycle than more affluent students. First, the private costs of primary education are higher for poor students than for more affluent students. Second, the expected benefits of primary education are lower for poor students. Together, the higher costs and lower expected benefits of education mean that a poor family's rate of return from investment in a child's education is lower than it is for other families. The poor are therefore more likely to drop out during the early years of schooling.

2.5 Health Systems and Development

2.5.1 Health and Productivity

The devastating effects of poor health on child mortality are clear enough. But do poor health conditions in developing countries also harm the productivity of adults? Careful statistical methods have shown that a large part of the effect of health on raising earnings is due to productivity differences: It is not just the reverse causality that higher wages are used in part to purchase better health. For example, Nobel laureate Robert Fogel has found that citizens of developed countries are substantially taller today than they were two centuries ago and has argued that tallness is a useful index of the health and general well-being of a population. Increases in

height also have been found in developing countries in recent decades as health conditions have improved. In most cases, rapid increases in average height earlier in the twentieth century gave way to smaller increases by mid-century.

The existing evidences show that health and nutrition do affect employment, productivity, and wages and very substantially so among the poorest of the poor. This evidence magnifies the policy priority of health in development; not only is health a major goal in itself, but it has a significant impact on income levels as well. After their exhaustive review of the literature and its complex statistical and data problems, Strauss and Thomas conclude that "the balance of evidence points to a positive effect of elevated nutrient intakes on wages, at least among those who are malnourished." A healthy population is a prerequisite for successful development.

2.5.2 Health Systems Policy

In the WHO's (World Health Organization) definition, a health system is "all the activities whose primary purpose is to promote, restore, or maintain health." Health systems include the components of public health departments, hospitals and clinics, and offices of doctors and paramedics. Outside this formal system is an informal network used by many poorer citizens, which includes traditional healers, who may practice with the use of somewhat effective herbal remedies, or other methods that provide some medical benefits, such as acupuncture, but who also may practice techniques for which there is no evidence of effectiveness beyond the placebo effect.

In 2000, the WHO released its first detailed report comparing health systems around the world. The report indicated great variability in the performance of health systems at each income level. The study used five performance indicators to measure health systems in the 191 WHO member states: (1) the overall level of health of the population; (2) health inequalities within the population; (3) health-system responsiveness (a combination of patient satisfaction and system performance); (4) the distribution of responsiveness within the population (how well people of varying economic status find that they are served by the health system); and (5) the distribution, or fairness, of the health system's financial burden within the population.

Formal public health measures have played a very important role in developing countries. Ministries of health, sometimes complemented by the services of non-governmental organizations, have played vital roles in extending vaccines to remote rural areas, greatly reducing once lethal diseases such as smallpox. But like educational systems, public health operations have often favored the wealthy and well connected. Partly as a result, health systems often use public funds inefficiently. In effect, subsidies turn out to be focused on expensive curative measures for older (and generally richer) patients, such as those with heart disease or cancer, who are influential enough to get into the right hospitals. Too often ignored or at best underfunded are

cost-effective preventive health campaigns and basic medical care for those not currently attended to by any health professionals? Doctors trained with public subsidies often choose to practice a specialty in affluent areas of the cities or emigrate to developed countries. And as the World Bank concluded, "In some countries a single teaching hospital can absorb 20 percent or more of the budget of the ministry of health, even though almost all cost-effective interventions are best delivered at lower-level facilities."

In addition to its direct positive effect on national health standards, basic health is also an effective means to achieve goals of poverty reduction. Although both parents may be employed and self-employed long hours, parents are too weak, unhealthy, and unskilled to be productive enough to support their family, the children have to work. But if the children work, they cannot get the education they need, so when they grow up, they will have to send their own children to work. Thus, the bad equilibrium of child labor examined earlier in the chapter may extend across the generations, as a family is effectively locked into a vicious cycle of poverty. Calculations of benefits of health investments need to keep these long-term spillovers in mind.

An effective government role in health systems is critical for at least three important reasons. First, health is central to poverty alleviation, because people are often uninformed about health, a situation compounded by poverty. Second, households spend too little on health because they may neglect externalities (such as, literally, contagion problems). Third, the market would invest too little in health infrastructure and research and development and technology transfer to developing countries due to market failures. Government has different roles in different countries, but as the WHO concluded, "The careful and responsible management of the well-being of the population-stewardship-is the very essence of good government. The health of people is always a national priority: government responsibility for it is continuous and permanent."

2.6. The Gender Gap: Women and Education

Young females receive considerably less education than young males in almost every developing country. In a recent year, in 66 out of 108 countries, women's enrollment in primary and secondary education was lower than that of men by at least 10 percentage points. This **educational gender gap** is the greatest in the poorest countries and regionally in the Middle East and North Africa. Even though progress has been made recently, a substantial gender gap persists.

Why is female education important? Is it simply a matter of equity? The answer is that there exists ample empirical evidence that educational discrimination against women hinders economic development in addition to reinforcing social inequality. Closing the educational gender gap by expanding educational opportunities for women, a key plank of the Millennium Development Goals is economically desirable for four reasons:

1. The rate of return on women's education is higher than that on men's in most developing countries.
2. Increasing women's education not only increases their productivity on the farm and in the factory but also results in greater labor force participation, later marriage, lower fertility, and greatly improved child health and nutrition.
3. Improved child health and nutrition and more educated mothers lead to multiplier effects on the quality of a nation's human resources for many generations to come.
4. Because women carry a disproportionate burden of the poverty and landlessness that permeates developing societies, any significant improvements in their role and status via education can have an important impact on breaking the vicious cycle of poverty and inadequate schooling.

2.7. Consequences of Gender Bias in Health and Education

Studies from around the developing world consistently show that expansion of basic education of girls earns among the very highest rates of return of any investment-much larger, for example, than most public infrastructure projects. This is one reason why discrimination against girls in education, as well as health, is not just inequitable but very costly from the standpoint of achieving development goals.

Education of girls has also been shown to be one of the most cost-effective means of improving local health standards. Studies by the United Nations, the World Bank, and other agencies have concluded that the social benefits alone of increased education of girls is more than sufficient to cover its costs-even before considering the added earning power this education would bring. However, evidence from Pakistan, Bangladesh, and other developing countries shows that we cannot assume that education of girls will increase automatically with increases in family income. Inferior education and health care access for girls shows the interlinked nature of economic incentives and the cultural setting. Empirical studies demonstrate what we might guess from these perverse incentives: Often more strenuous efforts are made to save the life of a son than a daughter, and girls generally receive less schooling than boys.

Greater mother's education generally improves prospects for both her sons' and daughters' health and education. Studies show that mothers' education plays a decisive role in raising nutritional levels in rural areas. The level of child stunting, a valid indicator of child under nutrition, is much lower with higher education attainment of the mother at every income level. They note that this is almost ten times the projected impact of a 10% increase in per capita income. Coupled with the

result that in many countries mothers education tends to make a disproportionately larger health difference toward daughters than sons can expect major benefits for girls.

Taken together, evidences show that increases in family income do not automatically result in improved health status or educational attainment. If higher income cannot be expected to necessarily lead to higher health and education, as we will see in subsequent sections, there are no guarantees that higher health or education will lead to higher productivities and incomes. Much depends on the context, on whether gains from income growth and also the benefits of public investments in health and education and other infrastructure are shared equitably. Thus, in the remainder of this unit, we examine issues of education and health systems in turn. Even though health and education will be examined separately, it is important to keep their mutually reinforcing roles in mind.

2.8 Policies for Health, Education, and Income Generation

In the coming years, the clear evidence that health and education are joint investments may offer scope for a more integrated policy approach. It may be that one of the most effective investments we can make in education quality is to improve child health. Similarly, one of the most effective investments we could make in health may be to improve the quality of education. In fact, a number of prominent poverty programs in developing countries now explicitly integrate incentives for the development of health and education human capital among low income families. Integrated strategies seek linking highly successful micro credit systems with health and education.

Integrating program on education, health and nutrition is another good strategy to refer to. In this integrated strategy, there is cash transfer to poor family, family clinic visit, and other in-kind nutritional supplement, and other health benefits for pregnant and lactating women and their children under a specified age. Some of the benefits can be made conditional on child's regular school attendance. In this manner it is possible to integrated health, education and benefit packages to the households.

CHAPTER THREE

Rural-Urban Interaction, Migration, and Unemployment

Unit Introduction

In unit three, we focus on one of the most complex and nuanced dilemmas of the development process: the phenomenon of massive and historically unprecedented movements of people from the rural countryside to the burgeoning cities of the contemporary developing countries of Africa, Asia and Latin America.

This unit examines the potential role of cities—both the formal sector and the urban informal sector in fostering economic development. We then turn to a well-known theoretical model of rural-urban labor transfer in the context of rapid growth and high urban unemployment: The Todaro Model of rural — urban Migration. In the final section, we evaluate various policy options that developing countries governments may wish to pursue in their attempts to curtail the heavy flow of rural-to-urban migration and to ameliorate the serious unemployment problems that continue to plague their crowded cities.

Unit Objectives

After completing this unit, you will be able:

- To understand the potential roles of cities to enhance economic development in the contemporary developing economies
- To familiar with the well-known theoretical model of rural- urban migration (the Todaro Model of Migration)
- To examine the advantages and disadvantages of the informal sector in the modern cities of the contemporary developing countries
- To evaluate different policy options that the contemporary developing countries should pursue to solve problems associated with heavy in flow of rural to urban migration

3.1. The Role of Cities

To a large degree, cities are formed because they provide cost advantages to producers and consumers through what are called **agglomeration economies**. Agglomeration economies come in two forms. **Urbanization economies** are effects associated with the general growth of a concentrated geographic region. **Localization economies** are effects captured by particular sectors of the economy, such as finance or automobiles, as they grow within an area. When transportation costs are significant, users of the outputs of an industry may benefit from a nearby location to save on these costs. This benefit is a type of forward linkage. In addition, firms of the same or related industries may benefit from being located in the same city, so they can all draw on

a large pool of workers with the specific skills used in that sector or from specialized infrastructure. This is a type of backward linkage. Workers with specialized skills appropriate to the industry prefer to be located there as well so that they can easily find a new job or be in a position to take advantage of better opportunities.

3.1.1 Industrial Districts

Firms often also prefer to be located where they can learn from other firms doing similar work. Learning takes place in both formal relationships, such as joint ventures, and informal ones, such as from tips learned in evening social clubs. These spillovers are also agglomeration economies, part of the benefits called "industrial districts," and they play big role in "clusters" theory of competitive advantage. Firms located in such industrial districts also benefit from the opportunity to contract out work easily when an unusually large order materializes. Thus, a firm of modest size does not have to turn down a big job due to lack of capacity, an arrangement that provides "flexible specialization." Further, firms may wish to operate in well-known districts for the marketing advantages of locating where consumers of their goods know to shop to get the best selection.

A growing body of evidence shows that industrial clusters are now common in developing countries, at stages of industrial development ranging from cottage industry to advanced manufacturing techniques and appear to be significant factors in emerging industrial competitiveness. Nevertheless, the dynamism of these clusters has varied widely. Some of the identified districts are traditional clusters of artisans that have shown little ability to innovate, export, or expand.

3.1.2 Efficient Urban Scale

Localization economies do not imply that it would be efficient for all of a country's industries to be located together in a single city! These economies extend across closely related industries, such as those with strong backward and forward linkages, but there are fewer productivity benefits for unrelated industries to locate together. One notable exception is the potential spillover from technological progress in one industry to its adaptation for different uses in another industry. But there are also some important congestion costs. The higher urban density, the higher costs of real estate.

In large urban areas, workers may find themselves with longer and longer commutes and greater transportation costs and may demand higher wages to cover these costs. In addition, the costs of infrastructure such as water and sewer systems are higher in concentrated urban areas. In theory, if costs of transportation of finished goods are high and consumers wish to be located in the largest city to avoid paying those transportation costs as much as possible, economic activity could

become indefinitely concentrated within a city (called the "black hole" effect), but it is generally much less costly to improve the transportation system of a country than to pay the costs of maintaining a gargantuan urban complex. Under competitive forces and other things being equal, if workers are mobile, a worker in a large city with higher wages but higher costs of living (such as higher housing prices) is no better off in real material terms than a worker with comparable education, experience, ability, and health in a small city who has lower wages and lower costs of living.

Thus, the concentrating, or "centripetal," forces of urban agglomeration economies are opposed by the dispersing, or "centrifugal," forces of diseconomies featuring increasing costs with greater concentration, because some of the factors of production, most obviously land, are not mobile. Thus, it is normal for an economy to have a range of cities, with sizes dependent on the scale of the industries it sponsors, and the extent of agglomeration economies found for that industry or cluster of industries.

Two well-known theories of city size are the urban hierarchy model (central place theory) and the differentiated plane model. In the urban hierarchy model, plants in various industries have a characteristic market radius that results from the interplay of three factors: economies of scale in production, transportation costs, and the way the demand for land is spread over space. The larger the economies of scale in production and the lower the transportation costs, the larger the radius of territory that will be served by that industry to minimize costs. In contrast, if the price of real estate is bid up to high levels in the resulting cities; this will tend to create smaller areas.

As a result, small cities contain activities with short market radius, while large cities emerge to contain activities of both small and large radius. Generally speaking, activities of a national scope, such as government and finance, will be located in a single city (though not necessarily the same large city because of the effect of congestion costs). Clearly, the urban hierarchy approach applies better to non-export industries than to export industries. When countries have different specializations in the international market or are at different stages of economic development, the size distribution of cities may potentially differ. For example, a developing country that still overwhelmingly specializes in agriculture might reasonably have one or two large cities serving national industries such as finance and government and many smaller towns serving local agricultural areas. A country with a highly differentiated manufacturing and service base might have a large number of medium-size cities.

In the differentiated plane model, the limited number of transportation routes linking the industries within an economy plays a key role. The model predicts urban concentrations where the scarce transportation routes cross, called "internal nodes." The hierarchy of urban sizes depends

on the pattern of nodes and the industrial mix. Primary processing industries have few inputs and are usually located near the source of the primary resource. However, there will also be incentives for industries with strong backward or forward linkages to locate in the same city.

3.1.3 The Urban Giantism Problem

In the case of developing countries, the main transportation routes are often a legacy of colonialism. Theorists of the dependency school have compared colonial transportation networks to drainage systems, emphasizing ease of extraction of the country's natural resources. In many cases, the capital city will be located near the outlet of this system on the sea coast. This type of transportation system is also called a "hub-and-spoke system," which is especially visible when the capital city is located in the interior of the country.

Sometimes one urban core grows to become too large to keep the costs of the industries located there to a minimum. In developed countries, other cores are often developed within the broad metropolitan region, enabling the region as a whole to continue to receive benefits of agglomeration while lowering some of the costs; or new cities may develop in entirely different parts of the country. But this creation of new urban cores does not happen automatically if there are advantages to locating where other firms and residents are already located today. Who will be the pioneer if it is less costly to stay where you are and wait for other pioneers to settle in the new city first? In economic terms, the agglomeration economies of cities are externalities, which must somehow be internalized or the market will fail.

First-City Bias: A form of urban bias that has often caused considerable harm might be termed first-city bias. This means that the country's largest or "first" city receives a disproportionately large share of public investment and incentives for private investment, in relation to the country's second city and other smaller cities. As a result, the first city receives a disproportionately, and inefficiently, large share of population and economic activity.

3.1.3.1 Causes of Urban Giantism

Why have first cities often swelled to such a large multiple of second cities in developing countries? Overall, urban giantism probably results from a combination of a hub-and-spoke transportation system and the location of the political capital in the largest city, thus combining the effects of the urban hierarchy model with the differentiated plane model. This is further reinforced by a political culture of rent seeking and the capital market failures that make the creation of new urban centers a task that markets cannot complete.

Thus, firms wish to set up operations in the city where the most consumers already live, which attracts more people to the region in search of jobs and perhaps lower prices; this concentration in turn attracts still more firms and consumers in a circle of causation. However, when trade barriers

are reduced, the incentive to focus production on the home market is also reduced, and exporters and their suppliers have much less incentive to be located in the country's biggest population center. This moves production toward ports and borders, or elsewhere in the country, to escape the overly large congestion costs of the largest city.

In the developing world, until recently relatively few countries were effective democracies. In the 1970s, almost all developing countries had authoritarian governments of one form or another. To remain in power and prevent popular uprisings and coups, which were generally thought to be most threatening when launched from the capital city, governments had an incentive to "buy off" the population of the largest city. This focus of national government spending on the capital city is the bread-and-circuses effect, recalling the phrasing of "rent-sharing" policies in ancient Rome in its period of expansion. The availability of better opportunities, whether the equivalent of the grain handouts in ancient Rome jobs, wages, infrastructure, and other government services concentrated in the capital city of many of today's developing countries, attracts an ever-growing migrant population, in turn leading to larger precautionary government spending as the fear of political instability grows.

A final political economy factor contributes to capital city giantism: It becomes advantageous for firms to be located where they have easy access to government officials, to curry political favor from a regime that can be induced to give companies special favors for a price or that simply demands bribes to function at all. The resulting first-city giantism may be viewed as a form of underdevelopment trap, which may be escaped fully only with a return to democratic rule together with a better balance of incentives to compete for exports as well as home consumption. Democracy does not eliminate political benefits of location in the national capital, but while lobbyists still congregate in the political capital, there may be less incentive for production to become over concentrated there.

3.2 The Urban Informal Sector

As discussed in module one, a major focus of development theory has been on the dualistic nature of developing countries' national economies—the existence of a modern urban capitalist sector geared toward capital-intensive, large-scale production and a traditional rural subsistence sector geared toward labor-intensive, small-scale production. In recent years, this dualistic analysis has also been applied specifically to the urban economy, which has been decomposed into a formal and an informal sector.

The existence of an unorganized, unregulated, and mostly legal but unregistered **informal sector** was recognized in the 1970s, following observations in several developing countries that massive

additions to the urban labor force failed to show up in formal modern-sector unemployment statistics. The bulk of new entrants to the urban labor force seemed to create their own employment or to work for small-scale family- owned enterprises. The self-employed were engaged in a remarkable array of activities, such as hawking, street vending, and other jobs. Others found jobs as mechanics, carpenters, small artisans, barbers, and personal servants. Still others were highly successful small-scale entrepreneurs with several employees (mostly relatives) and high incomes. Some could even eventually graduate to the formal sector, where they become legally registered, licensed, and subject to government labor regulations. With the unprecedented rate of growth of the urban population in developing countries expected to continue and with the increasing failure of the rural and urban formal sectors to absorb additions to the labor force, more attention is being devoted to the role of the informal sector in serving as a panacea for the growing unemployment problem.

The informal sector continues to play an important role in developing countries, despite decades of benign neglect at best and outright hostility at worst. In many developing countries, about half of the employed urban population works in the informal sector. The informal sector is characterized by a large number of small-scale production and service activities that are individually or family-owned and use simple, labor-intensive technology. They tend to operate like monopolistically competitive firms with ease of entry, excess capacity, and competition driving profits (incomes) down to the average supply price of labor of potential new entrants. The usually self-employed workers in this sector have little formal education, are generally unskilled, and lack access to financial capital. As a result, worker productivity and income tend to be lower in the informal sector than in the formal sector.

Moreover, workers in the informal sector do not enjoy the measure of protection afforded by the formal modern sector in terms of job security, decent working conditions, and old -age pensions. Many workers entering this sector are recent migrants from rural areas unable to find employment in the formal sector. Their motivation is often to obtain sufficient income for survival, relying on their own indigenous resources to create work. Since many members of the household as possible are involved in income-generating activities, including women and children, and they often work very long hours. A large fraction inhabits shacks that they themselves have built in slums and squatter settlements, which generally lack minimal public services such as electricity, water, drainage, transportation, and educational and health services. Others are even less fortunate. They find sporadic temporary employment in the informal sector as day laborers and hawkers, but their incomes are insufficient to provide even the most rudimentary shelter.

3.2.1 Arguments in Favor of Urban Informal Sector

There are several arguments that are made in favor of promoting the informal sector: First, scattered evidence indicates that the informal sector generates surpluses even under the currently hostile policy environment, which denies it access to the advantages offered to the formal sector, such as credit, foreign exchange, and tax concessions. Thus, the informal sector's surplus could provide an impetus to growth in the urban economy.

Second, as a result of its low capital intensity, only a fraction of the capital needed in the formal sector is required to employ a worker in the informal sector, offering considerable savings to developing countries so often plagued with capital shortages. Third, by providing access to training and apprenticeships at substantially lower costs than that provided by formal institutions and the formal sector; the informal sector can play an important role in the formation of human capital

Fourth, the informal sector generates demand for semiskilled and unskilled labor whose supply is increasing in both relative and absolute terms and is unlikely to be absorbed by the formal sector with its increasing demands for a skilled labor force. Fifth, the informal sector is more likely to adopt appropriate technologies and make use of local resources, allowing for a more efficient allocation of resources.

Sixth, the informal sector plays an important role in recycling waste materials, engaging in the collection of goods ranging from scrap metals to cigarette butts, many of which find their way to the industrial sector or provide basic commodities for the poor. Finally, promotion of the informal sector would ensure an increased distribution of the benefits of development to the poor, many of whom are concentrated in the informal sector.

Promotion of the informal sector is not, however, without its disadvantages. One of the major disadvantages in promoting the informal sector lies in the strong relationship between rural-urban migration and labor absorption in the informal sector. Migrants from the rural sector have both a lower unemployment rate and a shorter waiting period before obtaining a job in the informal sector. Promoting income and employment opportunities in the informal sector could therefore aggravate the urban unemployment problem by attracting more labor than either the desirable parts of the informal or the formal sector could absorb.

Furthermore, there is concern over the environmental consequences of a highly concern- treated informal sector in the urban areas. Many informal-sector activities cause pollution and congestion or inconvenience to pedestrians. Moreover, increased densities in slums and low-income neighborhoods, coupled with poor urban services, could cause enormous problems for urban areas. Any policy measures designed to promote the informal sector must be able to cope with

these various problems. Finally, it is an almost universal observation that when regular formal-sector employment becomes available, informal-sector micro entrepreneurs switch sectors to take these jobs clear evidence of "revealed preference."

Because access to skills plays an important role in determining the structure of the informal sector, governments should facilitate training in the areas that are most beneficial to the urban economy. In this way, the government can play a role in shaping the informal sector so that it contains production and service activities that provide the most value to society. Specifically, such measures might promote legal activities and discourage illegal ones by providing proper skills and other incentives.

The lack of capital is a major constraint on activities in the informal sector. The provision of credit would therefore permit these enterprises to expand, produce more profit, and hence generate more income and employment. Access to improved technology would have similar effects. Providing infrastructure and suitable locations for work could help alleviate some of the environmental consequences of an expanded informal sector. Most important, better living conditions must be provided, if not directly, then by promoting growth of the sector on the fringes of urban areas or in smaller towns where the population will settle close to its new area of work, away from the urban density. Promotion of the informal sector outside the urban areas may also help redirect the flow of rural-urban migration, especially if carried out in conjunction with the policies discussed at the end of this unit.

3.3 Urban Unemployment

One of the major consequences of the rapid urbanization process has been the burgeoning supply of job seekers into both the modern (formal) and informal sectors of the urban economy. In many developing countries, the supply of workers far exceeds the demand, the result being extremely high rates of unemployment and underemployment in urban areas. Because a major contributing factor to both high rates of urban growth and high rates of unemployment and underemployment is rural-urban migration, it is essential to investigate this issue in some detail.

3.4 Migration and Development

As we have seen earlier in the unit, rural-urban migration has been dramatic, and urban development plays an important role in economic development. Rates of rural-urban migration in developing countries have exceeded rates of urban job creation and thus surpassed greatly the absorption capacity of both industry and urban social services. Migration today, particularly to the largest LDC cities, must be seen as the major factor contributing to the ubiquitous phenomenon of urban surplus labor, a force that continues to exacerbate already serious urban unemployment problems.

Migration worsens rural-urban structural imbalances in two direct ways. First, on the supply side, internal migration disproportionately increases the growth rate of urban job seekers relative to urban population growth, which itself is at historically unprecedented levels, because of the high proportion of well-educated young people in the migrant system. Their presence tends to swell the urban labor supply while depleting the rural countryside of valuable human capital. Second, on the demand side, urban job creation is generally more difficult and costlier to accomplish than rural job creation because of the need for substantial complementary resource inputs for most jobs in the industrial sector. Moreover, the pressures of rising urban wages and compulsory employee fringe benefits in combination with the unavailability of appropriate, more labor-intensive production technologies mean that a rising share of modern-sector output growth is accounted for by increases in labor productivity. Together this rapid supply increase and lagging demand growth tend to convert a short-run problem of resource imbalances into a long-run situation of chronic and rising urban surplus labor.

But the impact of migration on the development process is much more pervasive than its exacerbation of urban unemployment and underemployment. In fact, the significance of the migration phenomenon in most developing countries is not necessarily in the process itself or even in its impact on the sectoral allocation of human resources. Rather, its significance lies in its implications for economic growth in general and for the character of that growth, particularly its distributional manifestations.

We must therefore recognize that migration in excess of job opportunities is both a symptom of and a contributor to underdevelopment. Understanding the causes, determinants, and consequences of internal rural-urban labor migration is thus central to understanding the nature and character of the development process and to formulating policies to influence this process in socially desirable ways. A simple yet crucial step in underlining the centrality of the migration phenomenon is to recognize that any economic and social policy that affects rural and urban real incomes will directly or indirectly influence the migration process. This process will in turn itself tend to alter the pattern of sectoral and geographic economic activity, income distribution, and even population growth. Because all economic policies have direct and indirect effects on the level and growth of urban or rural incomes or both, they all will tend to influence the nature and magnitude of the migration stream.

Although some policies may have a more direct and immediate impact, there are many others that, though less obvious, may in the long run be no less important. Included among these policies are land tenure arrangements; commodity pricing; credit allocation; taxation; export promotion; import substitution; commercial and exchange-rate policies; the geographic distribution of social

services; the nature of public investment programs; location of new industries and others. There is thus a clear need to recognize the central importance of internal and, for many countries, even international migration and to integrate the two-way relationship between migration and population distribution on the one hand and economic variables on the other into a more comprehensive framework designed to improve development policy formulation.

In addition, we need to understand better not only why people move and what factors are most important in their decision-making process but also what the consequences of migration are for rural and urban economic and social development. If all development policies affect migration and are affected by it, which ones are the most significant, and why? What are the policy options and trade-offs among different and sometimes competing objectives? Part of our task in the following sections will be to seek answers to these and other questions relating to migration, unemployment, and development.

Migration patterns are complex. The most important type of migration from the standpoint of long-run development is rural-urban migration, but a great deal of rural-rural, urban-urban, and even urban-rural migration also takes place. Rural-urban migration is most important because the population share of cities is growing, despite the fact that fertility is much lower in urban areas, and the difference is accounted for by rural-urban migration. It is also important because of the potential development benefits of economic activity of cities, due to agglomeration economies and other factors. However, urban-rural migration is important to understand because it usually occurs when hard times in cities coincide with increases in output prices from the country's cash crops; the recent experience of Ghana is a notable example. Thus, the overall picture is one of a remarkable continuous movement of many people within developing countries, especially over short distances. These movements contradict the popular image of stasis in traditional societies.

In addition to wage differentials, age, and education, migration is also explained partly by relocation upon remarrying; prior emigration of family members; distance and costs of relocation, occurrence of famine, disease, violence, and other disasters; and relative standing in the origin community, with those lower on the social order more likely to migrate. Migration can also be a form of portfolio diversification of families who seek to settle some members in areas where they may not be affected by economic shocks in the same way as if they had stayed at home.

3.4.1. Toward an Economic Theory of Rural-Urban Migration

The economic development of Western Europe and the United States was closely associated with the movement of labor from rural to urban areas. For the most part, with a rural sector dominated by agricultural activities and an urban sector focusing on industrialization, overall economic development in these countries was characterized by the gradual reallocation of labor out of

agriculture and into industry through rural-urban migration, both internal and international. Urbanization and industrialization were in essence synonymous.

But the overwhelming evidence of the past several decades, when developing nations witnessed a massive migration of their rural populations into urban areas despite rising levels of urban unemployment and underemployment, lessens the validity of the Lewis two-sector model of development. An explanation of the phenomenon, as well as policies to address the resulting problems must be sought elsewhere. As a result, the **Todaro migration model** was developed.

3.4.2 Description of the Todaro Model

Starting from the assumption that migration is primarily an economic phenomenon, the Todaro model postulates that migration proceeds in response to urban-rural differences in **expected income** rather than actual earnings. The fundamental premise is that migrants consider the various labor market opportunities available to them in the rural and urban sectors and choose the one that maximizes their expected gains from migration. In essence, the theory assumes that members of the labor force, both actual and potential, compare their expected incomes for a given time horizon in the urban sector (the difference between returns and costs of migration) with prevailing average rural incomes and migrate if the former exceeds the latter.

Consider the following illustration. Suppose that the average unskilled or semiskilled rural worker has a choice between being a farm laborer (or working his own land) for an annual average real income of, say, 50 Birr or migrating to a city, where a worker with his skill or educational background can obtain wage employment yielding an annual real income of 100 Birr. The more commonly used economic models of migration, which place exclusive emphasis on the income differential factor as the determinant of the decision to migrate, would indicate a clear choice in this situation. The worker should seek the higher-paying urban job. It is important to recognize, however, that these migration models were developed largely in the context of advanced industrial economies and hence implicitly assume the existence of full or near-full employment. In a full-employment environment, the decision to migrate can be based solely on the desire to secure the highest-paid job wherever it becomes available. Simple economic theory would then indicate that such migration should lead to a reduction in wage differentials through the interaction of the forces of supply and demand, in areas of both emigration and immigration. Unfortunately, such an analysis is not realistic in the context of the institutional and economic framework of most developing nations.

First, developing countries are overwhelmed by a chronic unemployment problem that a typical migrant cannot expect to secure a high-paying urban job immediately. In fact, it is much more likely that on entering the urban labor market, many uneducated, unskilled migrants will either

become totally unemployed or will seek casual and part-time employment as vendors, hawkers, repairmen, or any other job at relatively competitive price and wage. In the case of migrants with considerable human capital in the form of a secondary or university certificate, opportunities are much better, and many will find formal-sector jobs relatively quickly. But they constitute only a small proportion of the total migration stream.

Consequently, in deciding to migrate, the individual must balance the probabilities and risks of being unemployed or underemployed for a considerable period of time against the positive urban-rural real income differential. The fact that a typical migrant who gains a modern-sector job can expect to earn twice the annual real income in an urban area than in a rural environment may be of little consequence if the actual probability of his securing the higher-paying job within, say, a one-year period is one chance in five. Thus the actual probability of his being successful in securing the higher-paying urban job is 20%, and therefore his expected urban income for the one-year period is in fact, 20 Birr ($= 100 \text{ Birr} \times 0.2$) and not the 100 Birr that an urban worker in a full employment environment would expect to receive. So, with a one-period time horizon and a probability of success of 20%, it would be irrational for this migrant to seek an urban job, even though the differential between urban and rural earnings capacity is 100%. However, if the probability of success were 60% and the expected urban income therefore 60 Birr, it would be entirely rational for our migrant with his one-period time horizon to try his luck in the urban area, even though urban unemployment may be extremely high.

If we now approach the situation by assuming a considerably longer time horizon—a more realistic assumption, especially in view of the fact that the vast majority of migrants are between the ages of 15 and 24—the decision to migrate should be represented on the basis of a longer-term, more permanent income calculation. If the migrant anticipates a relatively low probability of finding regular wage employment in the initial period but expects this probability to increase over time as he is able to broaden his urban contacts, it would still be rational for him to migrate, even though expected urban income during the initial period or periods might be lower than expected rural income. As long as the **present** value of the net stream of expected urban income over the migrant's planning horizon exceeds that of the expected rural income, the decision to migrate is justifiable.

Rather than equalizing urban and rural wage rates, as would be the case in a competitive model, we see that rural-urban migration in our model acts as an equilibrating force that equates rural and urban expected incomes. For example, if average rural income is 60 and urban income is 120, a 50% urban unemployment rate would be necessary before further migration would no longer be profitable. Because expected incomes are defined in terms of both wages and employment

probabilities, it is possible to have continued migration despite the existence of sizable rates of urban unemployment. In our example, migration would continue even if the urban unemployment rate were 30% to 40%.

3.4.3 Five Policy Implications of the Model

Although the Todaro theory might at first seem to devalue the critical importance of rural-urban migration by portraying it as an adjustment mechanism by which workers allocate themselves between rural and urban labor markets, it does have important policy implications for development strategy with regard to wages and incomes, rural development, and industrialization.

First, imbalances in urban-rural employment opportunities caused by the urban bias, particularly first-city bias, of development strategies must be reduced. Because migrants are assumed to respond to differentials in expected incomes, it is vitally important that imbalances between economic opportunities in rural and urban sectors be minimized. Permitting urban wage rates to rise at a greater pace than average rural incomes will stimulate further rural-urban migration in spite of rising levels of urban unemployment. This heavy influx of people into urban areas not only gives rise to socioeconomic problems in the cities but may also eventually create problems of labor shortages in rural areas, especially during the busy seasons. These social costs may exceed the private benefits of migration.

Second, urban job creation is an insufficient solution for the urban unemployment problem. The traditional (Keynesian) economic solution to urban unemployment can result in the paradoxical situation where more urban employment leads to higher levels of urban unemployment! Once again, the imbalance in expected income-earning opportunities is the crucial concept. Because migration rates are assumed to respond positively to both higher urban wages and higher urban employment opportunities, it follows that for any given positive urban-rural wage differential, higher urban employment rates will widen the expected differential and induce even higher rates of rural-urban migration. For every new job created, two or three migrants who were productively occupied in rural areas may come to the city. Hence a policy designed to reduce urban unemployment may lead not only to higher levels of urban unemployment but also to lower levels of agricultural output due to **induced migration**.

Third, indiscriminate educational expansion will lead to further migration and unemployment. The Todaro model also has important policy implications for curtailing investment in excessive educational expansion, especially at the higher levels. The heavy arrival of rural migrants into urban areas at rates much in excess of new employment opportunities has necessitated a rationing device in the selection of new employees. Although within each educational group such selection may be largely random, many observers have noted that employers tend to use educational

attainment typical rationing device. For the same wage, they will hire people with more education in preference to those with less, even though extra education may not contribute to better job performance. It follows that for any given urban wage, if the probability of success in securing a modern-sector job is higher for people with more education, their expected income differential will also be higher, and they will be more likely to migrate to the cities. The basic Todaro model therefore provides an economic rationale for the observed fact in most Developing countries that rural inhabitants with more education are more likely to migrate than those with less. From the viewpoint of educational policy; it is safe to predict that as job opportunities become scarce in relation to the number of applicants, students will experience increasing pressure to proceed further up the educational ladder. The private demand for education, which in many ways is a derived demand for urban jobs, will continue to exert tremendous pressure on governments to invest in post primary school facilities. But for many of these students, the specter of joining the ranks of the “educated unemployed” becomes more of a reality with each passing year.

Fourth, wage subsidies and traditional scarcity factor pricing can be counterproductive. Because actual urban wages generally exceed the market or “correct” wage as a result of a variety of institutional factors, it is often argued that the elimination of wage distortions through price adjustments or a subsidy system will encourage more labor- intensive modes of production, they can also lead to higher levels of unemployment in accordance with our argument about induced migration. The overall welfare impact of a **wage subsidy** policy when both the rural and urban sectors are considered is not immediately clear. Much will depend on the level of urban unemployment, the size of the urban-rural expected-income differential, and the magnitude of induced migration as more urban jobs are created.

Finally, programs of integrated rural development should be encouraged. Policies that operate only on the demand side of the urban employment picture, such as wage subsidies, direct government hiring, elimination of factor price distortions, and employer tax incentives, are probably far less effective in the long run in alleviating the unemployment problem than policies designed directly to regulate the supply of labor to urban areas. Clearly, however, some combination of both kinds of policies is most desirable.

Policies of rural development are crucial to this aim. Many informed observers agree on the central importance of rural and agricultural development if the urban unemployment problem is to be solved. Most proposals call for the restoration of a proper balance between rural and urban incomes and for changes in government policies that currently give development programs a strong bias toward the urban industrial sector (e.g., policies in the provision of health, education, and social services). Conceptually, it may be useful to think of cities and their surrounding rural

areas as integrated systems. There are significant complementarities between town and country. Agricultural and raw materials grown and extracted in rural areas are inputs for urban industry.

Although there is some urban agriculture, most food consumed in urban areas is grown in agricultural regions. Towns are needed to allow sufficient agglomeration economies, as well as economies of scale, to produce and exchange many goods and services that are needed in rural areas. In turn, when rural incomes grow, markets for urban manufactures expand. People come from their rural residences to work in the city by the day or the week. City residents temporarily migrate to nearby agricultural regions during peak planting and harvesting seasons. Thus rural-urban linkages are extensive. And while investment in urban areas can accelerate migration to cities, investment in agriculture can raise productivity and incomes, making labor redundant, and also accelerate migration. As a result, for policy purposes, it may make a great deal of sense to take account of rural impacts when devising urban policies and vice versa.

At the same time, as globalization proceeds, cities tend to trade more with other cities, often in distant parts of the world, and less with nearby rural areas. Moreover, cities generally get the upper hand when urban and rural areas are treated as a bloc, reinforcing urban bias. And rural hinterlands, far from significant cities and from the attention of distant governments, whether national or regional, often suffer from benign neglect at best and systematic exploitation at worst, such as forced sale of food at low prices. Thus, rural areas need to retain their own autonomy, and poverty programs need to be tailored to the needs of rural citizens.

Every effort must be made to broaden the economic base of the rural economy. The present unnecessary economic incentives for rural-urban migration must be minimized through creative and well-designed programs of integrated rural development. These should focus on farm and non-farm income generation, employment growth, health-care delivery, educational improvement, infrastructure development and so on. Successful rural development programs adapted to the socioeconomic and environmental needs of particular countries and regions seem to offer the only viable long-run solution to the problem of excessive rural-urban migration.

To assert, however, that there is an urgent need for policies designed to curb the excessive influx of rural migrants is not to imply an attempt to reverse what some observers have called inevitable historical trends. Rather, the implication of the Todaro migration model is that there is a growing need for a policy package that does not exacerbate these historical trends toward urbanization by artificially creating serious imbalances in economic opportunities between urban and rural areas.

All in all, the Todaro migration model has four basic characteristics:

1. Migration is stimulated primarily by rational economic considerations of relative benefits and costs, mostly financial but also psychological.
2. The decision to migrate depends on expected rather than actual urban-rural real-wage differentials where the expected differential is determined by the interaction of two variables, the actual urban-rural wage differential and the probability of successfully obtaining employment in the urban sector.
3. The probability of obtaining an urban job is directly related to the urban employment rate and thus inversely related to the urban unemployment rate.
4. Migration rates in excess of urban job opportunity growth rates are not only possible but also rational and even likely in the face of wide urban-rural expected income differentials. High rates of urban unemployment are therefore inevitable outcomes of the serious imbalance of economic opportunities between urban and rural areas in most underdeveloped countries.

CHAPTER FOUR

Agriculture and Economic Development

4. 1 Agricultural progress and Rural Development

Despite real progress, nearly 2 billion people in the developing world grind out a meager and often inadequate existence in agricultural pursuits. Over 3.1 billion people lived in rural areas in developing countries in 2010, a quarter of them in extreme poverty. People living in the countryside make up more than half of the population of such diverse Latin American and Asian nations as Haiti, Guatemala, India, Indonesia, Myanmar, Honduras, Sri Lanka, Pakistan, Bangladesh, the Philippines, Thailand, and China. In sub-Saharan Africa, the ratios are much higher, with rural dwellers constituting 65% of the total population.

Over two thirds of the world's poorest people are also located in rural areas and engaged primarily in subsistence agriculture. Their basic concern is survival. The United Nations Food and Agriculture Organization estimated that in 2009, over 1 billion people did not have enough food to meet their basic nutritional needs.

If development is to take place and become self-sustaining, it will have to include the rural areas in general and the agricultural sector in particular. The core problems of widespread poverty, growing inequality, and rapid population growth all originate in the stagnation and often retrogression of economic life in rural areas, particularly in Africa. Traditionally in economic development, agriculture has been assumed to play a passive and supportive role. Its primary purpose was to provide sufficient low-priced food and manpower to the expanding industrial economy, which was thought to be the dynamic “leading sector” in any overall strategy of economic development.

Lewis's famous two-sector model, discussed in your development economics one course, is an example of a theory of development that places heavy emphasis on rapid industrial growth with an agricultural sector fueling this industrial expansion by means of its cheap food and surplus labor. Nobel laureate Simon Kuznets introduced an early schema, noting that agriculture made four “contributions to economic development”:

- **The product contribution** of inputs for industry such as textiles and food processing,
- **The foreign-exchange contribution** of using agricultural export revenues to import capital equipment.
- **The market contribution** of rising rural incomes creating more demand for consumer products, and finally,
- **The factor market contribution**, divided between the labor contribution (Lewis's manpower) - workers not needed on farms after agricultural productivity was raised could then work in

industry - and the capital contribution (some farm profits could be reinvested in industry as agriculture became a steadily smaller fraction of national income). The framework implicitly and ironically still treats industrialization rather than rural modernization as the core development goal.

Today, most development economists share the consensus that far from playing a passive, supporting role in the process of economic development, the agricultural sector in particular and the rural economy in general must play an indispensable part in any overall strategy of economic progress, especially for the low-income developing countries. An agriculture - and employment-based strategy of economic development requires three basic complementary elements:

1. Accelerated output growth through technological, institutional, and price incentive changes designed to raise the productivity of small farmers;
2. Rising domestic demand for agricultural output derived from an employment-oriented urban development strategy; and
3. Diversified, nonagricultural, labor-intensive rural development activities that directly and indirectly support and are supported by the farming community.

To a large extent, therefore, agricultural and rural development has come to be regarded by many economists as the *sine qua non* (pre-requisite) of national development. Without such *integrated rural development*, in most cases, industrial growth either would be stultified or, if it succeeded, would create severe internal imbalances in the economy.

4.2. The structure of Agrarian Systems in Developing World

Although there is considerable diversity among developing nations, as well as within developing countries, each region tends to have a number of characteristics in common. The agricultural patterns of Africa, Asia and Latin America are agriculture-based economies, agriculturally transforming economies and urbanized economies respectively.

Relatedly, agriculture in these regions often typifies the stages of subsistence, mixed, and commercial farming, with important regional exceptions and varying success at inclusion of the poor. With successful development, countries tend to move toward commercialized agriculture, though with different trajectories and differing economic, social, and technical problems to solve along the way. Regions that have high concentrations of poverty also often reflect patterns of traditional agriculture (in Africa), high population density and subdivided smallholdings (in Asia), and the sharp inequalities of very large and very small farms (in Latin America).

We will identify the various challenges facing each group of countries and look at countries that are typical of their region and some countries and districts that deviate from the pattern. Over two-thirds of the world's extreme poor are involved in agricultural activities. We will therefore examine the

economics of peasant subsistence agriculture and discuss the stages of transition from subsistence to commercial farming in developing nations. Our focus is not only the economic factors but also the social, institutional, and structural requirements of small-farm modernization. We will then explore the meaning of integrated rural development and review alternative policies designed to raise levels of living in rural areas.

4.2.1. Peasant Agriculture in Latin America, Asia and Africa

In many developing countries, various historical circumstances have led to a concentration of large areas of land in the hands of a small class of powerful landowners. This is especially true in Latin America and parts of the Asian subcontinent. In Africa, both historical circumstances and the availability of relatively more unused land have resulted in a different pattern and structure of agricultural activity.

Agrarian Patterns in Latin America: Progress and Remaining Poverty Challenges

In Latin America, as in Asia and Africa, agrarian structures are not only part of the production system but also a basic feature of the entire economic, social and political organization of rural life. The agrarian structure that has existed in Latin America since colonial times and is still widespread in a substantial part of the region is a pattern of agricultural dualism known as *latifundio* - *minifundio*. Basically, *latifundios* are very large landholdings. They are usually defined as farms large enough to provide employment for more than 12 people, though some employ thousands. In contrast, *minifundios* are the smallest farms. They are defined as farms too small to provide employment for a single family (two workers) with the typical incomes, markets, and levels of technology and capital prevailing in each country or region.

Using Gini coefficients to measure the degree of land concentration, researchers report that the coefficient for Colombia is 0.86, for Costa Rica 0.82, for Uruguay 0.82, for Peru 0.91, and for Venezuela 0.91. Although estimates vary, changes in land inequality are limited in the case of Latin America. Other countries are even more unequal; the Gini for Paraguay is an almost perfectly unequal 0.94. These are the highest regional Gini coefficients in the world, and they dramatically reflect the degree of land ownership inequality (and thus, in part, income inequality) throughout Latin America. But *latifundios* and *minifundios* do not constitute the entirety of Latin American agricultural holdings. A considerable amount of production occurs on family farms and medium-size farms. The former provide work for two to four people (recall that the *minifundio* could provide work for fewer than two people), and the latter employ 4 to 12 workers (just below the *latifundio*).

In Venezuela, Brazil, and Uruguay, these intermediate farm organizations account for almost 50% of total agricultural output and employ similar proportions of agricultural labor. These farms use a more efficient balance between labor and land, and studies show that they have much higher total factor

productivity than either latifundios or minifundios, as the law of diminishing returns would suggest. Indeed, evidence from a wide range of developing countries, demonstrates that smaller farms are more efficient (lower-cost) producers of most agricultural commodities.

A major explanation for the relative economic inefficiency of farming the fertile land on the latifundios is simply that the wealthy landowners often value these holdings not for their potential contributions to national agricultural output but rather for the considerable power and prestige that they bring. Much of the land is left idle or farmed less intensively than on smaller farms. Also, latifundio transaction costs, especially the cost of supervising hired labor, are much higher than the low effective cost of using family labor on family farms or minifundios. It follows that raising agricultural production and improving the efficiency of Latin American agrarian systems in traditional areas will require much more than direct economic policies that lead to the provision of better seeds, more fertilizer, less distorted factor prices, higher output prices, and improved marketing facilities. It will also require a reorganization of rural social and institutional structures to provide Latin American peasants, particularly indigenous people who find it more challenging to migrate, a real opportunity to lift themselves out of their present state of economic subsistence and social subservience.

Despite the fact that many minifundio owners remain in poverty, especially among indigenous and mixed-race populations, and many latifundios continue to operate well below their productivity potential, a more dynamic sector, including some larger farms, has emerged. Efficient family and medium-size farms are found throughout the region. At an aggregate level the agricultural sector in Latin America appears to be doing fairly well. Chile has led the way in “nontraditional exports,” notably fresh fruits for the northern hemisphere winter markets and also aquaculture, vegetables, and wines. Diversification has reduced variance in export earnings. Productivity growth in cereals has been quite solid. Sugarcane-based biofuels and soybeans have played important roles in agricultural growth in Brazil.

And in traditional exports, particularly coffee, Latin America has led the way in taking advantage of niche opportunities for higher-value-added activities such as organic and Fair Trade markets. Some Latin American countries, such as Guatemala and Honduras, are still in the mixed transition phase, and in such countries, the latifundio-minifundio pattern tends to remain particularly dominant. But much of this pattern still prevails in many other areas. The extreme rural inequalities in Latin America typically stem from the Spanish and Portuguese colonial period, in which indigenous peoples were exploited in what often amounted to slavery, and African slaves were forcibly brought to the region. Overcoming this legacy has been a long and painful process, with much remaining to

be achieved. Social discrimination continues, and improved access for the poor to agricultural land in countries such as Colombia is still in all too many cases suppressed.

Areas with less favorable agricultural conditions, often with a concentration of minority populations, such as northeast Brazil, the Andean region, and parts of Mexico and Central America, tend to have persistently high poverty levels. Extreme rural inequality inhibits progress in these areas both because of reduced access by the poor to credit and other inputs and because elites effectively continue to block political participation by the poor, who often receive low levels of government services. Moreover, rural-to-urban migration has been disproportionately among more educated people, and the result is that rural populations are becoming older, more female, and more indigenous. These are factors in poverty rates that remain high for middle-income countries and will require sustained action by government and civil society.

Transforming Economies: Problems of Fragmentation and Subdivision of Peasant Land in Asia

The basic problem in Asia is one of too many people crowded on too little land. For example, the per capita availability of arable land as early as 1994 in India, China, and Bangladesh was 0.19, 0.08, and 0.07 hectares, respectively. The land is distributed more equally in Asia than in Latin America but still with substantial levels of inequality. The estimated Gini coefficients for land distribution in Asia range from 0.42 in Bangladesh and 0.45 in Thailand to 0.62 in India and 0.56 in Indonesia and Nepal. (Later estimates show no significant change in Thailand, a rise in inequality to about 0.48 in Bangladesh, and a smaller level of inequality in India of about 0.45. Throughout much of the twentieth century, rural conditions in Asia typically deteriorated. Nobel laureate Gunnar Myrdal identified three major interrelated forces that molded the traditional pattern of land ownership into its present fragmented condition: (1) the intervention of European rule, (2) the progressive introduction of monetized transactions and the rise in power of the moneylender, and (3) the rapid growth of Asian populations.

The traditional Asian agrarian structure before European colonization was organized around the village. Local chiefs and peasant families each provided goods and services—produce and labor from the peasants to the chief in return for protection, rights to use community land, and the provision of public services. Decisions on the allocation, disposition, and use of the village's most valuable resource, land, belonged to the tribe or community, either as a body or through its chief. Land could be redistributed among village members as a result of either population increase or natural calamities such as drought, flood, famine, war, or disease. Within the community, families had a basic right to cultivate land for their own use, and they could be evicted from their land only

after a decision by the whole village.

The arrival of the Europeans (mainly the British, French, and Dutch) led to major changes in the traditional agrarian structure, some of which had already begun. As Myrdal points out, “Colonial rule acted as an important catalyst to change, both directly through its effects on property rights and indirectly through its effects on the pace of monetization of the indigenous economy and on the growth of population. In the area of property rights, European land tenure systems of private property ownership were both encouraged and reinforced by law. One of the major social consequences of the imposition of these systems was, as Myrdal explains, the breakdown of much of the earlier cohesion of village life with its often elaborate, though informal, structure of rights and obligations.

The landlord was given unrestricted rights to dispose of the land and to raise the tribute from its customary level to whatever amount he was able to extract. He was usually relieved of the obligation to supply security and public amenities because these functions were taken over by the government. Thus his status was transformed from that of a tribute receiver with responsibilities to the community to that of an absolute owner unencumbered by obligations toward the peasants and the public, other than the payment of land taxes.

Contemporary landlords (The proprietor of a freehold interest in land with rights to lease out to tenants in return for some form of compensation for the use of the land) in India and Pakistan are able to avoid much of the taxation on income derived from their ownership of land. There are variations, but landlords in South Asia are often absentee owners who live in the town and turn over the working of the land to sharecroppers (A tenant farmer whose crop has to be shared with the landlord, as the basis for the rental contract) and other tenant farmers (One who farms on land held by a landlord and therefore lacks ownership rights and has to pay for the use of that land, for example, by giving a share of output to the owner). Sharecropping is widespread in both Asia and Latin America but more pervasive in Asia.

It has been estimated that of all tenanted land, some 84.5% is sharecropped in Asia but only 16.1% in Latin America. The institution is almost unknown in Africa, where the typical arrangement continues to be farms operated under tribal or communal tenures. For example, as about 48% of all tenanted land is sharecropped in India, 60% in Indonesia, and 79% in the Philippines. Though common in Colombia, sharecropping is unusual elsewhere in Latin America; for example, it has all but disappeared in Peru. The creation of individual titles to land made possible the rise to power of another dubious agent of change in Asian rural socioeconomic structures, the Money lender (A person who lends money at high rates of interest, for example to peasant farmers to meet their needs

for seeds, fertilizers, and other inputs). Once private property came into effect, land became a negotiable asset that could be offered by peasants as security for loans and, in the case of default, could be forfeited and transferred to the often unscrupulous moneylender.

At the same time, Asian agriculture was being transformed from a subsistence to a commercial orientation, both as a result of rising local demand in new towns and, more important, in response to external food demands of colonial European powers. With this transition from subsistence to commercial production, the role of the moneylender changed drastically. In the subsistence economy, his activities had been restricted to supplying the peasant with money to tide him over a crop failure or to cover extraordinary ceremonial expenditures such as family weddings or funerals. Most of these loans were paid in kind (in the form of food) at very high rates of interest. With the development of commercial farming, however, the peasant's cash needs grew significantly.

Money was needed for seeds, fertilizer, and other inputs. It was also needed to cover his food requirements if he shifted to the production of cash crops such as tea, rubber, or jute. Often moneylenders were more interested in acquiring peasant lands as a result of loan defaults than they were in extracting high rates of interest. By charging exorbitant interest rates or inducing peasants to secure larger credits than they could manage, moneylenders were often able to drive the peasants off their land. They could then reap the profits of land speculation by selling this farmland to rich and acquisitive landlords. Largely as a consequence of the moneylenders' influence, Asian peasant cultivators saw their economic status deteriorate. In addition to this rapid population growth is the other contributor for fragmentation and impoverishment. So that today average peasant holdings in many areas of these countries are less than 1 hectare.

Colonial practices often had long-lasting influences. In the case of India, regions in which property rights to land were given to landlords had significantly lower productivity and agricultural investments and significantly lower investments in health and education in the post-independence period than regions in which property rights were given to cultivators.

Subsistence Agriculture and Extensive Cultivation in Africa

Subsistence farming on small plots of land is the way of life for the majority of African people living in agriculture-based economies. The great majority of farm families in tropical Africa still plan their output primarily for their own subsistence. (The main exceptions are in former colonial settlement areas like the White Highlands of Kenya; sugar, cocoa, coffee, and other plantations in East and West Africa; and farms devoted to such export crops as green beans in Niger, cut flowers in Kenya and Ethiopia, and legumes in Tanzania.) Since the basic variable input in African agriculture is farm family and village labor, African agriculture systems are dominated by three major characteristics:

1. The importance of subsistence farming in the village community;
2. The existence of some (though rapidly diminishing) land in excess of immediate requirements, which permits a general practice of shifting cultivation and reduces the value of land ownership as an instrument of economic and political power; and
3. The rights of each family (both nuclear and extended) in a village to have access to land and water in the immediate territorial vicinity, excluding from such access use by families that do not belong to the community even though they may be of the same tribe. Where traditional systems are breaking down, inequality is often increasing further.

The low-productivity subsistence farming characteristic of most traditional African agriculture results from a combination of three historical forces restricting the growth of output:

1. In spite of the existence of some unused and potentially cultivable land, only small areas can be planted and weeded by the farm family when it uses only traditional tools such as the short-handled hoe, the ax, and the long-handled knife or panga. In some countries, use of animals is impossible because of the tsetse fly or a lack of fodder in the long dry seasons, and traditional farming practices must rely primarily on the application of human labor to small parcels of land.

2. Given the limited amount of land that a farm family can cultivate in the context of a traditional technology, these small areas tend to be intensively cultivated. As a result, they are subject to rapidly diminishing returns to increased labor inputs. In such conditions, shifting cultivation is the most economic method of using limited supplies of labor on extensive tracts of land. Under shifting cultivation, once the minerals are drawn out of the soil as a result of numerous crop pings, new land is cleared, and the process of planting and weeding is repeated. In the meantime, formerly cropped land is allowed to recover fertility until it can be used again. Under such a process, manure and chemical fertilizers have been unnecessary, although in most African villages; some form of manure (mostly animal waste) is applied to nearby plots that are intensively cultivated in order to extend their period of fertility.

3. Labor is scarce during the busiest part of the growing season, planting and weeding times. At other times, much of the labor is underemployed. Because the time of planting is determined by the onset of the rains and because much of Africa experiences only one extended rainy season, the demand for workers during the early weeks of this rainy season usually exceeds all available rural labor supplies.

As long as population size remained relatively stable, this historical pattern of low productivity and shifting cultivation enabled most African tribes to meet their subsistence food requirements. But the feasibility of shifting cultivation has now broken down as population densities increase. It has largely been replaced by sedentary cultivation on small owner-occupied plots. As a result, the need for other

nonhuman productive inputs and new technologies grows, especially in the more densely populated agricultural regions of Kenya, Nigeria, Ghana, and Uganda. Farm size has also fallen in countries such as Malawi and Tanzania. Moreover, with the growth of towns, the penetration of the monetary economy, soil erosion and deforestation of marginal lands, and the introduction of land taxes, pure subsistence-agricultural practices are no longer viable. And as land becomes increasingly scarce, land degradation is increasing in scope.

By 2007, only 4% of the cropland in sub-Saharan Africa was irrigated, in sharp contrast to 39% in South Asia and 29% in the East Asia and Pacific region. Despite some recent progress, just 22% of the cereal-growing farmland in sub-Saharan Africa is sown with improved varieties, which are used on a large majority of the land in all other developing regions. Dependence on unimproved seeds sown on unfertilized, rain-fed fields is a worsening problem for the region given both the depletion of soils and the unreliability of rainfall.

In general of all the major regions of the world, Africa has suffered the most from its inability to expand food production at a sufficient pace to keep up with its rapid population growth. As a result of declining production, African per capita food consumption fell dramatically during the 1980s and 1990s while dependence on imports particularly wheat and rice increased.

4.3 The Important Role of women in agriculture

A major and until recently often overlooked feature of agrarian systems in the developing world, particularly in Africa and Asia, is the crucial role played by women in agricultural production. In Africa, where subsistence farming is predominant and shifting cultivation remains important, nearly all tasks associated with subsistence food production are performed by women. Women are typically responsible for subsequent operations, including removing and burning felled trees, sowing or planting the plot, weeding, harvesting, and preparing the crop for storage or immediate consumption.

Women do much of the labor for cash crop production, cultivate food for household consumption, raise and market livestock, generate additional income through cottage industries, collect firewood and water, and perform household chores, including the processing and cooking of food. It has become clear that since women produce a large share of agricultural output and supply a large share of labor that has actually been increasing over time successful agricultural reform will require raising women's productivity and ensuring that gender-specific policies are at the core of rural development strategy. The necessity of starting with women's activity when agricultural policy is designed is captured by the maxim of feminist economists that "you cannot just add women and stir."

The diversity of women's duties makes it difficult to determine their share of agricultural production,

much less place an economic value on their work. However, current estimates underscore the importance of women's agricultural labor. It is estimated that in addition to work in the household, women provide 60% to 80% of agricultural labor in Africa and Asia and about 40% in Latin America. Much of this work, however, is statistically "invisible" in that women often receive no payment for the work they perform. Women make an important contribution to the agricultural economy through the labor they supply in the cultivation of cash crops.

In addition to commercial crops, women frequently cultivate small vegetable gardens that provide food for family consumption. Though the cash value of produce from these gardens may be small, it often represents an important component of the total resources available to women.

Women's work in the low-income household involves a range of demanding tasks including processing and pounding raw grains, tending livestock, cooking, and caring for children. Collecting increasingly scarce firewood and water from distant sources may add several hours to the workday. To raise additional income, it is common for women to engage in household production of goods for sale in village markets. These items are specific to each region, but a few examples are homemade beer, processed foods, handicrafts, and textile

Perhaps the most important role of women is providing food security for the household. This is accomplished through the supplementation of household earnings, diversification of household income sources, and rising of livestock to augment household assets. The production of vegetables for household consumption helps insulate households from swings in food prices and reduces cash outlays for the purchase of household necessities. Women's investments in revenue-generating projects and livestock are crucial to stabilizing household income, especially but not only in female-headed households, where resource constraints are the most severe. However, financial investments are inherently risky, and the poorer the household, the more averse its members are to taking any kind of risk.

When credit and resources are unavailable, reducing the variability of household earnings generally entails choosing less efficient methods of production and thus lowers average income. This trade-off occurs most frequently in female headed households, where resource constraints are greatest. Thus as a consequence of their restricted range of choices, women tend to retain traditional modes of economic activity. The upshot is that their productivity has stagnated while that of men has continued to improve. Where the structure of agriculture is becoming more commercialized, women's roles and hence their economic status are changing. In many developing regions, women are still unremunerated for the long hours they contribute to the tending of commercial crops. As revenue-generating cash cropping rises in importance, the proportion of resources controlled by women tends to diminish. This is largely due to the fact that household resources, such as land and

inputs, are transferred away from women's crops in order to promote the production of cash crops. Nonfarm activities are growing in importance and represent an important path for rural women's economic and social advancement.

Government extension programs that provide resources exclusively to men tend to exacerbate existing disparities between men's and women's access to resources. Consequently, it is important that the design of government extension programs reflect the interests of all household members.

When men control income from cash crops after development leads to new marketing opportunities, the perverse result can be to increase men's already high bargaining power. The differing use of funds affects not only adults but also the children. Again, the evidence is clear that in most contexts, a larger fraction of income provided and controlled by the wife tends to be used for children's health and education than that by husbands. Moreover, evidence is growing that agricultural households could earn more by reallocating inputs such as manure from husbands' to wives' plots. Thus, gender inequality also leads to significant losses in efficiency.

In addition to the obvious social justice concerns, this efficiency argument forms part of the economic case for supporting programs that empower rural women. Yet many government-sponsored programs effectively continue to exclude women, often because women lack collateral for loans or are barred from owning property or conducting financial transactions without their husbands' permission. Agricultural inputs and training are rarely provided to female applicants. Even efforts to reduce poverty through land reforms have been found to reduce female income and economic status because they distribute land titles only to male heads of household.

Cultural and social barriers to women's integration into agricultural programs remain strong because in many countries, women's income is perceived as a threat to men's authority. While men are taught new agricultural techniques to increase their productivity, women, if involved at all, are trained to perform low-productivity tasks that are considered compatible with their traditional roles, such as sewing, cooking, and basic hygiene. Women's components of development projects are frequently little more than welfare programs that fail to improve economic well-being. Furthermore, these projects tend to depend on the unpaid work of women, while men are remunerated for their efforts.

Although efforts to increase the income of women by providing direct access to credit and inputs have experienced considerable success, programs that work indirectly with women have frequently fallen short of their stated goals. Studies have found that projects are most likely to elicit the engagement of women when resources are placed directly under their control. Clearly, projects that depend on the unremunerated labor of women are likely to obtain only minimal support. Adoption of new crops and technologies will be more effective where patterns of production are consistent with

the interests of female household members. Because the active participation of women is critical to agricultural prosperity, policy design should ensure that women benefit equally from development efforts.

4.4 The Economics of agricultural development and farmer's behavior

The Transition from Peasant Subsistence to Specialized Commercial Farming

For expository convenience, we can identify three broad stages in the evolution of agricultural production. The first is the pure, low-productivity, mostly subsistence-level peasant farm, still prevalent in Africa. The second stage is what might be called diversified or mixed family agriculture, where a small part of the produce is grown for consumption and a significant part for sale to the commercial sector, as in much of Asia. The third stage represents the modern farm, exclusively engaged in high-productivity specialized agriculture geared to the commercial market, as in developed countries and often found in the highly urbanized developing countries.

Agricultural modernization in mixed-market developing economies may be described in terms of the gradual but sustained transition from subsistence to diversified and specialized production. Transforming traditional agriculture often requires, in addition to adapting the farm structure to meet the demand for increased production, profound changes affecting the entire social, political, and institutional structure of rural societies. We first consider the evolution of the agricultural system of a developing nation over time from a predominantly subsistence-level and small-scale peasant orientation to more diversified operations and eventually to the rise of fully commercial enterprises, though still often family-based.

Subsistence Farming: Risk Aversion, Uncertainty, and Survival

On the classic peasant subsistence farm, most output is produced for family consumption (although some may be sold or traded in local markets), and a few staple foods (A main food consumed by a large portion of a country's population usually including cassava, wheat, barley, sorghum, rice, potatoes, or corn) are the chief sources of nutrition. Output and productivity are low, and only the simplest traditional methods and tools are used. Capital investment is minimal; land and labor are the principal factors of production. The law of diminishing returns is in operation as more labor is applied to shrinking parcels of land.

The failure of the rains, the appropriation of the land, and the appearance of the moneylender to collect outstanding debts are the banes of the peasant's existence. Labor is underemployed for most of the year, although workers may be fully occupied at seasonal peak periods such as planting and harvest. The peasant usually cultivates only as much land as his family can manage without the need

for hired labor, although many peasant farmers intermittently employ one or two landless laborers. Much of the cash income that is generated comes from nonfarm wage labor.

In much of sub-Saharan Africa, agriculture is still largely in this subsistence stage, as it is in pockets in Asia and even Latin America. The green revolution has bypassed much of Africa. But in spite of the relative backwardness of production technologies and the misguided convictions of some foreigners who attribute the peasants' resistance to change as a sign of incompetence or irrationality, the fact remains that given the nature of the peasants' environment, the uncertainties that surround them, the need to meet minimum survival levels of output, and the rigid social institutions into which many peasants, but particularly women, are locked, most farmers do behave in an economically rational manner when confronted with alternative opportunities. Some insight into the economics of subsistence agriculture is provided by the traditional two-factor neoclassical theory of production in which land (and perhaps capital) is fixed, labor is the only variable input, and profit is maximized. Specifically, the theory provides an economic rationale for the observed low productivity of traditional agriculture in the form of the law of diminishing marginal productivity.

Unfortunately, this theory does not satisfactorily explain why small-scale farmers are often resistant to technological innovation in farming techniques or to the introduction of new seeds or different cash crops. According to the standard theory, a rational income or profit-maximizing farm or firm will always choose a method of production that will increase output for a given cost (in this case, the available labor time) or lower costs for a given output level. But the theory is based on the crucial assumption that farmers possess "perfect knowledge" of all technological input-output relationships as well as current information about prevailing factor and product prices. This is the point at which the simple theory loses a good deal of its validity when applied to the environment of subsistence agriculture. Furthermore, when access to information is highly imperfect, the transaction costs of obtaining this information are usually very high. Given price uncertainty, peasant farmers often face a wide range of possible prices rather than a single input price.

Along with limited access to credit and insurance, such an environment is not conducive to the type of behavior posited by neoclassical theory and goes a long way toward explaining the actual risk-averse behavior of peasant farmers, including their caution in the use of purchased inputs such as fertilizer. Subsistence agriculture is thus a highly risky and uncertain venture. It is made even more so by the fact that human lives are at stake.

In regions where farms are extremely small and cultivation is dependent on the uncertainties of variable rainfall, average output will be low, and in poor years, the peasant family will be exposed to the very real danger of starvation. In such circumstances, the main motivating force in the peasant's

life may be the maximization not of income but of the family's chances of survival. Accordingly, when risk and uncertainty are high, small farmers may be very reluctant to shift from a traditional technology and crop pattern that over the years they have come to know and understand to a new one that promises higher yields but may entail greater risks of crop failure. When sheer survival is at stake, it is more important to avoid a bad year (total crop failure) than to maximize the output in better years.

Risk-avoiding peasant farmers are likely to prefer a technology of food production that combines a low mean per-hectare yield with low variance (fluctuations around the average) to alternative technologies and crops that may promise a higher mean yield but also present the risk of a greater variance. Many programs to raise agricultural productivity among small farmers in Africa and elsewhere have suffered because of failure to provide adequate insurance (both financial credit and physical "buffer" stocks) against the risks of crop shortfalls, whether these risks are real or imagined.

Moreover, in parts of Asia and Latin America where agriculture has performed poorly, a closer examination of why peasant farmers have apparently not responded to an "obvious" economic opportunity will often reveal that (1) the landlord secured much if not all of the gain, (2) the moneylender captured the profits, (3) the government's "guaranteed" price was never paid, or (4) complementary inputs (fertilizers, pesticides, assured supplies of water, adequate nonusurious credit, etc.) were never made available or their use was otherwise more problematic than outsiders understood.

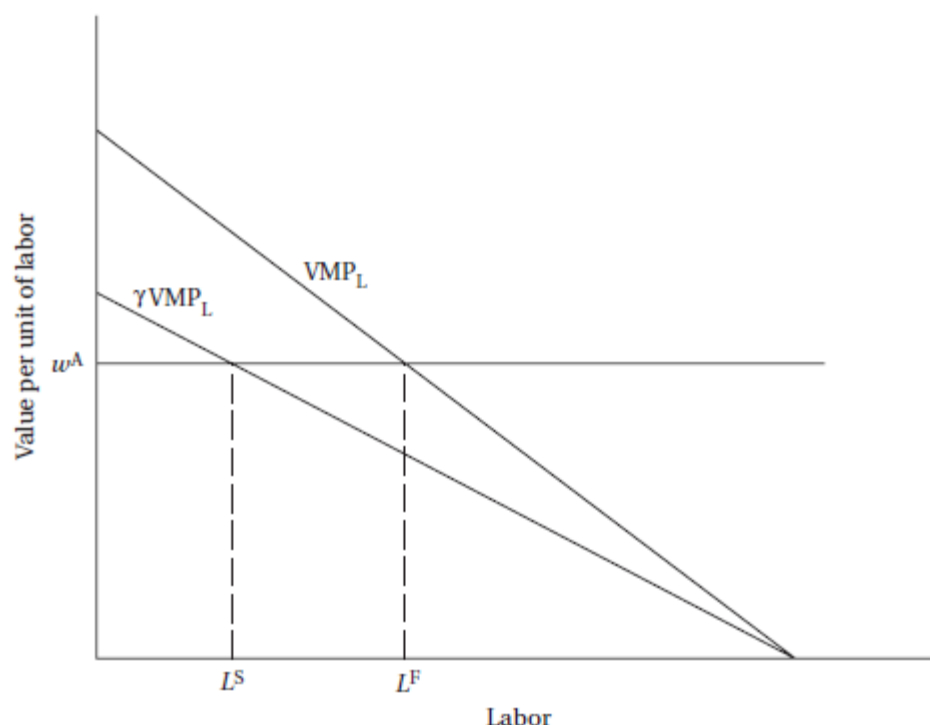
Farmers will consider the expected value of the marginal product of any inputs they apply such as fertilizer, which will be lowered in relation to the probability they place on expropriation. For example, if fertilizer lasts for two growing seasons but the peasant is sure her land will be expropriated as soon as someone with the power to do so sees that the land has already been fertilized, then too little fertilizer will be used from the social point of view, because the peasant will consider the benefits of the fertilizer as if it disappeared after just one season (while its price is not lowered). This type of effect has been confirmed by careful econometric evidence from China.

The Economics of Sharecropping and Interlocking Factor Markets

The phenomenon of risk aversion among peasant farmers in the presence of high land inequality also helps explain the prevalence of sharecropping throughout much of Asia and parts of Latin America. Although different types of relationships may arise between the owners of land and the people who work on them (e.g., the farmers could rent or act as wage laborers), sharecropping is widespread. Sharecropping occurs when a peasant farmer uses the landowner's farmland in exchange for a share of food output, such as half of the rice or wheat grown. The landlord's share may vary from less than

a third to more than two-thirds of output, depending on local labor availability and the other inputs (such as credit, seeds, and tools) that the landlord provides. The poor incentive structure of sharecropping lends itself to inefficiency.

The poor incentive structure of sharecropping lends itself to inefficiency. Alfred Marshall observed that the farmer was in effect paid only part, rather than all, of his marginal product and would rationally reduce work effort accordingly. This effect can be seen graphically in Figure 4.1. Labor input is found along the x-axis, which may be interpreted as number of hours of work, or of total effort; output per unit of labor is found along the y-axis.



A farmer who owned his own farm would work until his value marginal product of labor (VMPL) was equal to his alternative wage, or opportunity cost of labor, w^A , and so would put in an efficient amount of labor effort L^F . However, a sharecropper receives only a fraction, γ , of his effort; for example, under 50–50 sharecropping, the sharecropper's share would be $\gamma = 0.5$. Thus the sharecropper would receive only γ of his value marginal product, or γVMPL . As a result, the sharecropper would have an incentive to put in an inefficiently low level of effort L^S , as seen in Figure 4.1.

This view was challenged in the 1960s by Steven Cheung, who argued that profit-maximizing landlords would establish contracts requiring adequate work effort from the tenant as well as

stipulating each party's share of the output. If, as Cheung argued, effort is not too difficult to monitor, then if one tenant failed to live up to his part of the bargain, he would be replaced by another tenant who was willing to work harder; as a result, sharecropping would be as efficient as any other contractual form. Cheung's theory is known as the monitoring approach, in contrast to the Marshallian approach to the analysis of sharecropping illustrated in Figure 4.1; Cheung argued that labor effort LF would also obtain under sharecropping.

The monitoring approach was popular for two decades, and it was difficult to test because of endogeneity. For example, only low-productivity people may choose to enter into sharecropping contracts. In fact, some scholars believe that landlords may offer tenants an option of either sharecropping or pure rental contracts precisely because higher-ability people more often choose pure rental arrangements: High-ability farmers are able to get the full value of their high marginal product, while this is not as attractive to lower-ability farmers. If landlords are not sure which farmers have high ability, they may find out by observing which ones choose the pure rental contract. The motivation may be to enable landlords to squeeze more profits out of the renters, charging higher effective rents for pure rental contracts than for sharecropping contracts—but not too high or even high-ability farmers would choose sharecropping. This approach is known as the screening hypothesis of sharecropping.

However, Radwan Ali Shaban identified farmers who farmed plots that they owned and who also leased out additional farmland under a sharecropping contract. By comparing the same farmers' behavior under different contractual arrangements, Ali Shaban controlled for factors specific to individual farmers that cannot be easily observed. He found that farmers used fewer inputs and produced less output on the sharecropped land than on their own land, all else being equal. These results provide evidence that sharecropping is less efficient than farming one's own land, just as Marshall predicted.

A final approach suggests that sharecropping is relatively efficient after all, in that it makes the best out of an inherently uncertain and risky situation for both parties. If the landlord pays the tenant a straight wage, which would be efficient if the tenant always gave his full effort and it didn't cost the landlord anything to make sure of this, the tenant has every incentive to accept the money and not work hard. If the tenant pays a straight rent for the land, he faces the appalling risk that there will be a particularly lean year, such as a drought, and there will not be enough food left after the rent is paid to prevent starvation. Thus sharecropping represents a compromise between the risk to the landlord that the tenant will not do much work and the risk to the tenant that a fixed rent will in some years leave him no income.

So even though sharecropping, with its poor work incentives, would be inefficient in a world of perfect certainty, in the real world, with inequality in land ownership as well as uncertainty, it is “as efficient as we can get.” However, this arrangement is necessary only because of extreme inequality of land ownership. Farmers who own their own farms do not generally choose sharecropping contracts for themselves. As a result, the enormous efficiency loss, is not negated by this important explanation of why sharecropping arises.

Where tenancy reform is well designed and enforced, giving sharecroppers a larger share of the produce and security of tenure on the land, the result can be not only higher income for the tenants but also greater overall efficiency. A clear example is the tenancy reform policy implemented in the Indian state of West Bengal in the late 1970s. The explanation is clear from what we have just established: that a higher product share gives greater work effort incentives, and greater security of tenure gives greater investment incentives. Land reform that distributes ownership of “land to the tiller” can provide similar and superior improvements in incentives, if needed complementary inputs are provided. More broadly, the economic and social framework in which sharecropping takes place is one of extraordinary social inequality and far-reaching market failure. When the peasant faces his landlord, he often faces not only the individual whom he must persuade to rent him productive land but at the same time his prospective employer, his loan officer, and even his ultimate customer for any crops he wishes to sell.

Such conditions, an example of interlocking factor markets, provide the rural landlord with abundant sources of monopoly and monopsony power. Under some conditions—in particular, the availability of a perfectly elastic supply of tenants and the ability of the landlord to subdivide his land into as many plots as he chooses—the peasant is forced to his reservation utility level, or next-best income opportunity. (In practice, on one hand, peasants are sometimes prevented from learning about some of the alternatives available to them; on the other hand, subdivision may be restricted.) Interlocked-factor-market sharecropping does have the resource allocation advantage that it is in the landlord’s interest to see to it that his sharecropper receives credit from the lowest-cost source. At the same time, the personal nature of inters linkage gives the dominant party far-ranging leverage and acts as a barrier to entry that restricts competition that might ultimately benefit the peasant.

The Transition to Mixed or Diversified Farming

It is neither realistic nor necessarily desirable to think of instantly transforming a traditional agrarian system that has prevailed for many generations into a highly specialized commercial farming system. Attempts to introduce cash crops indiscriminately in subsistence farms have often resulted in the peasants’ loss of land to moneylenders or landlords. Subsistence living is merely substituted for

subsistence production. For small farmers, exclusive reliance on cash crops can be even more precarious than pure subsistence agriculture because the risks of price fluctuations are added to the uncertainty of nature.

Diversified or mixed farming therefore represents a logical intermediate step in the transition from subsistence to specialized production. In this stage, the staple crop no longer dominates farm output, and new cash crops such as fruits, vegetables, coffee, tea, and pyrethrum are established, together with simple animal husbandry. These new activities can take up slack in farm workloads during times of the year when disguised unemployment is prevalent. For example, if the staple crop occupies the land only during parts of the year, new crops can be introduced in the slack season to take advantage of both idle land and family labor.

And where labor is in short supply during peak planting seasons, simple laborsaving devices (such as small tractors, mechanical seeders, or animal-operated steel plows) can be introduced to free labor for other farm activities. Finally, the use of better seeds, fertilizers, and simple irrigation to increase yields of staple crops such as wheat, maize, and rice can free part of the land for cash crop cultivation while ensuring an adequate supply of the staple food. The farm operator can thus have a marketable surplus, which she can sell to raise her family's consumption standards or invest in farm improvements. Diversified farming can also minimize the impact of staple crop failure and provide a security of income previously unavailable.

The success or failure of such efforts to transform traditional agriculture will depend not only on the farmer's ability and skill in raising his productivity but also, even more important, on the social, commercial, and institutional conditions under which he must function. Specifically, if he can have reasonable and reliable access to credit, fertilizer, water, crop information, and marketing facilities; if he receives a fair market price for his output; and if he can feel secure that he and his family will be the primary beneficiaries of any improvements, there is no reason to assume that the traditional farmer will not respond to economic incentives and new opportunities to improve his standard of living.

Evidence from such diverse countries as Colombia, Mexico, Nigeria, Ghana, Kenya, India, Pakistan, Thailand, and the Philippines shows that under the proper conditions, small farmers are responsive to price incentives and economic opportunities and will make radical changes in what they produce and how they produce it. Lack of innovation in agriculture, as noted earlier, is usually due not to poor motivation or fear of change but to inadequate or unprofitable opportunities. In Africa, lack of information is often a constraint, but farmers learn from each other when valuable new crops and techniques are introduced locally.

From Divergence to Specialization: Modern Commercial Farming

The specialized farm represents the final and most advanced stage of individual holding in a mixed market economy. It is the most prevalent type of farming in advanced industrial nations. It has evolved in response to and parallel with development in other areas of the national economy. General rises in living standards, biological and technical progress, and the expansion of national and international markets have provided the main impetus for its emergence and growth. In specialized farming (The final and most advanced stage of the evolution of agricultural production in which farm output is produced wholly for the market), the provision of food for the family with some marketable surplus is no longer the basic goal. Instead, pure commercial profit becomes the criterion of success, and maximum per-hectare yields derived from synthetic (irrigation, fertilizer, pesticides, hybrid seeds, etc.) and natural resources become the object of farm activity. Production, in short, is entirely for the market. Economic concepts such as fixed and variable costs, saving, investment and rates of return, optimal factor combinations, maximum production possibilities, market prices, and price supports take on quantitative and qualitative significance.

The emphasis in resource utilization is on capital formation, technological progress, and scientific research and development in stimulating higher levels of output and productivity. Specialized farms vary in both size and function. They range from intensively cultivated fruit and vegetable farms to the vast wheat and corn fields of North America. In most cases, sophisticated laborsaving mechanical equipment, ranging from huge tractors and combine harvesters to airborne spraying techniques, permits a single family to cultivate many thousands of hectares of land.

The common features of all specialized farms, therefore, are their emphasis on the cultivation of one particular crop, their use of capital-intensive and in many cases laborsaving techniques of production, and their reliance on economies of scale to reduce unit costs and maximize profits. In some ways, specialized farming is no different in concept or operation from large industrial enterprises. In fact, some of the largest specialized farming operations in both the developed and the less developed nations are owned and managed by large multinational corporate agribusiness enterprises. Large, modern farms are now found in many middle-income countries such as Brazil.

But for smallholder farmers where subsistence farming predominates, strategies for dealing with risk, and in some cases overcoming coordination failures in specialization, remain prerequisites for successful specialization. Although we can find all three types of farms subsistence, mixed, and specialized commercial coexisting in almost all developing countries at any given time, for the majority of low-income countries, particularly in Africa, contemporary agricultural systems are still dominated by small-scale mixed and even subsistence-based family farms. The further transition to a

preponderance of commercial enterprises may be difficult to achieve, depending as it does on the solution to many other short- and intermediate-term problems.

But there is wide agreement that the improvement of small- and medium-scale mixed farming practices that will not only raise farm incomes and average yields but, if labor-intensive, also effectively absorb underutilized rural labor offers the major immediate avenue toward the achievement of real people-oriented rural development.

Core Requirements of a Strategy of Agricultural and Rural Development

If the major objective of agricultural and rural development in developing nations is the progressive improvement in rural levels of living achieved primarily through increases in small-farm incomes, output, and productivity, along with genuine food security, it is important to identify the principal sources of agricultural progress and the basic conditions essential to its achievement.

Improving Small-Scale Agriculture Technology and Innovation

In most developing countries, new agricultural technologies and innovations in farm practices are preconditions for sustained improvements in levels of output and productivity. In many parts of Africa, however, increased output in earlier years was achieved without the need for new technology simply by extending cultivation into unused but potentially productive lands. Almost all of these opportunities have by now been exploited, and there is little scope for further significant or sustainable expansion. Two major sources of technological innovation can increase farm yields. Unfortunately, both have somewhat problematic implications for agricultural development. The first is the introduction of mechanized agriculture to replace human labor. The introduction of laborsaving machinery can have a dramatic effect on the volume of output per worker, especially where land is extensively cultivated and labor is scarce. For example, one man operating a huge combine harvester can accomplish in a single hour what would require hundreds of workers using traditional methods.

But in the rural areas of many developing nations, where land parcels are small, capital is scarce, and labor is abundant, the introduction of heavily mechanized techniques is often ill suited to the physical environment and has the effect of creating more rural unemployment without necessarily lowering per-unit costs of food production. Importation of such machinery can require large tracts of land (and thus the consolidation of small holdings) and tends to exacerbate the already serious problems of rural poverty and underemployment. And if mechanized techniques exclude women, the male-female productivity gap could widen further, with serious repercussions. Biological hybrid seeds and biotechnology, water control (irrigation), and chemical fertilizer, pesticides, insecticides, etc.

innovations the second major source are not without their own problems. They are land-augmenting; that is, they improve the quality of existing land by raising yields per hectare.

Only indirectly do they increase output per worker. Improved seeds, advanced techniques of irrigation and crop rotation; the increasing use of fertilizers, pesticides, and herbicides; and new developments in veterinary medicine and animal nutrition represent major scientific advances in modern agriculture. These measures are often technologically scale-neutral; theoretically, they can be applied equally effectively on large and small farms. They do not necessarily require large capital inputs or mechanized equipment. They are therefore particularly well suited for tropical and subtropical regions and offer enormous potential for raising agricultural output in developing nations and have been highly effective in doing so, particularly in Asia. Again, the major challenge is to extend this success to sub Saharan Africa, which will in some cases need new innovations. There are also important environmental challenges in many parts of the developing world, including risks posed by a falling water table, for which well-designed government policy and in some cases restored collective action mechanisms are usually necessary.

Institutional and Pricing Policies: Providing the Necessary Economic Incentives

Unfortunately, although the green revolution varieties of wheat, corn, and rice, together with needed irrigation and chemicals, are scale neutral and thus offer the potential for continued small-farm progress, the social institutions and government economic policies that accompany their introduction into the rural economy are often not scale-neutral. On the contrary, they often merely serve the needs and vested interests of the wealthy landowners. Because the new hybrid seeds require access to complementary inputs such as irrigation, fertilizer, insecticides, credit, and agricultural extension services, if these are provided only to a small minority of large landowners, one impact of the green revolution can be (as in parts of South Asia and Mexico) the further impoverishment of many peasants. Large landowners, with their disproportionate access to these complementary inputs and support services, are able to gain a competitive advantage over smallholders and eventually drive them out of the market.

Large-scale farmers obtain access to low-interest government credit, while smallholders are forced to turn to moneylenders. The result has all too often been the further widening of the gap between rich and poor and the increased consolidation of agricultural land in the hands of a very few so-called progressive farmers. A developmental innovation with great potential for alleviating rural poverty and raising agricultural output can thus turn out to be anti developmental if public policies and social institutions militate against the active participation of the small farmer in the evolving agrarian structure. Another critical area of many past and some continued failures in government policies

relates to the pricing of agricultural commodities, especially food grains and other staples produced for local markets. Many governments in developing nations, in their headlong pursuit of rapid industrial and urban development, maintained low agricultural prices in an attempt to provide cheap food for the urban modern sector.

Farmers were paid prices below either world competitive or free-market internal prices. The relative internal price ratio between food and manufactured goods (the domestic terms of trade) thus turned against farmers and in favor of urban manufacturers. With farm prices so low—in some cases below the costs of production—there was no incentive for farmers to expand output or invest in new productivity raising technology. As a result, local food supplies continually fell short of demand, and many developing nations, especially in sub-Saharan Africa, that were once self-sufficient in food production had to import food. Many development economists therefore argue that if governments are to promote further increases in agricultural production that make a larger impact on poverty reduction through green revolution technologies, they must not only make the appropriate institutional and credit market adjustments but also make continued progress to provide incentives for small and medium size farmers by implementing pricing policies that truly reflect internal market conditions.

Adapting to New Opportunities and New Constraints

As a route out of poverty and toward genuine rural development, enhanced cereal productivity (the classic green revolution characteristic) represents only a small part of the agricultural opportunities. The best opportunities for sales to growing urban areas are generally found in higher value-added activities, particularly horticulture (fruits, vegetables, and cut flowers) and aquaculture. These products, along with organic and perhaps Fair Trade versions of some otherwise traditional developing country exports such as coffee and spices, also provide good opportunities for higher-value exports. But small farmers will need special organization and assistance to take advantage of new opportunities. As the 2008 World Development Report concluded, “Smallholders can bargain better as a group than as individuals. So a high priority is to facilitate collective action through producer organizations to reach scale in marketing and bargain for better prices.” Otherwise, the risk is large that these developments will benefit mainly the larger farmers.

One of the biggest constraints looking ahead is the looming environmental problems driven by global warming and climate change, which are expected to most negatively affect sub-Saharan Africa and South Asia. Smaller and poorer farmers are likely to be affected severely, because of their lower access to irrigation and other inputs and generally lesser capacity to adapt— although, ironically, with their smaller use of irrigation and different crop mix, their absolute income declines may be less

than those of richer farmers. Although the majority of global warming problems are caused by developed countries, to the extent that cultivated areas in developing countries continue to increase by means of eliminating remaining forested areas, climate change problems will only worsen. This “agricultural extensification,” not only in forests but also in drier and other sensitive lands, further brings the risk of local soil degradation and lost environmental services such as maintaining water and air quality. The losses of wetlands and of biodiversity also lead to substantial national (as well as international) costs. Moreover, intensification of agriculture has often brought with it the misuse of agrochemicals, which can entail large human and ecosystem costs.

Unit Summary

- Agricultural and rural development has come to be regarded by many economists as the pre requisite of national development. Without such *integrated rural development*, in most cases, industrial growth either would be stultified or, if it succeeded, would create severe internal imbalances in the economy.
- The agricultural patterns of developing nations of Africa, Asia and Latin America are agriculture-based economies, agriculturally transforming economies and urbanized economies respectively.
- A major and until recently often overlooked feature of agrarian systems in the developing world, particularly in Africa and Asia, is the crucial role played by women in agricultural production. In Africa, where subsistence farming is predominant and shifting cultivation remains important, nearly all tasks associated with subsistence food production are performed by women.
- The principal sources of agricultural progress and the basic conditions essential to its achievements are improving small-scale agriculture technology and innovation, institutional and pricing policies (providing the necessary economic incentives) and adapting to new opportunities and new constraints.

CHAPTER FIVE
International Trade and Economic Development
The Trade Policy Debate and Industrialization

5.1 Role of Foreign Trade in Development

Chapter objectives

After completing this chapter you will be able to:

- Understand the importance of export to different developing nation
- Understand the trade policy debate
- Understand the problem of balance of payment deficit and macroeconomic stabilization

The whole economic basis for international trade rests on the fact that countries do differ in their resource endowments, their preferences and technologies, their scale economies, their economic and social institutions, and their capacities for growth and development. Developing countries are no exception to this rule. Some are rapidly ascending through the income rankings as they expand their industrial capacities. Some are very populous yet deficient in both natural resources and human skills, at least in large regions of the country. Others are sparsely populated yet endowed with abundant mineral and raw material resources. Yet others are small and economically weak, still having at present neither adequate human capital nor the material resources on which to base a sustained and largely self-sufficient strategy of economic and social development.

We begin with the basic neoclassical theory of international trade and its effect on efficiency, equity, stability and growth. We then provide a critique of the relevance of pure free-trade theories for developing countries. Like free markets, free trade has many desirable theoretical features, of which is the promotion of static economic efficiency and optimal resource allocation. But also like free markets and perfect competition, free trade exists more in theory than in practice and today's developing nations have to function in the imperfect and often highly unequal real world of international commerce. Consequently, we will briefly discuss alternative trade models that focus on imperfect competition, unequal trade, and the dynamic effects of differential human resource and technological growth.

Next we examine the balance of payments, review some issues in international finance, engage in an in-depth analysis of debt crises, and explore the range of commercial policies (tariffs, subsidies, quotas, exchange-rate adjustments, etc.) that a developing country might wish to adopt

within the broader context of the ongoing debate about the relative merits of export promotion versus import substitution. We then examine a wide range of commercial policies used in developing countries, including import tariffs, physical quotas, and export promotion versus import substitution.

Importance of Exports to Different Developing Nations

Although the overall figures for export volumes and values of developing countries are important indicators of patterns of trade for the group as a whole, while large countries are understandably less dependent on trade than small countries, at any given size, developing countries tend to devote a larger share of their output as merchandise exports than developed countries do. Some large countries like Brazil and India, which have had unusually closed economies, tend to be less dependent on foreign trade in terms of national income than most relatively small countries.

Less developed nations are more dependent on foreign trade in terms of its share in national income than the very highly developed countries are. This is reflected in the case of traditionally export-oriented Japan, whose exports amount to roughly 16% of GDP, whereas developing countries with somewhat larger populations, such as Indonesia and Nigeria, export a higher share of output.

The greater recorded share of developing-country exports in GDP is probably due in part to the much higher relative prices of non-traded services in developed than in developing countries. Nevertheless, the point remains that developing countries are generally more dependent on trade in international economic relations because most trade is in merchandise, for which price disparities are smaller across countries. Moreover, in general, the exports of developing countries are much less diversified than those of the developed countries. While total exports and the share major new exporters such as China have emerged, it is important to keep this rise in perspective.

Demand Elasticities and Export Earnings Instability

Most statistical studies of world demand patterns for different commodity groups reveal that in the case of primary products, the income elasticity of demand is relatively low: The percentage increase in quantity of primary agricultural products and most raw materials demanded by importers (mostly rich nations) will rise by less than the percentage increase in their GNIs. By contrast, for fuels, certain raw materials, and manufactured goods, income elasticity is relatively high. For example, it has been estimated that a 1% increase in developed-country incomes will normally raise their imports of foodstuffs by a mere 0.6% and of agricultural raw materials such as rubber and vegetable oils by 0.5% but raise imports of manufactures by about 1.9%. Consequently, when incomes rise in rich countries, their demand for food, food products, and raw

materials from the developing nations goes up relatively slowly, whereas demand for manufactures goes up relatively rapidly.

The net result of these low income elasticities of demand is the tendency for the relative price of primary products to decline over time. Moreover, since the price elasticity of demand for (and supply of) primary commodities also tends to be quite low (i.e., inelastic), any shifts in demand or supply curves can cause large and volatile price fluctuations. Together these two elasticity phenomena contribute to what has come to be known as export earnings instability, which has been shown to lead to lower and less predictable rates of economic growth. While almost all attention goes to merchandise exports, there has been a slow rise in the share of commercial services in the exports of both developed and developing countries. For the former, these are more likely to represent highly skilled activities such as investment banking and management consulting, while for the latter, construction and other less skill-intensive activities are more common.

5.2 The Terms of Trade and the Prebisch-Singer Hypothesis

The total value of export earnings depends not only on the volume of these exports sold abroad but also on the price paid for them. If export prices decline, a greater volume of exports will have to be sold merely to keep total earnings constant. Similarly, on the import side, the total foreign exchange expended depends on both the quantity and the price of imports. Clearly, if the price of a country's exports is falling relative to the prices of the products it imports, it will have to sell that much more of its exports and enlist more of its scarce productive resources merely to secure the same level of imported goods that it purchased in previous years. In other words, the real or social opportunity costs of a unit of imports will rise for a country when its export prices decline relative to its import prices.

The ratio between the price of a typical unit of exports and the price of a typical unit of imports is

called the commodity terms of trade, and it is expressed as $\frac{p_x}{p_m}$, where P_x and P_m represent the export and import price indexes, respectively, calculated on the same base period (e.g., 2012 = 100). The

commodity terms of trade are said to deteriorate for a country if $\frac{p_x}{p_m}$ falls, that is, if export prices decline relative to import prices, even though both may rise. Historically, the prices of primary commodities have declined relative to manufactured goods. As a result, the terms of trade have on the average tended to worsen over time for the nonoil-exporting developing countries while showing a relative improvement for the developed countries.

The main theory for the declining commodity terms of trade is known as the Prebisch-Singer hypothesis, after two famous development economists who explored its implications in the 1950s. They argued that there was and would continue to be a secular (long-term) decline in the terms of trade of primary-commodity exporters due to a combination of low income and price elasticities of demand. This decline would result in an ongoing transfer of income from poor to rich countries that could be combated only by efforts to protect domestic manufacturing industries through a process that has come to be known as import substitution.

Both because of this theory and because of the unfavorable terms-of-trade trends, developing countries have been doing their utmost over the past several decades to diversify into manufactures exports. After a slow and costly start, these efforts have resulted in a dramatic shift in the composition of developing-country exports, especially among middle-income countries. Led at first by the East Asian Tiger economies of South Korea, Taiwan, Hong Kong, and Singapore and now followed by many other countries, including China, the share of merchandise exports accounted for by manufactured goods has risen strongly in many developing countries.

Unfortunately, this structural change has not brought as many benefits to most developing countries as they had hoped, because relative prices within manufactures have also diverged: Over the past few decades, the prices of the basic manufactured goods exported by poor countries fell relative to the advanced products exported by rich countries. The price of textiles fell especially precipitously, and low-skilled electronic goods are not far behind. Using alternative methods, the United Nations found that the real decline in developing-country export prices of manufactures in the 1980s was about 3.5% per year, or about 30% for the decade. In a detailed study, Alf Maizels discovered that the terms of trade in manufacturing goods for developing countries vis-à-vis the United States deteriorated over the 1981–1997 period. The declines in textile prices accelerated dramatically starting in the late 1990s. Having reviewed some of the international trade issues that developing countries face, we turn next to consider alternative theories of the role that trade plays in economic development.

5.3 The trade policy debate: Export promotion industrialization versus Import substitution Industrialization

A traditional way to approach the complex issues of appropriate trade policies for development is to set these specific policies in the context of a broader strategy of looking outward or looking inward. In the words of Paul Streeten, outward-looking development policies “encourage not only free trade but also the free movement of capital, workers, enterprises and students . . . , the multinational enterprise, and an open system of communications.”

By contrast, inward-looking development policies stress the need for nations to evolve their own styles of development and to control their own destiny. This means policies to encourage indigenous “learning by doing” in manufacturing and the development of technologies appropriate to a country’s resource endowments. According to proponents of inward-looking trade policies, greater self reliance can be accomplished, in Streeten’s words, only if “you restrict trade, the movement of people, and communications and if you keep out the multinational enterprise, with its wrong products and wrong want-stimulation and hence its wrong technology.”

Basically, the distinction between these two traditional trade-related development strategies is that advocates of import substitution (IS) believe that a developing economy should initially substitute domestic production of previously imported simple consumer goods (first-stage IS) and then substitute through domestic production for a wider range of more sophisticated manufactured items (second-stage IS) all behind the protection of high tariffs and quotas on these imports. In the long run, IS advocates cite the benefits of greater domestic industrial diversification (“balanced growth”) and the ultimate ability to export some previously protected manufactured goods as economies of scale, low labor costs, and the positive externalities of learning by doing cause domestic prices to become more competitive with world prices.

By contrast, advocates of export promotion (EP) of both primary and manufactured goods cite the efficiency and growth benefits of free trade and competition, the importance of substituting large world markets for narrow domestic markets, the distorting price and cost effects of protection, and the tremendous successes of such export-oriented economies as South Korea, Taiwan, Singapore, Hong Kong, China, and others in Asia. They stress that firms in these economies have learned a great deal from the firms in the United States, Japan, and other developed-country economies that have been their long-term customers. Sometimes a distinction is made between “strong export promotion,” in which policies are explicitly geared to expansion of exports (in general, such as through a weak currency), rather than production for the domestic market, and “weak export promotion,” which emphasizes free trade and a level playing field and is viewed by advocates as likely to promote exports by comparison with previous import substitution policies (which tend to discourage exports in relative terms). Beyond this, many Asian countries also adopted a more nuanced approach that draws on some elements of both to develop targeted sectors.

In practice, the distinction between IS and EP strategies is much less pronounced than many advocates would imply. Most developing economies have employed both strategies with different degrees of emphasis at one time or another. For example, in the 1950s and 1960s, the inward-looking industrialization strategies of the larger Latin American and Asian countries such as Chile, Peru, Argentina, India, Pakistan, and the Philippines were heavily IS oriented. By the end of the 1960s,

some of the key sub-Saharan African countries like Nigeria, Ethiopia, Ghana, and Zambia had begun to pursue IS strategies, and some smaller Latin American and Asian countries also joined in.

However, since the mid-1970s, the EP strategy has been increasingly adopted by a growing number of countries. The early EP adherents—South Korea, Taiwan, Singapore, and Hong Kong—were thus joined by the likes of Brazil, Chile, Thailand, and Turkey, which switched from an earlier IS strategy. It must be stressed, however, that most successful East Asian export promoters have pursued protectionist IS strategies sequentially and simultaneously in certain industries, so it is inaccurate to call them free traders, even though they are outward-oriented.

5.3.1 Export Promotion: Looking Outward and Seeing Trade Barriers

The promotion of primary or secondary exports has long been considered a major ingredient in any viable long-run development strategy. The colonial territories of Africa and Asia, with their foreign-owned mines and plantations, were classic examples of primary outward-looking regions. It was partly in reaction to this enclave economic structure and partly as a consequence of the industrialization bias of the 1950s and 1960s that most developing countries put great emphasis on the production of manufactured goods initially for the home market (secondary inward) and then for export (secondary outward).

Primary-Commodity Export Expansion: Limited Demand

As noted earlier in this chapter, many low-income countries still rely on primary products for a majority of their export earnings. With the notable exception of petroleum exports and a few needed minerals, primary-product exports have grown more slowly than total world trade. On the demand side, there appear to be at least five factors working against the rapid expansion of primary-product and especially agricultural exports.

First, the income elasticities of demand for agricultural foodstuffs and raw materials are relatively low compared with those for fuels, certain minerals, and manufactures. For example, the income elasticities of demand for sugar, cacao, tea, coffee, and bananas have all been estimated at less than 1, with most in the range of 0.3–0.6. Inelastic demand means that only a sustained high rate of per capita income growth in the developed countries can lead to even modest export expansion of these particular commodities from the developing countries. (Many primary exporters have benefited from the boom in China since about 2002—excepting the 2008–2009 debacle—and this will be followed carefully.)

Second, developed-country population growth rates are now at or near the replacement level, so little expansion can be expected from this source. Third, the price elasticity of demand for most primary commodities is relatively low. When relative agricultural prices are falling, as they have been during most of the past five decades, such low elasticities mean less total revenue for exporting nations. With the exception of oil and a few minor commodities, international commodity agreements have not fared well. Such agreements are intended to set overall output levels, stabilize world prices, and assign quota shares to various producing nations for such items as coffee, tea, copper, lead, and sugar.

To work effectively, they require cooperation and compromise among participants. Commodity agreements can also provide greater protection to individual exporting nations against excessive competition and the overexpansion of world production. Such overexpansion of supply tends to drive down prices and curtail the growth of earnings for all countries. In short, commodity agreements attempt to guarantee participating nations a relatively fixed share of world export earnings and a more stable world price for their commodity. But proposals by the United Nations Conference on Trade and Development (UNCTAD) for the establishment of a common fund to finance “buffer stocks” to support the prices of some 19 primary commodities (including sugar, coffee, tea, bauxite, jute, cotton, tin, and vegetable oil) produced by various developing nations have made little progress. Most existing non-oil commodity agreements have either failed (tin) or been largely ignored by producers (coffee, sugar). Even in the best scenarios, such agreements cannot be effective for perishable commodities. Imagine trying to operate a buffer stock of bananas!

The fourth and fifth factors working against the long-run expansion of primary-product export earnings the development of synthetic substitutes and the growth of agricultural protection in the developed countries are perhaps the most important. Synthetic substitutes for commodities like cotton, rubber, sisal, jute, hide, skins, and copper (replaced by glass fiber optics for communication networks) act both as a brake against higher commodity prices and as a direct source of competition in world export markets. The synthetic share of world market export earnings has generally risen over time, while the share of natural products has fallen. In the case of agricultural protection, which usually takes the form of tariffs, quotas, and nontariff barriers such as sometimes arbitrary sanitary laws regulating food and fiber imports, the effects can be devastating to developing countries’ export earnings. The common agricultural policy of the European Union, for example, has resulted in greater subsidies that have harmed the competitiveness of developing countries.

On the supply side, a number of factors also work against the rapid expansion of primary-product export earnings. The most important is the structural rigidity of many rural production systems in developing countries. We discussed rigidities—such as limited resources; poor climate; bad soils;

antiquated rural institutional, social, and economic structures; and nonproductive patterns of land tenure. Whatever the international demand situation for particular commodities (which will differ from commodity to commodity), little export expansion can be expected when rural economic and social structures militate against positive supply responses from peasant farmers who are averse to risk. Furthermore, in developing nations with markedly dualistic farming structures (i.e., large corporate capital-intensive farms existing side by side with thousands of fragmented, low-productivity peasant holdings), any growth in export earnings is likely to be distributed very unevenly among the rural population. Small farmers have been further disadvantaged in countries (mostly in Africa) in which agricultural marketing boards act as middlemen between the farmers and export markets. Fortunately, these boards have been largely dismantled in recent years.

Primary export growth has remained modest, partly due to the pernicious effects of developed-country trade policies (such as the United States' sugar and cotton subsidies) and foreign-aid policies that depress agricultural prices in the least developed countries and discourage production. For example, the EU's policy of selling subsidized beef to the nations of West Africa in the guise of foreign assistance devastated cattle prices in those countries. As summarized by Kevin Watkins and Joachim von Braun of the International Food Policy Research Institute: Small farmers in developing countries suffer on several counts from rich-country farm policies. Northern production subsidies lower prices for farm produce. Unable to compete against subsidized competition, the world's poorest farmers are often pushed out of international and even domestic markets. The upshot is an agricultural trading system in which success depends less on comparative advantage than on comparative access to subsidies. Small farmers are efficient, innovative, and potentially competitive, and creatively combine farming with off-farm work. But the world's poorest farmers cannot compete against the world's richest treasuries, nor should they have to.

We may conclude, therefore, that the successful promotion of primary product exports in low-income countries and for the benefit of the poor cannot occur unless there is a reorganization of rural social and economic structures to raise total agricultural productivity and distribute the benefits more widely. The primary objective of any rural development strategy is widely accepted to be first to provide sufficient food to feed local people and only then to be concerned about export expansion. Given the structure of world demands for primary products, the threat of local food shortages and thus the desire of potential importers to focus on agricultural self-sufficiency, the inevitability of the development of further synthetic substitutes, and the (tragic) unlikelihood of significantly lower levels of agricultural protection among developed nations in light of the stalled trade talks, the real scope for primary-product export expansion in individual developing nations seems limited.

Expanding Exports of Manufactured Goods

The expansion of manufactured exports has been encouraged by the spectacular export performances of countries like South Korea, Singapore, Hong Kong, Taiwan, and China. For example, for decades, Taiwan's total exports grew at an annual rate of over 20%, and exports from South Korea grew even faster. In both cases, this export growth was led by manufactured goods, which contributed over 80% of both nations' foreign exchange earnings. For the developing world as a whole, manufactured exports grew from 6% of their total merchandise exports in 1950 to almost 64% by 2000. Taken together, by 2002, the low- and middle-income countries accounted for 25.7% of the world's manufactured exports; China commanded a fast-growing share. However, the low-income countries accounted for just over 1% of the world total. The export successes of recent decades, especially among the Asian Tigers, have provided impetus for arguments by market fundamentalists that economic growth is best served by allowing market forces, free enterprise, and open economies to prevail while minimizing government intervention. However, evidence from East Asia does not support this view of how export success was achieved. In South Korea, Taiwan, and Singapore (as in Japan earlier and to a large degree China more recently), the production and composition of exports was not left to the market but resulted from planned intervention by the government while making ample use of the profit incentive.

We return to this consideration later in the chapter. The demand problems for export expansion of many manufactured goods, though different in basic economic content from those for primary products, can still pose similar problems for developing countries. For many years, there was widespread protection in developed nations against the manufactured exports of developing countries, which was in part the direct result of the successful penetration of low-cost labor-intensive manufactures from countries like Taiwan, Hong Kong, and South Korea during the 1960s and 1970s. And as noted earlier, relative prices of the most basic manufactured goods have also fallen.

Industrial-nation trade barriers have been extensive. During the 1980s, for example, 20 of the 24 industrialized countries increased their protection against developing countries' manufactured or processed products. Moreover, their rates of protection were considerably higher against developing country exports than against those of high-income countries. Then there are the nontariff barriers, which came to form the main protection against manufactured exports from developing countries, affecting at least one-third of them. A major example was the Multi fiber Arrangement (MFA), in effect until 2005, a complex system of mostly bilateral quotas against exports of cotton, wool, and synthetic fiber products. The United Nations Development Program estimated that the MFA cost the developing world \$24 billion a year in lost textile and clothing export earnings.

The end of the MFA has benefited China most, though some other developing countries, notably Bangladesh, have been able to hold their market share. Much-publicized initiatives for opening

markets to the least developed countries, most prominently through the African Growth and Opportunities Act in the United States and Everything but Arms in the European Union, noted earlier, are bilateral offers that can later be withdrawn. These programs also have impediments such as a time horizon that is too short to be effective at encouraging investment or requiring costly and cumbersome documentation, as stressed by Africa expert Paul Collier. Whether displaced high-wage workers in developed-country manufactures (and now services) will continue to permit the unimpeded entry of low-wage products remains to be seen. WTO rules eliminated many formal barriers, but many implicit barriers remain. The encouraging pace of tariff reductions at the time of the Uruguay Round and the early years of the WTO has in recent years slowed almost to a halt. Antidumping “investigations” increased significantly, reaching a peak in 1999, with the United States the largest user of these protectionist measures.

Although the number of new investigations subsequently declined in the early years of the new century, they remain an important weapon in the protectionist arsenal. For example, as the global recession got underway in 2007, antidumping investigations surged until the end of 2009. Countervailing duty investigations are also on the rise: “Buy American” and analogous legislation that garnered much publicity in stimulus packages following the 2008 crisis are of dubious legality but can have major impacts on developing-country investments, at least for as long as they remain in place, and can also function in the protectionist arsenal as a deterrent.

Multi fiber Arrangement (MFA) A set of nontariff quotas established by developed countries on imports of cotton, wool, synthetic textiles, and clothing from individual developing countries.

Regional trading agreements, including the North American Free Trade Agreement (NAFTA) and the EU, may also have the effect of discriminating against exports from nonmember developing countries. Analysts also questioned how long the United States could continue to act as the “consumer of last resort” in the wake of its large and chronic trade deficits and how developing countries would respond to the apparently inevitable decline in the value of the U.S. dollar; the rebounding U.S. trade deficit after the financial crisis surprised many analysts, but at some point, this export opportunity for developing nations may well be reduced. It was also widely doubted how many other developed-country markets would open to the extent seen in the United States during this period. As in the case of agricultural and other primary production, the uncertain export outlook should be no cause for curtailing the needed expansion of manufacturing production to serve local markets.

There is also great scope for mutually beneficial trade in manufactures among developing countries themselves within the context of the gradual economic integration of their national economies.

South-South trade in minerals and agriculture has been rising much more quickly than South-South manufactures trade. China's primary goods investments in and exports from Africa are the most visible, but the emergence of manufacturing zones in Africa working under contract with Chinese firms is also significant. On the other hand, antidumping and other trade complaints against China by other developing nations are rising rapidly.

5.3.2 Import Substitution: Looking Inward but Still Paying Outward

Observing weak world markets for their primary products and subscribing to the widespread belief in the magic of industrialization and the Prebisch Singer hypothesis, developing nations turned to an import substitution strategy of urban industrial development in the post–World War II decades. Some countries still follow this strategy for both economic and political reasons, although pressure from the WTO, IMF, and World Bank impose high opportunity costs on such endeavors. As noted earlier, import substitution entails an attempt to replace commodities that are being imported, usually manufactured consumer goods, with domestic sources of production and supply. The typical strategy is first to erect tariff barriers or quotas on certain imported commodities and then to try to set up a local industry to produce these goods—items such as radios, bicycles, or household appliances.

Typically, this involves joint ventures with foreign companies, which are encouraged to set up their plants behind the wall of tariff protection and given all kinds of tax and investment incentives. Although initial costs of production may be higher than former import prices, the economic rationale put forward for the establishment of import-substituting manufacturing operations is either that the industry will eventually be able to reap the benefits of large-scale production and lower costs (the so-called infant industry argument for tariff protection) or that the balance of payments will be improved as fewer consumer goods are imported. Often a combination of both arguments is advanced. Eventually, it is hoped, the infant industry will grow up and be able to compete in world markets. It will then be able to generate net foreign exchange earnings once it has lowered its average costs of production. Let us see how the theory of protection can be used to demonstrate this process.

Tariffs, Infant Industries, and the Theory of Protection

A principal mechanism of the import substitution strategy is the erection of protective tariffs (taxes on imports) or quotas (limits on the quantity of imports) behind which IS industries are permitted to operate. The basic economic rationale for such protection is the infant-industry argument. Tariff protection against the imported commodity is needed, so the argument goes, in order to allow the now higher-priced domestic producers enough time to learn the business and to achieve the economies of scale in production and the external economies of learning by doing that are necessary to lower unit costs and prices. With enough time and sufficient protection, the infant will eventually

grow up, be directly competitive with developed-country producers, and no longer need this protection.

Ultimately, as actually seen in the case of many formerly protected IS industries in South Korea and Taiwan, domestic producers hope to be able not only to produce for the domestic market without a tariff wall or government subsidies but also to export their now lower-cost manufactured goods to the rest of the world. Thus for many developing-country industries, in theory, an IS strategy becomes the prerequisite for an EP strategy. It is for this reason, among others (including the desire to reduce dependence and attain greater self-reliance, the need to build a domestic industrial base, and the ease of raising substantial tax revenue from tariff collections), that import substitution has been appealing to so many governments. The basic theory of protection is an old and controversial issue in the field of international trade. It is relatively simple to demonstrate.

In the short run, it is clear that the impact of such a prohibitive tariff is to penalize consumers, who are in effect subsidizing domestic producers and their employees through higher prices and lower consumption. Alternatively, we can say that a tariff redistributes income from consumers to producers. However, in the longer run, advocates of IS protection for infant industries argue that everyone will benefit as domestic and other shoe manufacturers reap the benefits of economies of scale and learning by doing so that ultimately the domestic price falls below P_2 (the world price). Production will then occur for both the domestic and world markets, domestic consumers as well as domestic producers and their employees will benefit, protective tariffs can be removed, and the government will be able to replace any lost tariff revenue with taxes on the now very much higher incomes of domestic manufactures. It all sounds logical and persuasive in theory. But how has it performed in practice?

5.4 Balance of Payments and Macroeconomic Stabilization

5.4.1 Balance of payments

It is the summary statement of a nation's financial transactions with the outside world. A balance of payments table is designed to summarize a nation's financial transactions with the outside world. It is divided into three components current account, capital account and international reserve account. The current account focuses on the export and import of goods and services, investment income, debt service payments, and private and public net remittances and transfers. Specifically, it subtracts the value of imports from exports and then adds flows of the net investment income received from abroad (e.g., the difference between interest and dividend payments on foreign stocks, bonds, and bank deposit owned by developing-country nationals and brought into the country, as opposed to

being left overseas, and those securities, if any, of the developing country owned by foreigners plus repatriated profits of multinational corporations).

It subtracts debt service payments, which represents a major component of heavily indebted poor countries current account deficits, and adds net private and public remittances and transfers, such as money sent home by developing country nationals working abroad current account therefore allows us to analyze the impact of various commercial policies, primarily on merchandise trade but also indirectly on investment income, debt service payments, and private transfers. The capital account (financial account) records the value of private foreign direct investment (mostly by multinational corporations), foreign loans by private international banks, and loans and grants from foreign governments (as in the form of foreign aid) and multilateral agencies such as the IMF and the World Bank. It then subtracts an extremely important item, especially for the major debtor countries: what is called “resident capital outflow”.

To put its importance in perspective, during the 1980s debt crisis, wealthy nationals from many developing countries sent vast amounts of money into developed-nation bank accounts, real estate ventures, and stock and bond purchases; this capital flight is estimated to have had a value of up to half the total debt of some debtor nations at the peak of their debt problems. It dwarfed the receipt of private and public loans and investments and was a major contributor to the worsening balance of payments of many developing nations. Capital flight is also a chronic problem where autocratic governments have a shaky hold on power.

Finally, the cash account, or international reserve account, is the balancing item (along with the “errors and omissions,” which reconciles statistical inequalities, but is sometimes used as a proxy for disguised or unrecorded capital flows) that is lowered (shows a net outflow of foreign reserves) whenever total disbursements on the current and capital accounts exceed total receipts. Nations accumulate international cash reserves in any or all of the following three forms: (1) foreign hard currency (primarily U.S. dollars, but also Japanese yen, pounds sterling, or the European euro) whenever they sell more abroad than they purchase; (2) gold, mined domestically or purchased; and (3) deposits with the IMF, which acts as a reserve bank for individual nations’ central banks.

5.4 2. The issue of payment deficit

In the least developed nations of the world, which have to import food to feed a hungry population and possess limited monetary reserves, such payments deficits may spell disaster for millions of people. Facing existing or projected balance of payments deficits on combined current and capital accounts, developing nations have a variety of policy options. For one thing, they can seek to improve the balance on current account by promoting export expansion or limiting imports (or both). In the former case, there is the further choice of concentrating on primary or secondary product

export expansion. In the latter case, policies of import substitution (the protection and stimulus of domestic industries to replace previously imported manufactured goods in the local market) or selective tariffs and physical quotas or bans on the importation of specific consumer goods may be tried. Or countries can seek to achieve both objectives simultaneously by altering their official foreign exchange rates through a currency devaluation that lowers export prices and increases import prices.

Alternatively or concurrently, they can seek loans and assistance from the World Bank or the IMF. Traditionally, this has required that the countries follow very restrictive fiscal and monetary policies. These have been called stabilization policies by the IMF; and termed structural adjustment by the World Bank which has made structural adjustment loans as part of this process.

5.4.2 Attempts at Alleviation: Macroeconomic Instability, Classic IMF Stabilization Policies, The IMF Stabilization Program

Macroeconomic instability is the situation in which a country has high inflation accompanied by rising budget and trade deficits and a rapidly expanding money supply. One course of action that was increasingly but often reluctantly used by countries facing serious macroeconomic instability (high inflation and severe government budget and foreign-payments deficits) along with growing foreign-debt obligations was to renegotiate loans with private international banks.

The basic idea was to stretch out the payment period for principal and interest or to obtain additional financing on more favorable terms. Typically, however, such debtor countries had to deal with the IMF before a consortium of international banks would agree to refinance or defer existing loan schedules. Relying on the IMF to impose tough stabilization policies (A coordinated set of mostly restrictive fiscal and monetary policies aimed at reducing inflation, cutting budget deficits, and improving the balance of payments), a process known as conditionality, before it agreed to lend funds in excess of their legal IMF quotas, the private banks interpreted successful negotiations with the IMF as a sign that borrowing countries were making serious efforts to reduce payments deficits and earn the foreign exchange needed to repay earlier loans.

CHAPTER SIX

Foreign Aid, Debt, Financial Reform and Development

Unit objective:

At the end of this chapter the student will be able to understand:

- The meaning of foreign aid
- The reason or motivation of foreign aid
- Explain the notion behind foreign direct investment

6.1 Motivations for foreign aid

Foreign aid is the international transfer of public funds in the form of loans or grants either directly from one government to another (bilateral assistance) or indirectly through the vehicle of a multilateral assistance agency such as the World Bank. In addition to export earnings and private foreign direct and portfolio investment, developing countries receive two other major sources of foreign exchange: public (official) bilateral and multilateral development assistance and private (unofficial) assistance provided by nongovernmental organizations. Both of these activities are forms of foreign aid, although only public aid is usually measured in official statistics. In principle, all governmental resource transfers from one country to another should be included in the definition of foreign aid. Even this simple definition, however, raises a number of problems. For one thing, many resource transfers can take disguised forms, such as the granting of preferential tariffs by developed countries to exports of manufactured goods, particularly from the least developed countries.

This permits developing countries to earn more foreign exchange from selling their industrial products in developed-country markets at higher prices than would otherwise be possible. There is consequently a net gain for developing countries and a net loss for developed countries, which amounts to a real resource transfer to the developing world. Such implicit capital transfers, or disguised flows, should be counted in qualifying foreign-aid flows. Normally, however, they are not. However, we should not include all transfers of capital to developing countries, particularly the capital flows of private foreign investors. Private flows represent normal commercial transactions, prompted by commercial considerations of profits and rates of return, and therefore should not be viewed as foreign aid.

Commercial flows of private capital are not a form of foreign assistance, even though they may benefit the developing country in which they take place. Economists have defined foreign aid, therefore, as any flow of capital to a developing country that meets two criteria: (1) Its objective should be noncommercial from the point of view of the donor, and (2) it should be characterized by concessional terms; that is, the interest rate and repayment period for borrowed capital should be

softer (less stringent) than commercial terms. Even this definition can be inappropriate, for it could include military aid, which is both noncommercial and concessional.

Normally, however, military aid is excluded from international economic measurements of foreign-aid flows. The concept of foreign aid that is now widely used and accepted, therefore, is one that encompasses all official grants and concessional loans, in currency or in kind, that are broadly aimed at transferring resources from developed to less developed nations on development, poverty, or income distribution grounds. Unfortunately, there often is a thin line separating purely developmental grants and loans from sources ultimately motivated by security or commercial interests. Just as there are conceptual problems associated with the definition of foreign aid, there are measurement and conceptual problems in the calculation of actual development assistance flows.

In particular, three major problems arise in measuring aid. First, we cannot simply add up the dollar values of grants and loans; each has a different significance to both donor and recipient countries. Loans must be repaid and therefore cost the donor and benefit the recipient less than the nominal value of the loan itself. Conceptually, we should deflate or discount the dollar value of interest-bearing loans before adding them to the value of outright grants. Second, aid can be tied either by source (loans or grants have to be spent on the purchase of donor-country goods and services) or by project (funds can only be used for a specific project, such as a road or a steel mill). In either case, the real value of the aid is reduced because the specified source is likely to be an expensive supplier or the project is not of the highest priority (otherwise, there would be no need to tie the aid).

Furthermore, aid may be tied to the importation of capital-intensive equipment, which may impose an additional real resource cost, in the form of higher unemployment, on the recipient nation. Or the project itself may require the purchase of new machinery and equipment from monopolistic suppliers while existing productive equipment in the same industry is being operated at very low levels of capacity. Finally, we always need to distinguish between the nominal and real value of foreign assistance. Aid flows are usually calculated at nominal levels and tend to show a steady rise over time. However, when deflated for rising prices, the actual real volume of aid from most donor countries declined substantially in recent decades despite a recent uptick.

Concessional terms: Terms for the extension of credit that are more favorable to the borrower than those available through standard financial markets.

Amounts and Allocations: Public Aid

The money volume of **official development assistance (ODA)**, which includes bilateral grants, concessional loans, and technical assistance as well as multilateral flows, has grown from an annual rate of under \$5 billion in 1960 to \$50 billion in 2000 and to over \$128 billion in 2008. However, the

percentage of developed-country GNI allocated to official development assistance declined from 0.51% in 1960 to 0.23% in 2002 before improving to 0.33% by 2005 and to 0.45% in 2008 as part of a campaign to increase assistance in the wake of the continued lag in human development in sub-Saharan Africa a major initiative at the G8 meetings in Britain in 2005.

Although the full promise of these meetings was far from met, some significant progress was made. It remains to be seen how the long recession and fiscal crises in many high-income countries will affect these ratios in the coming years. ODA is allocated in some strange and arbitrary ways. South Asia, where nearly 50% of the world's poorest people live, receives \$8 per person in aid. The Middle East and North Africa, with well over triple South Asia's per capita income, receives nine times the per capita aid!

The patterns of aid become even clearer when examined at the individual country level. In 2008, by far the largest recipient was Iraq, with \$9.9 billion in aid, or approximately \$321 per capita. The second-largest recipient was Afghanistan, at \$4.9 billion or \$168 per capita. Some 20 countries received at least \$1 billion in aid. But India, with by far the largest number of extremely poor people in the world, received just \$2 per person in aid. And while Jordan, a middle-income country, received \$126 per person, Niger, considered the poorest country in the world, received just \$41 per person. Aid per capita to the least developed countries in Africa has increased significantly, however, since 2005. But these per capita receipts were still less than such middle-income countries as Serbia, Bosnia and Herzegovina, Albania, Macedonia, Lebanon, and Georgia, each of which received more than \$100 per capita.

It is clear that the allocation of foreign aid is only partly determined by the relative needs of developing countries. Much bilateral aid seems to be based largely on political and military considerations. Multilateral aid (e.g., from the World Bank and various UN agencies) is somewhat more economically rational, although here too, the rich often seem to attract more resources per than the poor. Because foreign aid is seen differently by donor and recipient countries, we must analyze the giving and receiving process from these two often contradictory viewpoints.

Why Donors Give Aid?

Donor-country governments give aid because it is in their political, strategic, or economic self-interest to do so. Some development assistance may be motivated by moral and humanitarian desires to assist the less fortunate (e.g., emergency food relief and medical programs) and certainly this has been the international rhetoric in the increases in aid in the first decade of the twenty-first century. Still, there is no historical evidence to suggest that over longer periods of time, donor nations assist others without expecting some corresponding benefits (political, economic, military,

counterterrorism, antinarcotics, etc.) in return. We focus here on the foreign-aid motivations of donor nations in two broad but often interrelated categories: political and economic.

Political Motivations Political motivations have been by far the more important for aid-granting nations, especially for the largest donor country, the United States. The United States has viewed foreign aid from its beginnings in the late 1940s under the Marshall Plan, which aimed at reconstructing the war-torn economies of Western Europe, as a means of containing the international spread of communism. When the balance of Cold War interests shifted from Europe to the developing world in the mid-1950s, the policy of containment embodied in the U.S. aid program dictated a shift in emphasis toward political, economic, and military support for “friendly” less developed nations, especially those considered geographically strategic.

Most aid programs to developing countries were therefore oriented more toward purchasing their security and propping up their sometimes shaky regimes than promoting long-term social and economic development. The successive shifts in emphasis from South Asia to Southeast Asia to Latin America to the Middle East and back to Southeast Asia during the 1950s and 1960s and then toward Africa and the Persian Gulf in the late 1970s, the Caribbean and Central America in the 1980s, and the Russian Federation, Bosnia, Ukraine, and the Middle East in the 1990s, with a renewed focus on the Islamic nations after 2001, reflect changes in U.S. strategic, political, security, and economic interests more than changing evaluations of poverty problems and economic need.

Recent increases in aid to African countries with public health crises including HIV assistance may be due in part to concerns that disease may spread internationally or lead to destabilizing state collapse and possible havens for terrorists. Even the Alliance for Progress, inaugurated in the early 1960s with great fanfare and noble rhetoric about promoting Latin American economic development, was formulated primarily as a direct response to the rise of Fidel Castro in Cuba and the perceived threat of communist takeovers in other Latin American countries.

As soon as the security issue lost its urgency and other more pressing problems came to the fore (the war in Vietnam, the rise in U.S. violence, etc.), the Alliance for Progress stagnated and began to fizzle out. Our point is simply that where aid is seen primarily as a means of furthering donor-country interests, the flow of funds tends to vary with the donor’s political assessment of changing international situations and not the relative need of potential recipients.

The behavior of other major donor countries such as Japan, Great Britain, and France has been similar to that of the United States. Although exceptions can be cited (Sweden, Denmark, the Netherlands, Norway, and perhaps Canada), by and large these Western donor countries have used

foreign aid as a political lever to prop up or underpin friendly political regimes in developing countries, regimes whose continued existence they perceived as being in their own national security interests.

It still remains to be seen how much the renewed rhetorical focus on extreme poverty in the period following in 2005 G8 summit in Britain portends a historic change in the prioritization of aid, but there is no doubt that political and business considerations will remain very important.

Economic Motivations: Two-Gap Models and Other Criteria Within the broad context of political and strategic priorities, foreign-aid programs of the developed nations have had a strong economic rationale. This is especially true for Japan, which directs most of its aid to neighboring Asian countries where it has substantial private investments and expanding trade. Even though political motivation may have been of paramount importance for other donors, the economic rationale was at least given lip service as the overriding motivation for assistance. Let us examine the principal economic arguments advanced in support of foreign aid.

Foreign-Exchange Constraints External finance (both loans and grants) can play a critical role in supplementing domestic resources in order to relieve savings or foreign-exchange bottlenecks. This is the so-called two-gap analysis of foreign assistance. The basic argument of the two-gap model is that most developing countries face either a shortage of domestic savings to match investment opportunities or a shortage of foreign exchange to finance needed imports of capital and intermediate goods. Basic two-gap and similar models assume that the savings gap (domestic real resources) and the foreign-exchange gap are unequal in magnitude and that they are essentially independent. The implication is that one of the two gaps will be “binding” for any developing economy at a given point in time.

If, for example, the savings gap is dominant, this would indicate that growth is constrained by domestic investment. Foreign savings may be used as a supplement to domestic savings. (However, decision-makers in a country with a shortage of savings may be unable or unwilling to divert purchasing power from consumption goods to capital goods, either bought domestically or from abroad. As a result, “excess” foreign exchange, including foreign aid, might be spent on the importation of luxury consumption goods.) An outstanding example of savings-gap nations would be the Arab oil exporters during the 1970s. When the foreign-exchange gap is binding, a developing economy has excess productive resources (mostly labor), and all available foreign exchange is being used for imports.

The existence of complementary domestic resources would permit them to undertake new investment projects if they had the external finance to import new capital goods and associated technical assistance. Foreign aid can therefore play a critical role in overcoming the foreign exchange constraint and raising the real rate of economic growth. Algebraically, the simple two-gap model can be formulated as follows:

1. The savings constraint or gap. Starting with the identity that capital inflows (the difference between imports and exports) add to investible resources (domestic savings), the savings-investment restriction can be written as

$$I \leq F + sY \dots\dots\dots (6.1)$$

Where F is the amount of capital inflows; if capital inflows, F , plus domestic saving, sY , exceeds domestic investment, I , and the economy is at full capacity, a savings gap is said to exist.

2. The foreign-exchange constraint or gap. If investment in a developing country has a marginal import share m_1 (typically ranging from 30% to 60%) and the marginal propensity to import out of a unit of noninvestment GNI (usually around 10% to 15%) is given by the parameter m_2 , the foreign exchange constraint or gap can be written as

$$(m_1 - m_2)I + m_2Y - E \leq F \dots\dots\dots (6.2)$$

Where E is the exogenous level of exports; the term F enters both inequality constraints and becomes the critical factor in the analysis. If F , E , and Y are initially assigned an exogenous current value, only one of the two inequalities will prove binding; that is, investment (and therefore the output growth rate) will be constrained to a lower level by one of the inequalities. Countries can therefore be classified according to whether the savings or foreign-exchange constraint is binding. More important from the viewpoint of foreign-aid analysis is the observation that the impact of increased capital inflows will be greater where the foreign-exchange gap (Equation 6.2) rather than the savings gap (Equation 6.1) is binding.

Two-gap models have been used to provide rough estimates of the relative impact of foreign aid on investment and growth in developing nations. The problem is that such gap forecasts are very mechanistic and are themselves constrained by the necessity of fixing import parameters and assigning exogenous values to exports and net capital inflows. In the case of exports, this is

particularly constricting because a liberalization of trade relations between the developed and the developing world would contribute more toward relieving foreign-exchange gaps than foreign aid. Although E and F are substitutable in Equation 6.2, they can have quite different indirect effects, especially in the case where F represents interest-bearing loans that need to be repaid. Thus the alteration of import and export parameters through government policy in both developed and developing countries can have a deep impact on whether the savings or foreign-exchange constraint is restricting the further growth of national output.

A third, “fiscal” gap may also be important, because domestic savings availability for investment and foreign exchange availability for capital goods imports may have little impact on private sector investment and growth without complementary public investments in roads and other forms of infrastructure, or in human capital. Three gap models have been used to account for this in understanding why growth has commonly to pick up during structural adjustment.

Growth and Savings External assistance is also assumed to facilitate and accelerate the process of development by generating additional domestic savings as a result of the higher growth rates that it is presumed to induce. Eventually, it is hoped, the need for concessional aid will disappear as local resources become sufficient to make development self-sustaining. In reality, much aid is not invested, and if it is, the productivity of that investment is often very low. However, among the main reasons for this are the very “strings” attached to foreign aid.

Technical Assistance Financial assistance needs to be supplemented by technical assistance in the form of high-level worker transfers to ensure that aid funds are used most efficiently to generate economic growth. This skill gap-filling process is thus analogous to the financial-gap-filling process mentioned earlier. Sustainable development impact requires a focus on training in recipient countries. Foreign aid (either bilateral or multilateral) that takes the form of the transfer of expert personnel, technicians, scientists, educators, and economic advisers, and particularly their use in training local personnel, rather than a simple transfer of funds.

Absorptive Capacity Finally, the amount of aid is considered in relation to the recipient country’s absorptive capacity, its ability to use aid funds wisely and productively (often meaning as donors want them to be used). Typically, the donor countries decide which developing countries are to receive aid, how much, in what form (loans or grants, financial or technical assistance), for what purpose, and under what conditions on the basis of the donor countries’ assessment of domestic absorptive capacities (particularly for the least developed countries). Absorptive capacity is the ability of a country to absorb foreign private or public financial assistance to use the funds in a productive manner.

But many types of assistance, such as resources for building infrastructure or for training (e.g., of government officials or health or education workers) itself increases absorptive capacity. It has been said that what one donor sees as a constraint on the ability of a country to use conventional aid, another sees as an opportunity to have more leveraged impact with new forms of assistance. In any case, in practice the total amount of aid rarely has much to do with developing-country absorptive capacities because typically, foreign aid is a residual and low-priority element in donor-country expenditures. In most instances, the recipient countries have little say in the matter.

Economic Motivations and Self-Interest The arguments on behalf of foreign aid as a crucial ingredient for successful development should not mask the fact that even at the strictly economic level, definite benefits accrue to donor countries as a result of their aid programs. The strong tendency toward providing interest-bearing loans instead of outright grants and toward tying aid to the exports of donor countries has saddled many countries, often among the least developed, with substantial debt repayment burdens. It has also increased their import costs because aid tied to donor-country exports limits the receiving nation's freedom to shop around for low-cost and suitable capital and intermediate goods. Tied aid in this sense is clearly a second-best option to untied aid (and perhaps also to freer trade through a reduction of developed-country import barriers). For example, a large fraction of U.S. aid has been spent on American consultants and other U.S. businesses.

Tied aid is Foreign aid in the form of bilateral loans or grants that require the recipient country to use the funds to purchase goods or services from the donor country.

Why Recipient Countries Accept Aid

The reasons why developing nations have usually been eager to accept aid, even in its most stringent and restrictive forms; have been given much less attention than the reasons why donors provide aid. The major reason is probably economic. Developing countries have often tended to accept the proposition typically advanced by developed-country economists and supported by reference to success stories such as Taiwan and South Korea to the exclusion of many more failures that aid is a crucial and essential ingredient in the development process. It supplements scarce domestic resources, it helps transform the economy structurally, and it contributes to economic growth.

Thus the economic rationale for aid is based in part on their acceptance of the donor's perceptions of what the poor countries require to promote economic development. Conflicts generally arise, therefore, not out of any disagreement about the role of aid but over its amount and conditions. Naturally, any developing country would like to have more aid in the form of outright grants or long term low-cost loans with a minimum of strings attached. This means not tying aid to donor exports

and granting greater latitude to recipient countries to decide for themselves what is in their best long-run development interests.

Unfortunately, a good deal of aid that comes in this form has either been wasted in showcase but unproductive projects (e.g., an elaborate parliamentary building, an oversize airport) or actually been plundered by corrupt government officials and their local cronies. Much of the criticism of the historical patterns of foreign aid—that it wastes resources, that it bolsters corrupt regimes, that it is appropriated by the rich at the expense of the poor—is justified. Some recipients in the past have accepted aid simply because it was there and they were not held accountable. A few leaders simply wish to leave no stone unturned in their quest for poverty alleviation, as perhaps describes Mozambique in the 1990s. They have been in the minority. The impact of the spread of democracy, press freedom, and the rule of law, including anticorruption drives, on the effectiveness of aid remains an open question.

Second, in some countries, aid is seen by both donor and recipient as providing greater political leverage to the existing leadership to suppress opposition and maintain itself in power. In such instances, assistance takes the form not only of financial-resource transfers but of military and, internal security reinforcement as well. This phenomenon was clearly at work in Central America in the 1980s. The problem is that once aid is accepted, the ability of recipient governments to extricate themselves from implied political or economic obligations to donors and prevent donor governments from interfering in their internal affairs can be greatly diminished.

Finally, whether on grounds of basic humanitarian responsibilities of the rich toward the welfare of the poor or because of a belief that the rich nations owe the poor nations reparations for past exploitation, many proponents of foreign aid in both developed and developing countries believe that rich nations have an obligation to support economic and social development, particularly in the least developed countries. They often link this moral obligation with the need for greater freedom of choice for recipient developing countries in the allocation and use of aid funds. In sum, while there is no doubt that the least developed countries will need more assistance to escape from the vicious circle of poverty, fresh approaches are needed to ensure effectiveness.

The Role of Nongovernmental Organizations in Aid

One of the fastest-growing and most significant forces in the field of development assistance is that provided through private nongovernmental organizations (NGOs). NGOs are voluntary organizations that work with and on behalf of mostly local grassroots organizations in developing countries. They also represent specific local and international interest groups with concerns as diverse as providing emergency relief, protecting child health, promoting women's rights, alleviating poverty, protecting

the environment, increasing food production, and providing rural credit to small farmers and local businesses. NGOs build roads, houses, hospitals, and schools. They work in family-planning clinics and refugee camps. They teach in schools and universities and conduct research on increasing farm yields.

NGOs include religious groups, private foundations and charities, research organizations, and federations of dedicated doctors, nurses, engineers, agricultural scientists, and economists. Many work directly on grassroots rural development projects; others focus on relief efforts for starving or displaced peoples. Some familiar NGOs include Save the Children, CARE, Oxfam, Planned Parenthood, Doctors without Borders, World Vision, the World Wildlife Fund, Habitat for Humanity, Africare, Heifer, Christian Aid, Project HOPE, and Amnesty International. Funding through developed country NGOs for aid activities in developing countries grew from just under \$1 billion in 1970 to over \$23 billion in 2008.

Many NGOs give local control to their developing-country affiliates or other local groups they support. Increasingly, indigenous NGOs such as BRAC in Bangladesh are becoming active in international assistance. NGOs have two important advantages. First, being less constrained by political imperatives, most NGOs are able to work much more effectively at the local level with the people they are trying to assist than massive bilateral and multilateral aid programs could. Second, by working directly with local people's organizations, many NGOs are better able to avoid the suspicion and cynicism on the part of the mostly poor people that they serve that their help is insincere or likely to be short-lived.

It is estimated that NGOs in developing countries are affecting the lives of some 250 million people; the fact that their voices are increasingly being listened to in the halls of developed-country governments and at international conferences on development makes it clear that the nature and focus of foreign aid are changing rapidly. NGOs have several other important comparative advantages in relation to government and the private sector but also some serious limitations, sometimes called "voluntary failure" (with reference to these private voluntary organizations). One critical question is whether international NGOs can sustainably transfer their knowledge and capabilities to domestic NGOs and other community-based organizations.

The Effects of Aid

The issue of the economic effects of aid, especially public aid, like that of the effects of private foreign investment, is fraught with disagreement. On one side are the economic traditionalists, who argue that aid has indeed promoted growth and structural transformation in many developing countries. On the other side are critics who argue that aid does not promote faster growth but may in

fact retard it by substituting for, rather than supplementing, domestic savings and investment and by exacerbating balance of payments deficits as a result of rising debt repayment obligations (when aid takes the form of loans, even if at reduced interest rates) and the linking of aid to donor-country exports. Official aid is further criticized for focusing on and stimulating the growth of the modern sector, thereby increasing the gap in living standards between the rich and the poor in developing countries.

Some critics on the left would even assert that foreign aid has been a force for antidevelopment in the sense that it both retards growth through reduced savings and worsens income inequalities. Rather than relieving economic bottlenecks and filling gaps, aid and for that matter private foreign investment not only widens existing savings and foreign-exchange resource gaps but may even create new ones (e.g., urban-rural or modern-sector-traditional-sector gaps). Critics on the right charge that foreign aid has been a failure because it has been largely appropriated by corrupt bureaucrats, has stifled initiative, and has generally engendered a welfare mentality on the part of recipient nations.

But one of the most promising developments of the new century has been the emphasis on rigorous testing of the impact of development assistance. In 2005, national and multilateral officials concerned with international development met in Paris and agreed to place greater emphasis on monitoring and systematically measuring aid effectiveness. Accompanying this policy emphasis is a growing acceptance of the value of evaluating programs with greater rigor. One major trend is to encourage evaluation through randomized trials. Clearly, not all valuable development activities can be studied with these methods; methods must follow from relevant development economics questions and cannot be the primary driver of the questions that are asked.

And it is often hard to generalize beyond the local experiment to other locations where conditions differ known as the external validity problem. But when feasible and appropriate, randomized trials are a powerful method. In recent years, randomization has been adapted to study a growing range of education, health, microfinance, and social welfare programs. After years of aid weariness, polls have shown that the public is increasingly willing to support increases in government aid budgets and to donate development assistance via NGOs, and the development crisis in many of the least developed countries, especially in sub-Saharan Africa has mobilized public opinion in support of greater development assistance. Poll numbers also suggest that the upturn in public support for aid was at least temporarily weakened in the aftermath of the recent global financial crisis.

The attention to improved assistance to reduce extreme poverty, particularly in its increased focus on the 49 least developed countries at the United Nations' 2010 Millennium Development Goals summit, the improvements in accountability and evaluation of aid that have taken more shape since

the Paris Declaration, and some enhancement of resources are hopeful signs that aid will become more effective and more targeted toward people living in poverty. And foreign aid has played a crucial role in assistance with conflict resolution, post conflict recovery, and making the transition to resumed development. We take up the problem of violent conflict in developing countries in the next section.

6.2 Foreign debt

The accumulation of external debt (Total private and public foreign debt owed by a country) is a common phenomenon of developing countries at the stage of economic development where the supply of domestic savings is low, current account payments deficits are high, and imports of capital are needed to augment domestic resources. Prior to the early 1970s, the external debt of developing countries was relatively small and primarily an official phenomenon, the majority of creditors being foreign governments and international financial institutions such as the IMF, the World Bank, and regional development banks. Most loans were on concessional (low-interest) terms and were extended for purposes of implementing development projects and expanding imports of capital goods.

However, during the late 1970s and early 1980s, commercial banks began playing a large role in international lending by recycling surplus OPEC “petrodollars” and issuing general-purpose loans to developing countries to provide balance of payments support and expansion of export sectors. Although foreign borrowing can be highly beneficial, providing the resources necessary to promote economic growth and development, when poorly managed, it can be very costly. In recent years, these costs have greatly outweighed the benefits for many developing nations. The main cost associated with the accumulation of a large external debt is debt service. Debt service is the payment of amortization (liquidation of the principal) and accumulated interest; it is a contractually fixed charge on domestic real income and savings.

As the size of the debt grows or as interest rates rise, debt service charges increase. Debt service payments must be made with foreign exchange. In other words, debt service obligations can be met only through export earnings, curtailed imports, or further external borrowing. Under normal circumstances, most of a country’s debt service obligations are met by its export earnings. However, should the composition of imports change or should interest rates rise significantly, causing a ballooning of debt service payments, or should export earnings diminish, debt-servicing difficulties are likely to arise. First, it is necessary to understand a fundamental concept known as the basic transfer. The basic transfer of a country is defined as the net foreign exchange inflow or outflow related to its international borrowing.

It is measured as the difference between the net capital inflow and interest payments on the existing accumulated debt. The net capital inflow is simply the difference between the gross inflow and the amortization on past debt. The basic transfer is an important concept because it represents the amount of foreign exchange that a particular developing country is gaining or losing each year from international capital flows. As you will soon discover, the basic transfer turned very negative for developing nations during the 1980s, causing a loss of foreign exchange and a net outflow of capital.

Basic transfer Net foreign exchange inflow or outflow related to a country's international borrowing; the quantitative difference between the net capital inflow (gross inflow minus amortization on past debt) and interest payments on existing accumulated debt.

6.3 Private Foreign Direct Investment and the Multinational Corporation

Few developments have played as critical a role in the extraordinary growth of international trade and capital flows during the past few decades as the rise of the multinational corporation (MNC). An MNC is most simply defined as a corporation or enterprise that conducts and controls productive activities in more than one country. These huge firms are mostly based in North America, Europe, and Japan; but a growing number are based in newly high-income economies such as South Korea and Taiwan. In recent years a much smaller but growing number of MNCs have emerged from upper middle-income countries such as Brazil and even some fast-growing lower-middle income countries, most notably China. MNCs and the resources they bring present a unique opportunity but may pose serious problems for the many developing countries in which they operate. The growth of private foreign direct investment (FDI) in the developing world has been extremely rapid though volatile in recent decades.

Private Foreign Investment: Some Pros and Cons for Development

Traditional Economic Arguments in Support of Private Investment: Filling Savings, Foreign-Exchange, Revenue, and Management Gaps

Therefore, the first and most often cited contribution of private foreign investment to national development (i.e., when this development is defined in terms of GDP growth rates an important implicit conceptual assumption) is its role in filling the resource gap between targeted or desired investment and locally mobilized savings.

A second contribution, analogous to the first, is its contribution to filling the gap between targeted foreign-exchange requirements and those derived from net export earnings plus net public foreign aid.

The third gap said to be filled by foreign investment is the gap between targeted governmental tax revenues and locally raised taxes. By taxing MNC profits and participating financially in their local operations, developing country governments are thought to be better able to mobilize public financial resources for development projects.

Fourth, there is a different type of gap in management, entrepreneurship, technology, and skill presumed to be partly or wholly filled by the local operations of private foreign firms.

Arguments against Private Foreign Investment: Widening Gaps

There are two basic arguments against private foreign investment in general and the activities of MNCs in particular the strictly economic and the more philosophical or ideological. On the economic side, the four gap-filling pro-foreign-investment positions just outlined are countered by the following arguments:

1. Although MNCs provide capital, they may lower domestic savings and investment rates by substituting for private savings, stifling competition through exclusive production agreements with host governments, failing to reinvest much of their profits, generating domestic incomes for groups with lower savings propensities, and inhibiting the expansion of indigenous firms that might supply them with intermediate products by instead importing these goods from overseas affiliates. MNCs also raise a large fraction of their capital locally in the developing country itself, and this may lead to some crowding out of investment of local firms.
2. Although the initial impact of MNC investment is to improve the foreign-exchange position of the recipient nation, its long-run impact may be to reduce foreign-exchange earnings or at least make the net increase smaller than it appeared, as a result of substantial importation of intermediate products and capital goods and because of the overseas repatriation of profits, interest, royalties, management fees, and other funds.
3. Although MNCs do contribute to public revenue in the form of corporate taxes, their contribution is considerably less than it might appear as a result of liberal tax concessions, the practice of transfer pricing, excessive investment allowances, disguised public subsidies, and tariff protection provided by the host government.
4. The management, entrepreneurial skills, ideas, technology, and overseas contacts provided by MNCs may have little impact on developing local sources of these scarce skills and resources and may in fact inhibit their development by stifling the growth of indigenous entrepreneurship as a result of the MNCs' dominance of local markets.

Government policies in developing countries may be directed toward mitigating some of these concerns. Many academic and political thought leaders in developing countries have commonly raised a number of more fundamental objections.

First, the impact of MNCs on development is very uneven, and in many situations, MNC activities reinforce dualistic economic structures and exacerbate income inequalities. They tend to promote the interests of a small number of local factory managers and relatively well-paid modern-sector workers against the interests of the rest by widening wage differentials. They divert resources away from needed food production to the manufacture of sophisticated products catering primarily to the demands of local elites and foreign consumers. And they tend to worsen the imbalance between rural and urban economic opportunities by locating primarily in urban export enclaves and contributing to excessive rural-urban migration.

Second, it is argued that multinationals typically produce inappropriate products (those demanded by a small, rich minority of the local population), stimulate inappropriate consumption patterns through advertising and their monopolistic market power, and do this all with inappropriate (capital-intensive) technologies of production that as a result create comparatively little employment. This is perhaps the major criticism of MNCs in light of the substantial employment problems of developing nations.

Third, as a result of the first two points, local resources tend to be allocated for socially undesirable projects. This in turn tends to aggravate the already sizable inequality between rich and poor and the serious imbalance between urban and rural economic opportunities.

Fourth, multinationals use their economic power to influence government policies in directions unfavorable to development. They are able to extract sizable economic and political concessions from competing governments of other developing countries in the form of excessive protection, tax rebates, investment allowances, and the cheap provision of factory sites and essential social services. This phenomenon is often referred to as a “race to the bottom.” As a result, the private profits of MNCs may exceed social benefits. In some cases, these social returns to host countries may even be negative. Alternatively, an MNC can avoid much local taxation in high-tax countries and shift profits to affiliates in low-tax countries by artificially inflating the price it pays for intermediate products purchased from overseas affiliates so as to lower its stated local profits. This phenomenon, known as transfer pricing, is a common practice of MNCs and one over which host governments can exert little control as long as corporate tax rates differ from one country to another. Some estimates place the lost revenue as a result of transfer pricing in the scores of billions of dollars.

Fifth, MNCs may damage host economies by suppressing domestic entrepreneurship and using their

superior knowledge, worldwide contacts, advertising skills, and range of essential support services to drive out local competitors and inhibit the emergence of small-scale local enterprises. Through the privatization of public corporations and the use of debt-for-equity swaps to reduce debt burdens, MNCs have been able to acquire some of the best and potentially most lucrative local businesses. They can thereby crowd out local investors and appropriate the profits for themselves. For example, in a quantitative study of developing countries outside the Pacific Basin, higher foreign direct investment was accompanied by lower domestic investment, lower national saving, larger current account deficits, and lower economic growth rates.

Finally, at the political level, the fear is often expressed that powerful multinational corporations can gain control over local assets and jobs and can then exert considerable influence on political decisions at all levels. In extreme cases, they may even; either directly by payoffs to corrupt public officials at the highest levels or indirectly by contributions to “friendly” political parties subverts the very political process of host nations (as occurred with International Telephone and Telegraph in the 1970s in Chile).

6.4 Reforming the Financial Systems

Financial Liberalization, Real Interest Rates, Savings, and Investment

The restriction of loans to a few large borrowers, together with the widespread existence of high inflation, growing budget deficits, and negative real interest rates, led to a serious “credit crunch” among developing countries during the 1980s. The global recessions of 1981–1982 and 1987 exposed the frailty of many development bank loans so that by the end of the decade, almost half of these banks were reporting 50% or more of their loans in arrears and another quarter had delinquency rates in excess of 25%. With real interest rates on savings deposits in the negative and expectations of continued inflation and exchange-rate devaluation contributing to substantial capital flight, it is not surprising that few individuals were willing to save.

In addition, commercial banks and other financial intermediaries were subject to numerous lending restrictions and faced mandatory interest-rate ceilings on loan able funds at levels well below market-clearing rates. These artificial interest-rate ceilings were often set by governments seeking to finance their budget deficits through the sale of low-interest bonds to private commercial banks. These banks in turn had to resort to rationing the available credit beyond the normal credit rationing observed in developed economies as a response to adverse selection.

One suggested solution to the problem is to liberalize the financial sector by allowing nominal interest rates to rise to market-clearing levels. This would cause real interest rates to rise to positive

levels and thus remove the explicit interest-rate subsidy accorded to preferred borrowers (rent seekers) who are powerful enough to gain access to the rationed credit. Higher real rates should also generate more domestic saving and investment and permit some borrowers to shift from the unorganized to the organized credit market. The World Bank cites evidence from countries such as Thailand, Turkey, and Kenya, where the liberalization of interest rates generated more saving and investment. However, evidence of the effects of financial reform in Chile during the 1970s revealed many shortcomings of the process. These included the acquisition of numerous banks by large conglomerates, or grupos, who used their new financial resources to buy recently privatized firms or to expand their own companies. When many of their firms faced financial losses, these grupos had to resort to additional funding to avoid bankruptcy. This made the Chilean financial system particularly vulnerable when the debt crisis struck in the 1980s.

Reform and liberalization of the organized money sector is therefore no panacea for the financial systems of developing nations. The early success of South Korea and Taiwan (and before them, Japan) with financial systems that exhibited many of the attributes of repression demonstrates that judicious and selective government intervention can be a stimulus to industrial development. Although there is evidence that the elimination of substantial interest rate distortions can promote greater saving and more rapid economic growth, financial reform must always be accompanied by other more direct measures to make sure that small farmers and investors have access to needed credit. Furthermore, careful supervision of the banking and financial sectors is needed to prevent undue concentration by local elites.

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